

RELION® PROTECTION AND CONTROL

REX615

Installation Manual





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Conformity

This product complies with following directive and regulations.

Directives of the European parliament and of the council:

- Electromagnetic compatibility (EMC) Directive 2014/30/EU
- Low-voltage Directive 2014/35/EU
- RoHS Directive 2011/65/EU
- RoHS Directive (EU) 2015/863 amending Annex II

UK legislations:

- Electromagnetic Compatibility Regulations 2016
- Electrical Equipment (Safety) Regulations 2016
- The Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012

These conformities are the result of tests conducted by the third-party testing in accordance with the product standard EN / BS EN 60255-26 for the EMC directive / regulation, and with the product standards EN / BS EN 60255-1 and EN / BS EN 60255-27 for the low voltage directive / safety regulation.

The product is designed in accordance with the international standards of the IEC 60255 series.

Safety information



Dangerous voltages can occur on the connectors, even though the auxiliary voltage has been disconnected.



Non-observance can result in death, personal injury or substantial property damage.



Only a competent electrician is allowed to carry out the electrical installation.



National and local electrical safety regulations must always be followed.



The frame of the protection relay has to be carefully earthed.



When the plug-in unit has been detached from the case, do not touch the inside of the case. The relay case internals may contain high voltage potential and touching these may cause personal injury.



The protection relay contains components which are sensitive to electrostatic discharge. Unnecessary touching of electronic components must therefore be avoided.



Whenever changes are made in the protection relay, measures should be taken to avoid inadvertent tripping.

Contents

1	Introduction.....	11
1.1	This manual.....	11
1.2	Intended audience.....	11
1.3	Product documentation.....	12
1.3.1	Product documentation set.....	12
1.3.2	Document revision history.....	12
1.3.3	Related documentation.....	12
1.4	Symbols and conventions.....	13
1.4.1	Symbols.....	13
1.4.2	Document conventions.....	13
2	Environmental aspects.....	15
2.1	Sustainable development.....	15
2.2	Disposal of a device	15
3	Unpacking, inspecting and storing.....	16
3.1	Removing transport packaging.....	16
3.2	Inspecting product and delivery items.....	17
3.2.1	Identifying product.....	17
3.2.2	Checking delivery items.....	17
3.2.3	Inspecting product.....	17
3.2.4	Returning a product damaged in transportation.....	17
3.3	Storing.....	17
4	Mounting.....	18
4.1	Housing.....	18
4.2	Checking environmental conditions and mounting space.....	19
4.2.1	Handling LHMI.....	19
4.3	Detaching and installing plug-in unit.....	20
4.3.1	Detaching plug-in unit.....	20
4.3.2	Installing plug-in unit.....	21
4.3.3	Sealing handle.....	25
4.3.4	Securing handle.....	26
4.4	Mounting protection relay.....	26
4.4.1	Required tools.....	26
4.4.2	Flush mounting protection relay	27
4.4.3	Semi-flush mounting protection relay	32
4.4.4	Semi-flush mounting protection relay inclined	36

4.4.5	Rack mounting protection relay	38
4.4.6	Wall mounting protection relay.....	42
4.4.7	Rack mounting protection relay and test switch RTXP into 19" equipment frame	49
4.4.8	Rack mounting protection relay into combiflex 19" equipment frame (Type RHGT 19" 4U variant C)	51
4.4.9	Replacing a Strömberg SP_J40 series relay by a standard size REX615	56
4.4.10	Replacing two Strömberg SP_J3 series relays by a standard size REX615	58
4.4.11	Replacing a Strömberg SP_J3/J6 series relay in 19" rack by a standard size REX615.....	59
4.4.12	Replacing a Strömberg SP_J6 series relay by a standard size REX615	61
4.4.13	Replacing three BBC S_ series relays by a standard size REX615	62
4.4.14	Replacing a SPA 300 series relay by a standard size REX615	64
4.4.15	Replacing a SPA 300 series relay in 19" rack by a standard size REX615	66
4.4.16	Flush mounting the standard size relay with protection cover.....	66
4.4.17	Flush mounting the wide size relay with protection cover.....	68
5	Connecting.....	71
5.1	Required tools.....	71
5.1.1	Tools for connector installation.....	71
5.2	Connecting wires	72
5.2.1	Connecting wires to screw-compression type connectors.....	72
5.2.2	Connecting ring-lug type wires.....	73
5.3	Connecting protective earthing	73
5.4	Connecting analog signals.....	74
5.4.1	Connecting current and voltage inputs.....	75
5.4.2	Connecting sensors (non-conventional instrument transformers).....	84
5.4.3	Connecting RTD and mA inputs.....	85
5.4.4	Connecting protection relay with a test switch	87
5.4.5	Shielding RTD/mA instrument cables.....	87
5.5	Connecting binary signals.....	90
5.6	Connecting power supply.....	94
5.7	Connecting communication.....	94
5.8	Mounting lens sensors for an arc protection system	95
5.9	Energizing protection relay	95
6	Removing, repairing and exchanging.....	97
6.1	Product life cycle.....	97
6.2	Checking protection relay information.....	97
6.3	Removing protection relay	98
6.4	Sending protection relay for repair	98
6.5	Exchanging protection relay	98

7 Technical data.....100

 7.1 Case and LHMI display variants..... 100

 7.1.1 Front side of the protection relay..... 100

 7.1.2 Rear side of the protection relay 102

 7.2 Dimensions.....104

 7.3 Degree of protection of the protection relay.....107

8 Accessories and ordering data.....108

9 Glossary..... 110

1 Introduction

1.1 This manual

The installation manual contains instructions on how to install the protection relay. The manual provides procedures for mechanical and electrical installation. The chapters are organized in the chronological order in which the relay should be installed.

1.2 Intended audience

This manual addresses the personnel responsible for installing the product hardware.

The installation personnel must have basic knowledge of handling electronic equipment.

1.3 Product documentation

1.3.1 Product documentation set

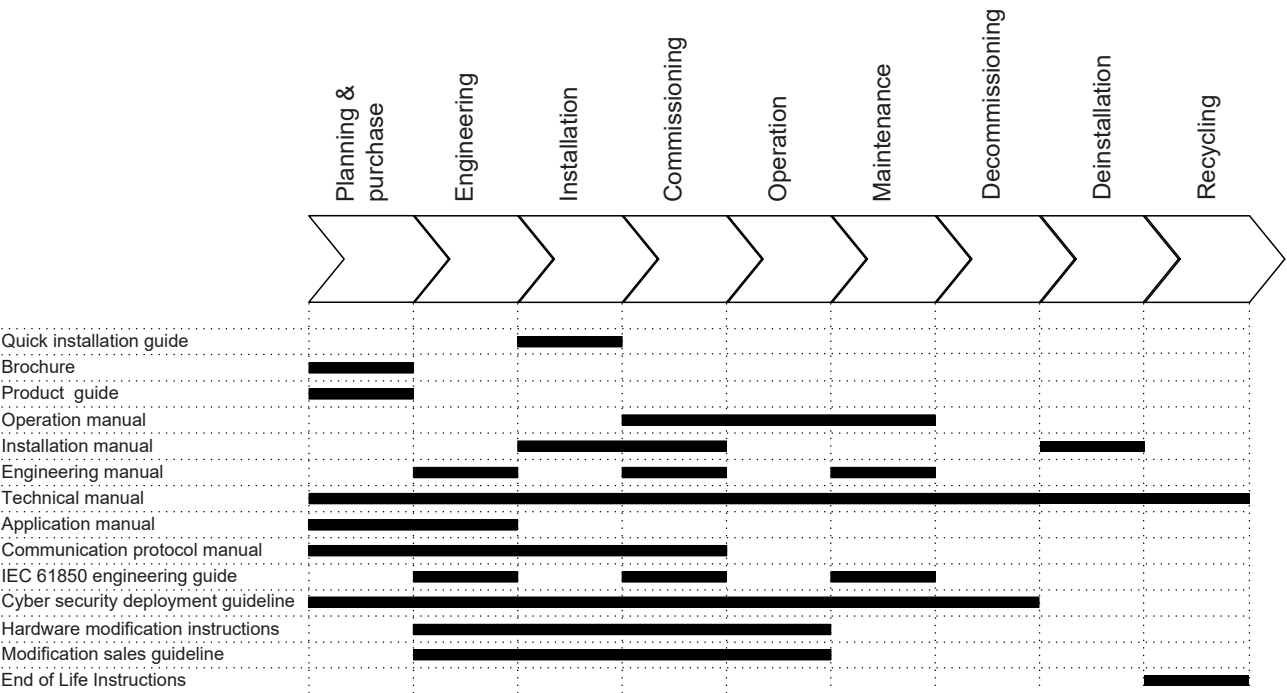


Figure 1: The intended use of documents during the product life cycle

1.3.2 Document revision history

Document revision/date	Product release	History
A/2024-04-05	PCL1	First release
B/2025-10-29	PCL1	Content updated
C/2026-02-25	PCL2	Content updated to correspond to the product connectivity level



Download the latest documents from the ABB Web site www.abb.com/medium-voltage.

1.3.3 Related documentation



Download the latest documents from the ABB Web site www.abb.com/medium-voltage.

1.4 Symbols and conventions

1.4.1 Symbols



The electrical warning icon indicates the presence of a hazard which could result in electrical shock.



The warning icon indicates the presence of a hazard which could result in personal injury.



The caution icon indicates important information or warning related to the concept discussed in the text. It might indicate the presence of a hazard which could result in corruption of software or damage to equipment or property.



The information icon alerts the reader of important facts and conditions.



The tip icon indicates advice on, for example, how to design your project or how to use a certain function.

Although warning hazards are related to personal injury, it is necessary to understand that under certain operational conditions, operation of damaged equipment may result in degraded process performance leading to personal injury or death. Therefore, comply fully with all warning and caution notices.

1.4.2 Document conventions

A particular convention may not be used in this manual.

- Abbreviations and acronyms are spelled out in the glossary. The glossary also contains definitions of important terms.
- Push-button navigation in the LHMI menu structure is presented by using the push-button icons.

To navigate between the options, use  and .

- Menu paths are presented in bold.
- Select **Main menu** > **Settings**.
- LHMI messages are shown in Courier font.

To save the changes in nonvolatile memory, select *Yes* and press .

- Parameter names are shown in italics.

The function can be enabled and disabled with the *Operation* setting.

- Parameter values are indicated with quotation marks.

The corresponding parameter values are "On" and "Off".

- Input/output messages and monitored data names are shown in Courier font.

When the function starts, the `START` output is set to `TRUE`.

- This document assumes that the parameter setting visibility is "Advanced".
- Values of quantities are expressed with a number and an SI unit. The corresponding imperial units may be given in parentheses.
- Protective earthing is indicated in figures with the symbol .

2 Environmental aspects

2.1 Sustainable development

Sustainability has been taken into account from the beginning of the product design including the pro-environmental manufacturing process, long life time, operation reliability and disposing of the device.

The choice of materials and suppliers has been made according to the EU RoHS directive (2011/65/EU) and EU RoHS Directive (2015/863/EU). These directives limit the use of hazardous substances.

Operational reliability and long life time have been ensured with extensive testing during the design and manufacturing processes. Moreover, long life time is supported by maintenance and repair services as well as by the availability of spare parts.

Design and manufacturing have been done under a certified environmental system. The effectiveness of the environmental system is constantly evaluated by an external auditing body. We follow environmental rules and regulations systematically to evaluate their effect on our products and processes.

A life cycle assessment has been done according to ISO 14040 and ISO 14044 to analyze the environmental impacts of the product throughout its entire lifecycle. The results are available in the environmental product declaration (EPD) according to ISO 14025, and this also includes a description of the constituent materials.

2.2 Disposal of a device

Before disposal, the protection relay needs to be decommissioned, and all sensitive information needs to be deleted from the relay. Instructions how to securely take the product out of use are available in the cyber security deployment guideline. After decommissioning the relay can be de-installed according to the installation manual.

End of life instructions are provided to show how to safely dismantle the relay and enable responsible recycling and disposal. Definitions and regulations of hazardous materials are country-specific and change when the knowledge of materials increases.

3 Unpacking, inspecting and storing

3.1 Removing transport packaging

Protection relays require careful handling.

1. Examine the delivered products to ensure that they have not been damaged during the transport.
2. Remove the transport packaging carefully without force.
3. Attach the protective film (supplied with the protection relay) on the top side of the unit for the installation phase.



Before connecting the auxiliary power, remove the protective film from top of the protection relay.



Figure 2: Protective film



The cardboard packaging material is 100% recyclable.

3.2 Inspecting product and delivery items

3.2.1 Identifying product

1. Locate the protection relay's order number from the label on top of the plug-in unit.
2. Compare the protection relay's order number with the ordering information to verify that the received product is correct.

3.2.2 Checking delivery items

- Check that all items are included in the delivery in accordance with the delivery documents.

3.2.3 Inspecting product

Protection relays require careful handling before installation on site.

- Check the protection relay to see if any damage occurred during transportation.

If the protection relay has been damaged during transportation, make a claim against the transport contractor, and notify the local ABB representative.

3.2.4 Returning a product damaged in transportation

If damage has occurred during transportation, appropriate actions must be taken against the latest carrier. Inform the nearest ABB office or representative.

ABB should be notified immediately if there are any discrepancies in relation to the delivery documents.

3.3 Storing

If the protection relay is stored before installation, it must be done in the original transport packaging in a dry and dust free place.

Observe the environmental requirements stated in the technical manual.

4 Mounting

4.1 Housing

REX615 is available with two different housings sizes, standard and wide. The wide version offers more module slots for binary I/O, RTD/mA and analogue inputs compared to standard version. Otherwise, the mechanical construction is the same in both versions. Mounting options and technical details are described in following chapters.



Figure 3: REX615 standard



Figure 4: REX615 wide

4.2 Checking environmental conditions and mounting space

The mechanical and electrical environmental conditions at the installation site must be within the limits described in the technical manual.

- Avoid installation in dusty or damp places.
- Avoid places susceptible to rapid temperature variations, powerful vibrations and shocks, surge voltages of high amplitude and fast rise time, strong induced magnetic fields or similar extreme conditions.
- Check that sufficient space is available.

Sufficient space is needed around the protection relay to allow space and access to wires and optical fibers and also to provide sufficient ventilation to the protection relay and to enable maintenance and future modifications.

- Ensure that flush-mounted protection relays can be added and replaced without excessive dismantling.
- Check the allowed minimum bending radius from the optical cable manufacturer.

4.2.1 Handling LHMI

- More information about plug-in unit handling can be found in REX615 Quick installation guide.



It is not allowed to push or touch the screen during the plug-in unit handling.

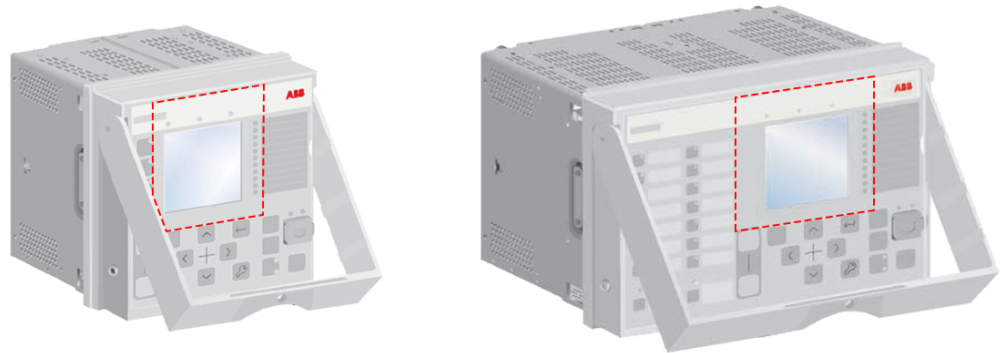


Figure 5: An area that is not allowed to be pushed or touched during the plug-in unit handling

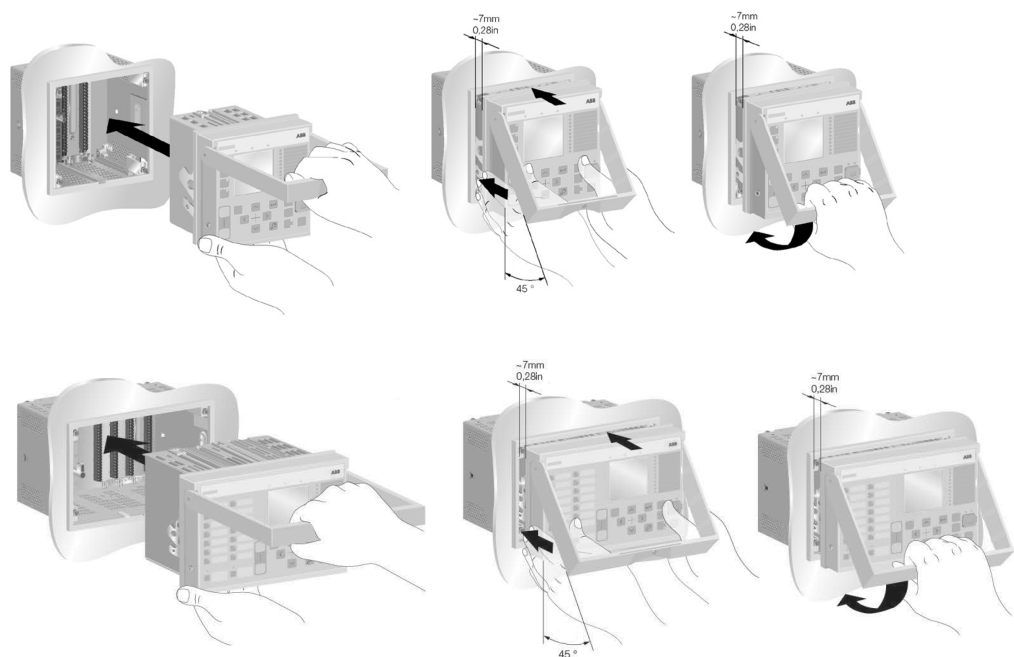


Figure 6: Example of hand positions when handling the plug-in unit

4.3 Detaching and installing plug-in unit

4.3.1 Detaching plug-in unit



Before detaching the plug-in unit from the case, the auxiliary voltage must be disconnected.

1. Turn off the power.
2. Open the seal on the front panel by removing the sealing wire and screw the sealing screw all the way in.
3. Lift the handle to 90 degrees to release the latching mechanism.
The plug-in unit is pushed about 7 mm (0.28 inches) out of the case and the connectors are separated.
4. Pull the unit out of the case.

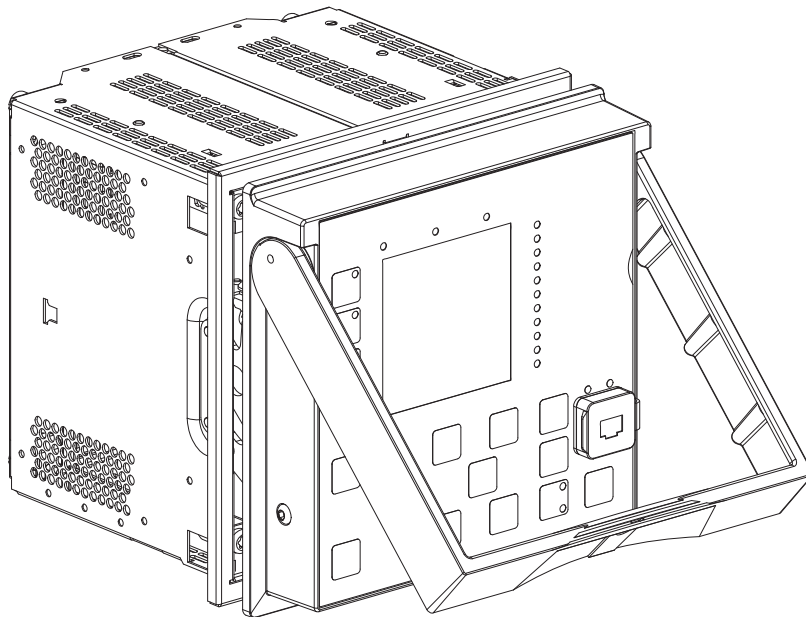


Figure 7: Detaching a plug-in unit from the case



The protection relay features an automatic short-circuit mechanism in the CT connector. Therefore, detaching the plug-in unit will not open the secondary circuit of the CT which could cause dangerously high voltages.



Do not touch terminals inside the case after removing the plug-in unit. Live terminals can be inside the case.



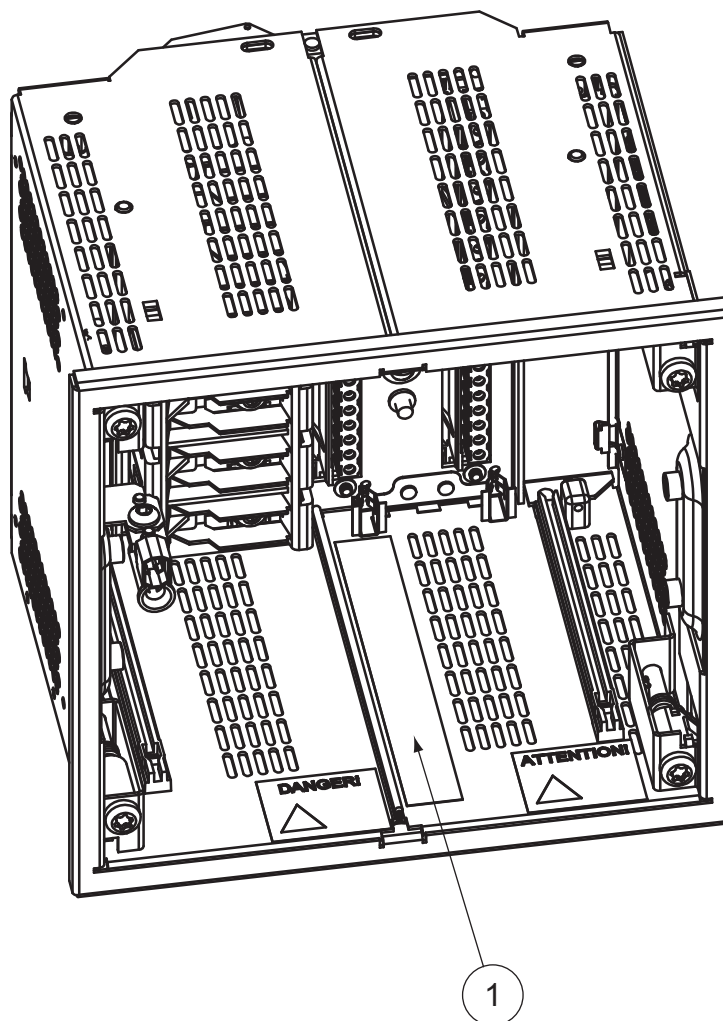
The signal connectors are left open when the plug-in unit is detached.

4.3.2 Installing plug-in unit

The protection relay is constructed in a way that a plug-in unit with voltage- or current-measuring inputs can only be plugged into a corresponding case. This prevents fitting an unsuitable plug-in unit into a wrong case.



Before fitting the plug-in unit into the case, check that the unit and the case have the same serial number.



1 Rating label with serial number

Figure 8: Serial number in the case



Forcing an unsuitable plug-in unit into the case can break both the plug-in unit and the case and may cause danger.

1. Lift the handle 90 degrees and push the plug-in unit into the case.

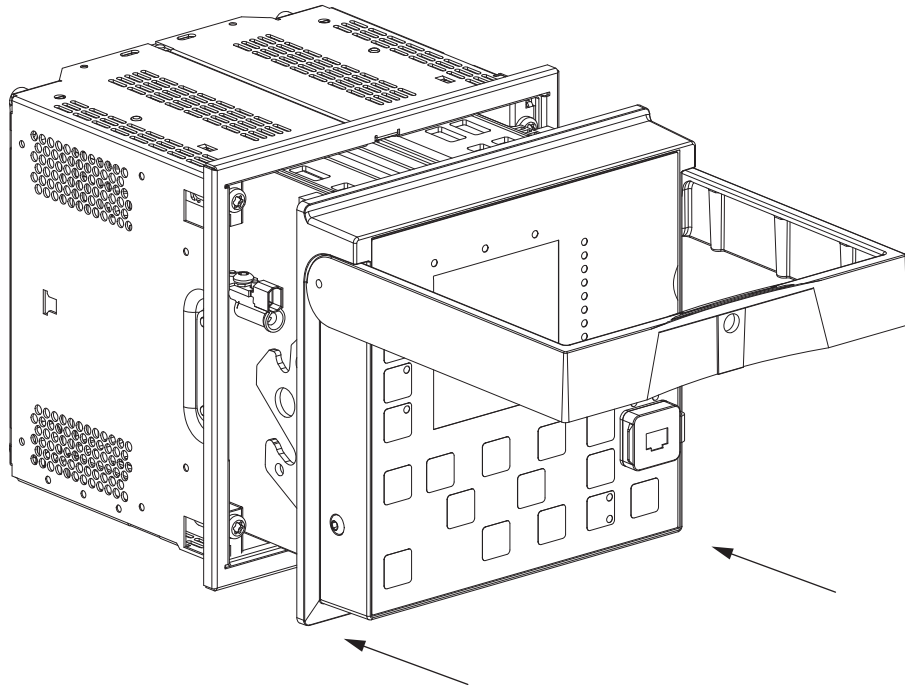
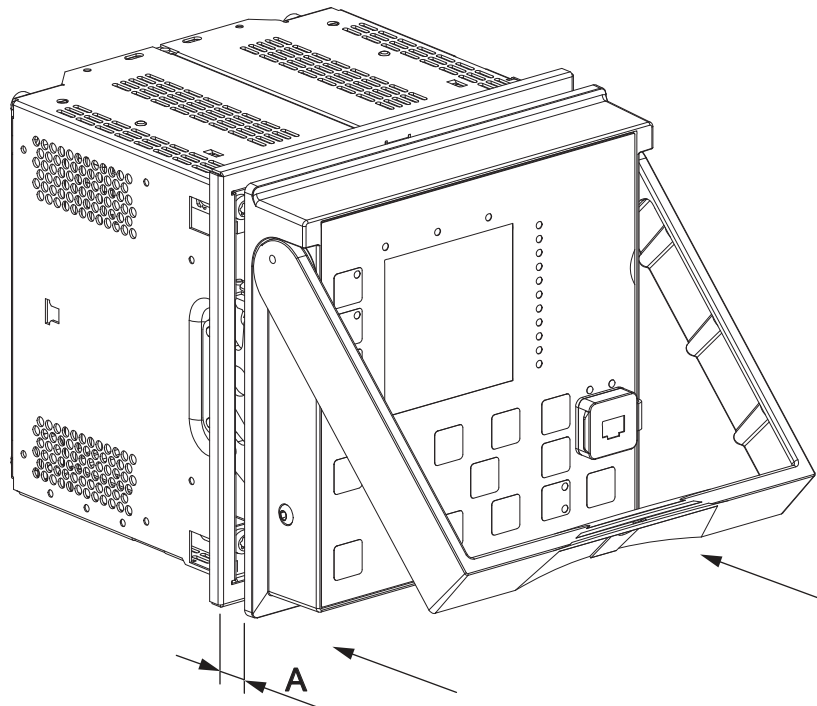


Figure 9: Installing a plug-in unit into the case

2. Let the handle swing down about 45 degrees. At the same time, push the plug-in unit into the case as far as it goes. Plug-in unit stops at about 7 mm (0.28 in) distance from the case.



A 7 mm (0.28 in)

Figure 10: Pushing the plug-in unit into the case

3. Turn down the handle to push the plug-in unit into its final position in the case.

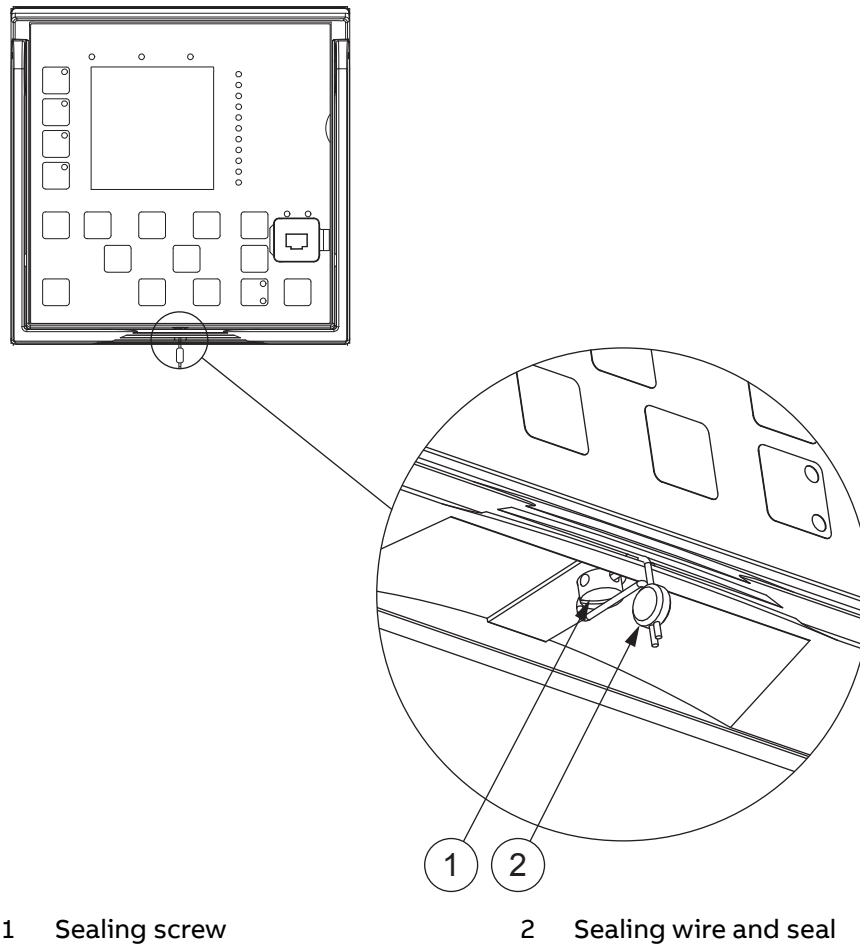


The handle must be locked or sealed to secure the mechanical performance under any conditions caused by vibration, pressure, shock or bump, seismic activity or other equivalent circumstances.

4.3.3 Sealing handle

The front panel of the protection relay has an integrated sealing screw. By default the screw is screwed all the way in and is not used when installing or detaching the plug-in unit.

1. Open the sealing screw about nine turns.
2. Thread a sealing wire through the holes in the sealing screw and the handle.



1 Sealing screw

2 Sealing wire and seal

Figure 11: Sealing screw

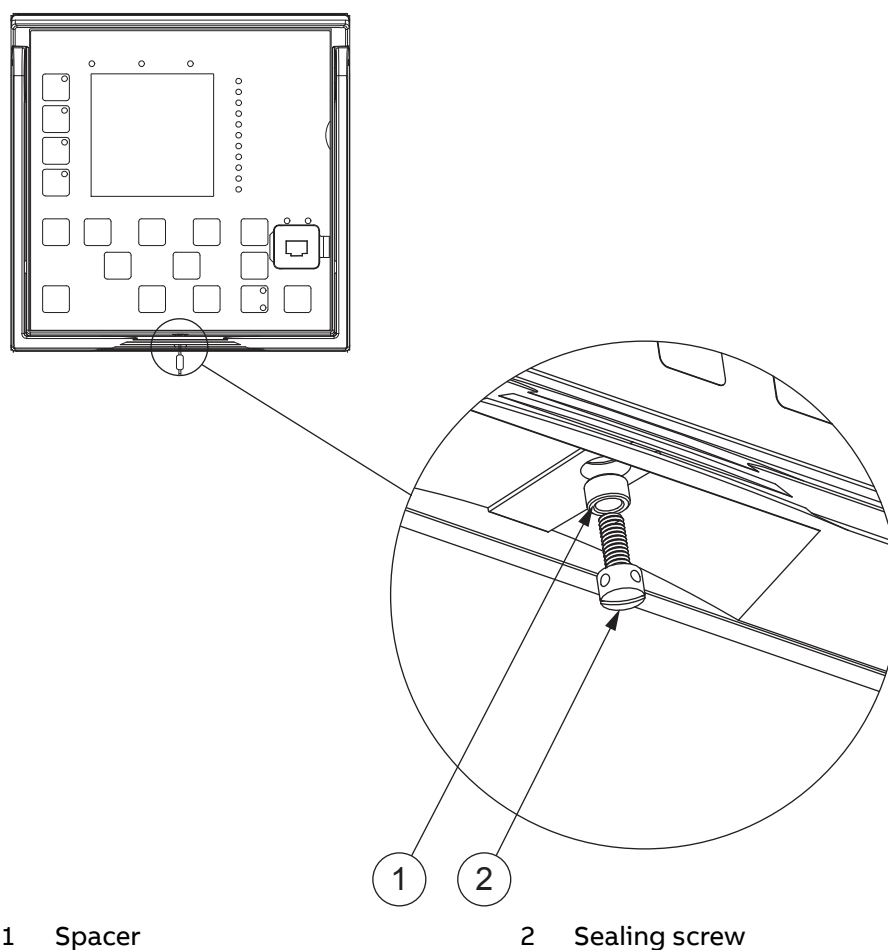
4.3.4 Securing handle

The front panel of the protection relay has an integrated sealing screw. By default, the screw is screwed all the way in and not used when installing or detaching the plug-in unit. Instead of sealing the plug-in unit, the sealing screw and the spacer supplied with the protection relay can be used for securing the handle in place.

1. Fully open the sealing screw and remove it.
2. Re-insert the sealing screw with the spacer.



The protection relay packaging includes a plastic bag containing loose parts such as the spacer.



1 Spacer

2 Sealing screw

Figure 12: Sealing screw with spacer

4.4 Mounting protection relay

4.4.1 Required tools

- T25 Torx screwdriver for mounting the case
- T20 Torx screwdriver for connecting the protective earthing



Only use adjustable torque screwdrivers.

4.4.2 Flush mounting protection relay

All the mounting elements are integrated in the protection relay.

Requirements for installation:

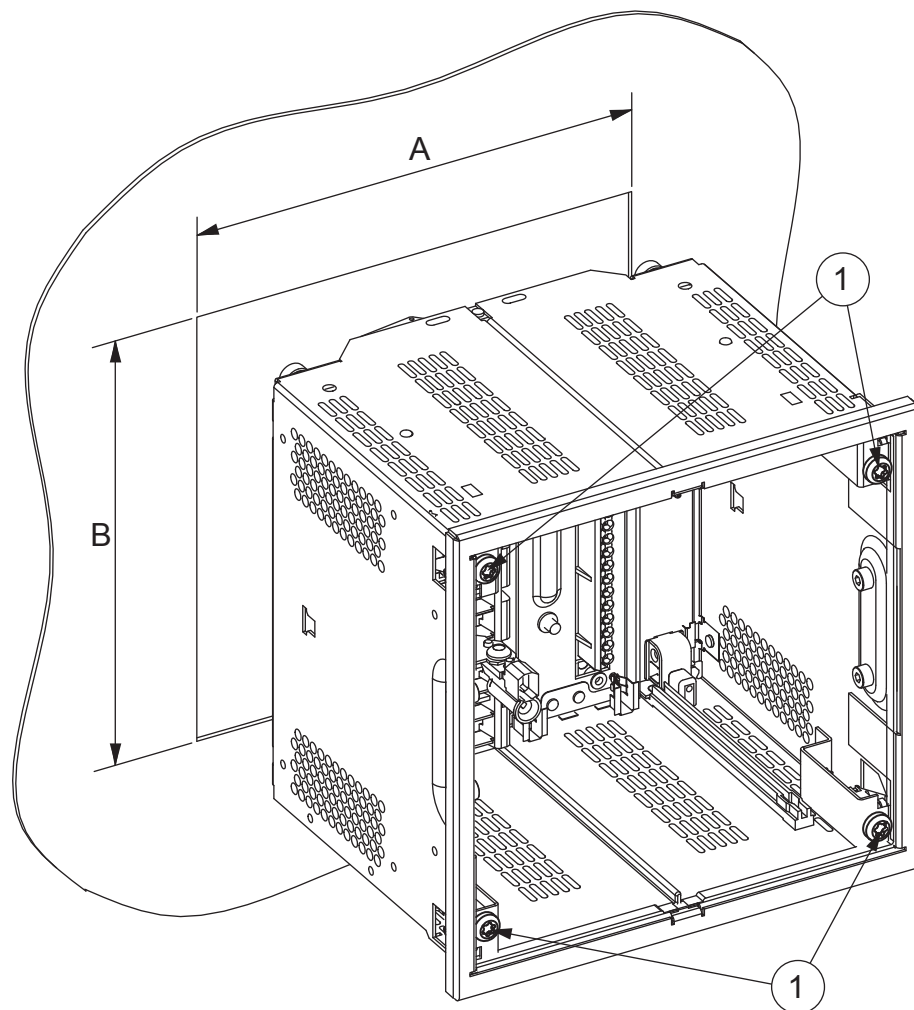
- Panel cut-out of 165.5 × 161.5 mm (6.52 x 6.36 inches) standard case
- Panel cut-out of 248 x 162 mm (9.76 x 6.38 inches) wide case
- Depth behind the panel 153 mm (6.02 inches)



A protection relay equipped with optical connections requires a minimum depth of 180 mm (7.09 inches). The allowed minimum bending radius has to be checked from the optical cable manufacturer.

1. Loosen the four M5 fixing screws in the case to fit the case into the panel cut-out.

2. Mount the case to the panel cut-out.



A 165.5 ±1 mm (6.52 inches)

B 161.5 ±1 mm (6.36 inches)

1 M5 fixing screws

Figure 13: Flush mounting a standard case into a panel cut-out

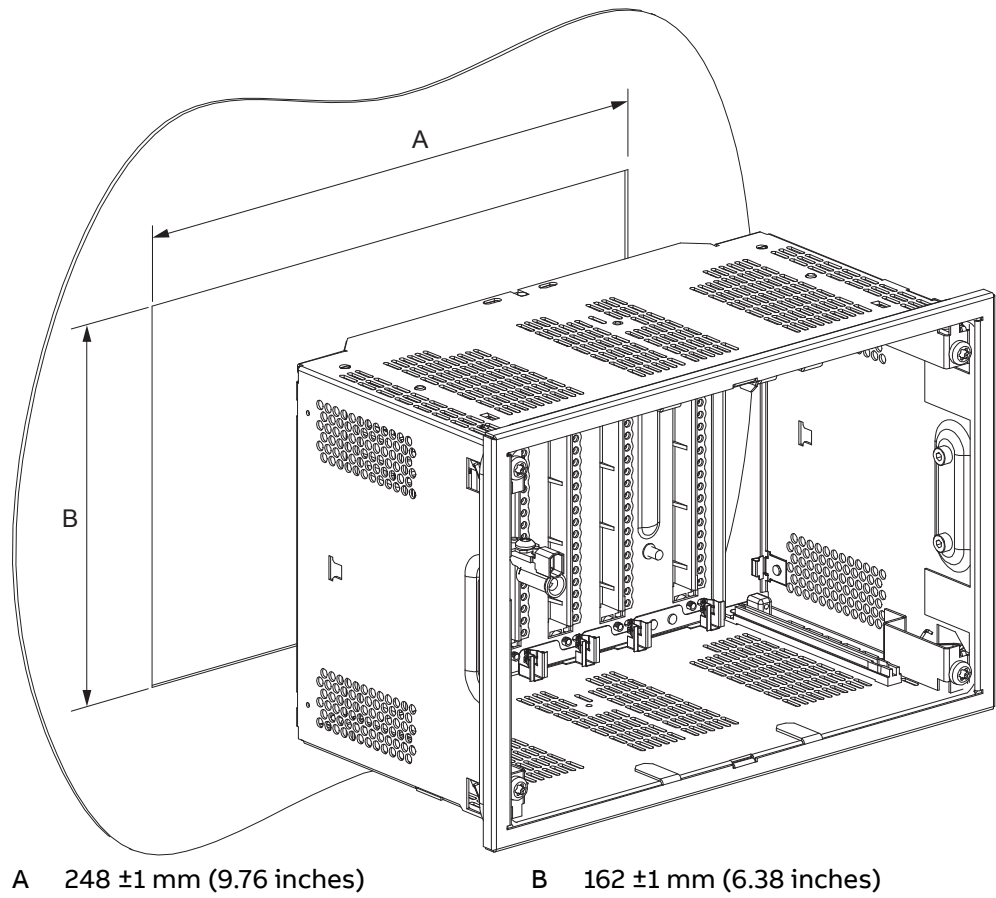


Figure 14: Flush mounting a wide case into a panel cut-out

3. Tighten the M5 (T25) screws.



The allowed range for the fixing screws' tightening torque is 0.7...1 Nm (0.52...0.74 foot-pounds).

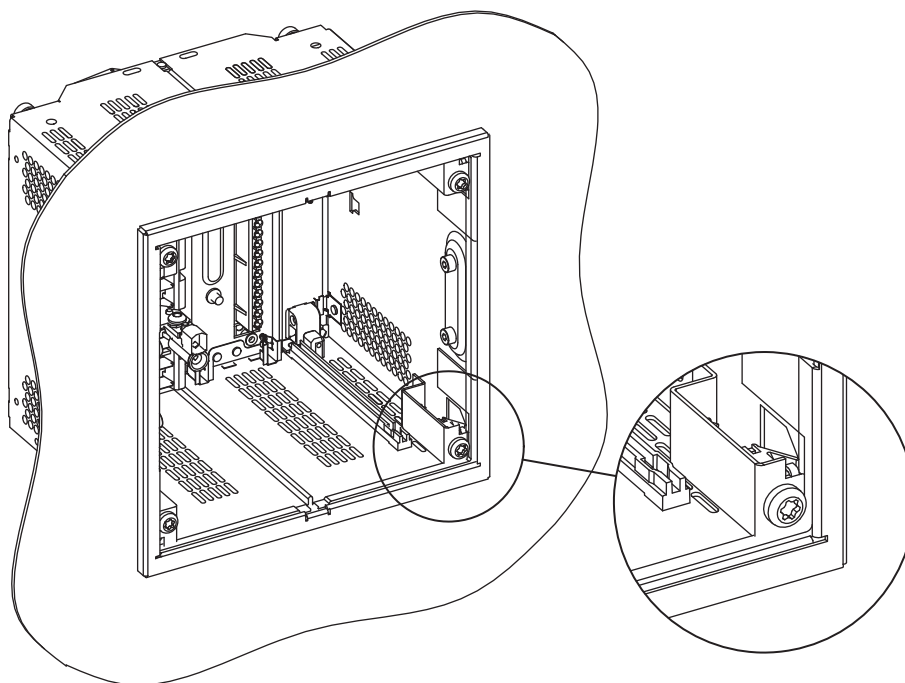
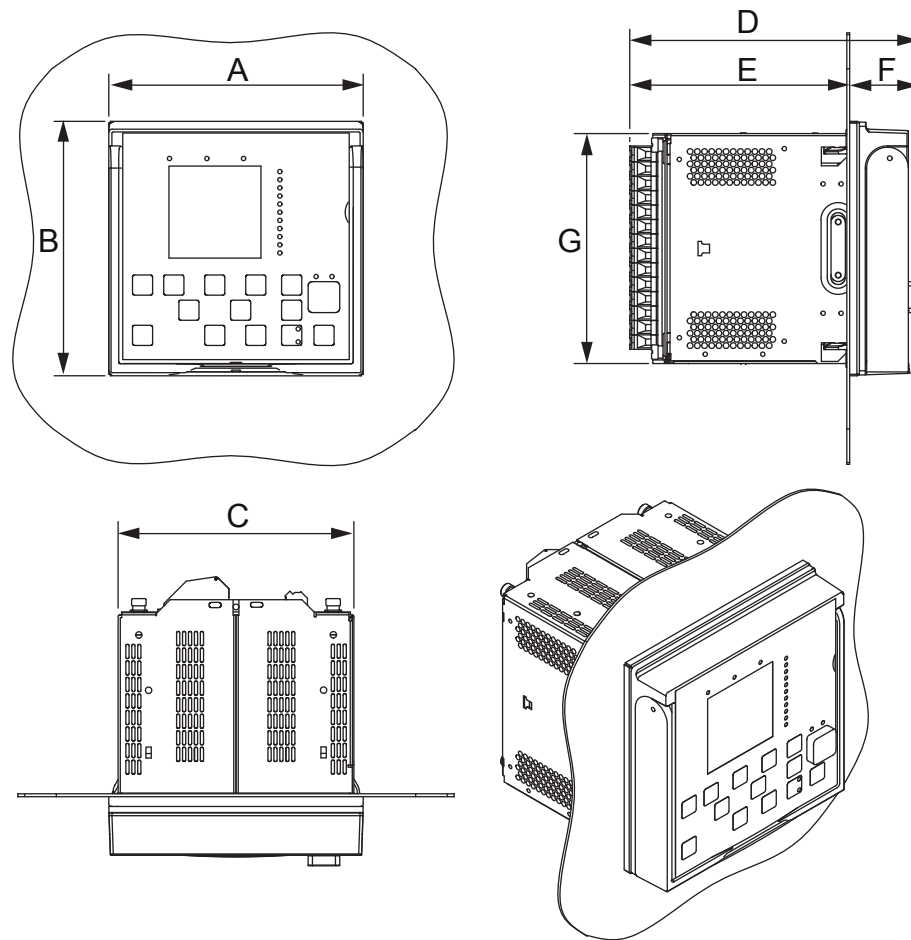


Figure 15: Flush mounted case, tightening the M5 fixing screws

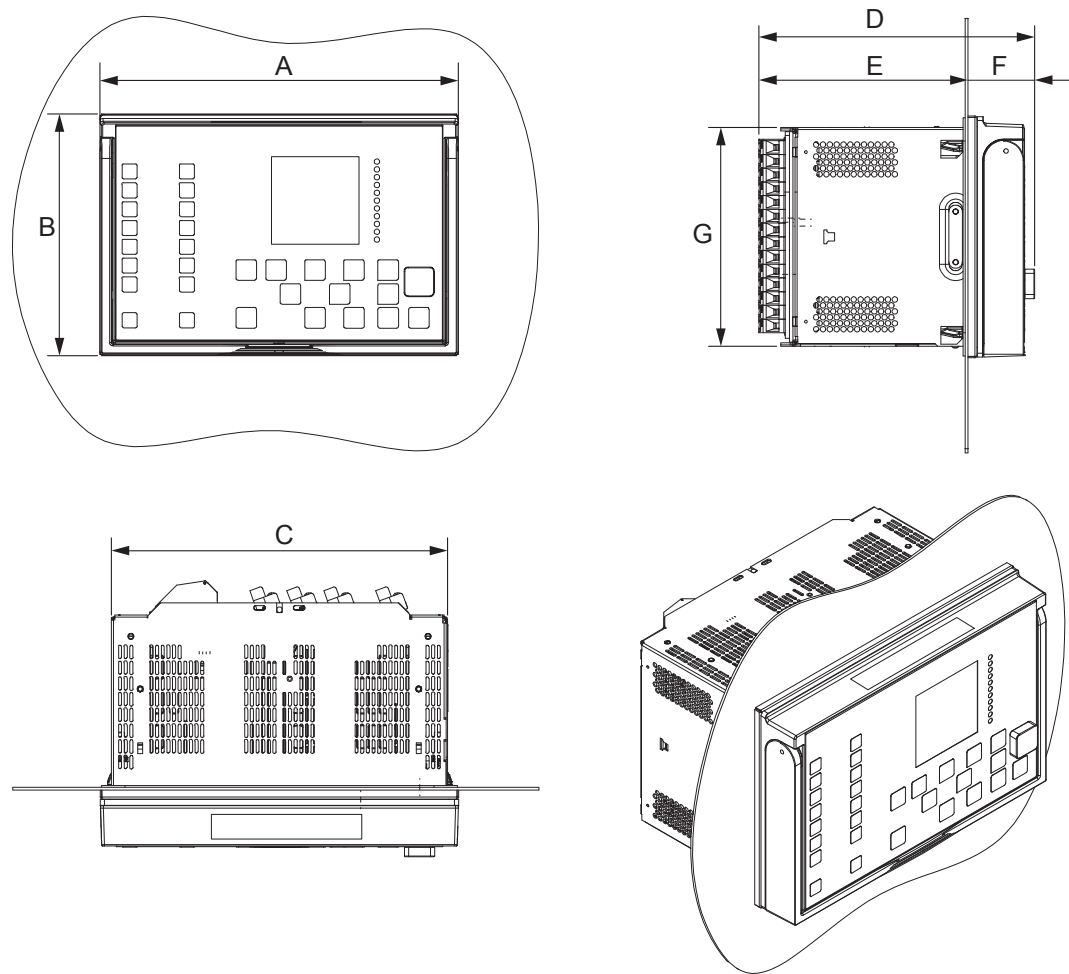
4. Install the plug-in unit into the case.
There is a protective film on the top side of the protection relay. Its purpose is to prevent debris falling inside the unit while installing electrical wiring. Remove the protective film before energizing the protection relay.
5. Seal or secure the handle.



A 177 mm (6.97 inches)
 B 177 mm (6.97 inches)
 C 164 mm (6.46 inches)

E 153 mm (6.02 inches)
 F 48 mm (1.89 inches)
 G 160 mm (6.30 inches)

Figure 16: Flush mounted standard case and plug-in unit



A 262 mm (10.31 inches)
 B 177 mm (6.97 inches)
 C 246 mm (9.69 inches)
 D 201 mm (7.91 inches)

E 153 mm (6.02 inches)
 F 48 mm (1.89 inches)
 G 160 mm (6.30 inches)

Figure 17: Flush mounted wide case and plug-in unit

4.4.3 Semi-flush mounting protection relay

A mounting kit is needed for semi-flush mounting the protection relay. In addition to the detailed mounting instructions, the mounting kit includes:

- Raising frame
- Gasket
- Screws

Requirements for installation:

- Panel cut-out of 165.5 × 161.5 mm (6.52 x 6.36 inches) standard case with mounting holes
- Panel cut-out of 248 x 162 mm (9.76 x 6.38 inches) wide case with mounting holes
- Depth behind the panel 103 mm (4.06 inches)
- When IP 54 degree of protection (according to IEC 60529) is required for the front side, a gasket has to be used in the installation.

1. Mount the raising frame into the panel cut-out with four screws.

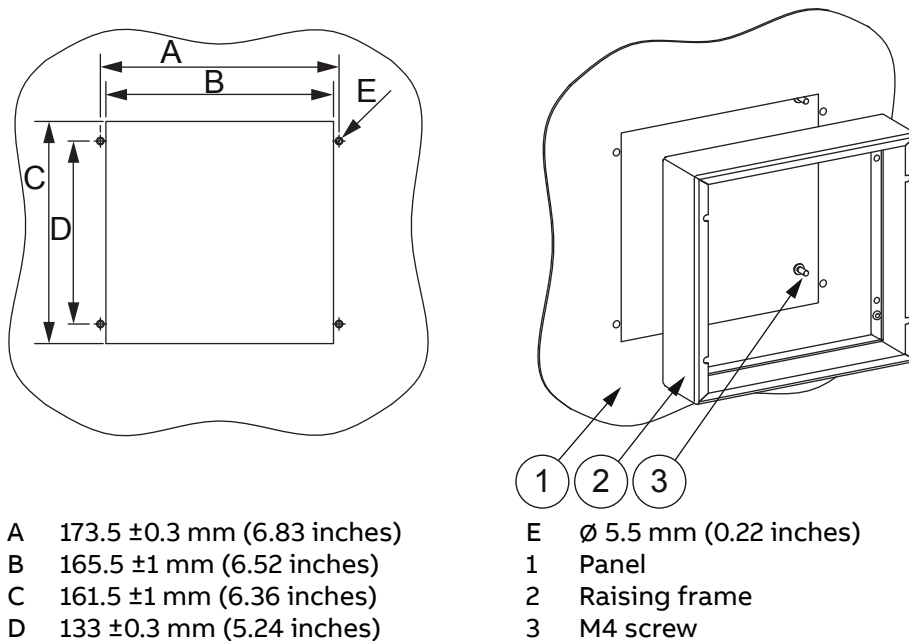


Figure 18: Mounting the standard size raising frame

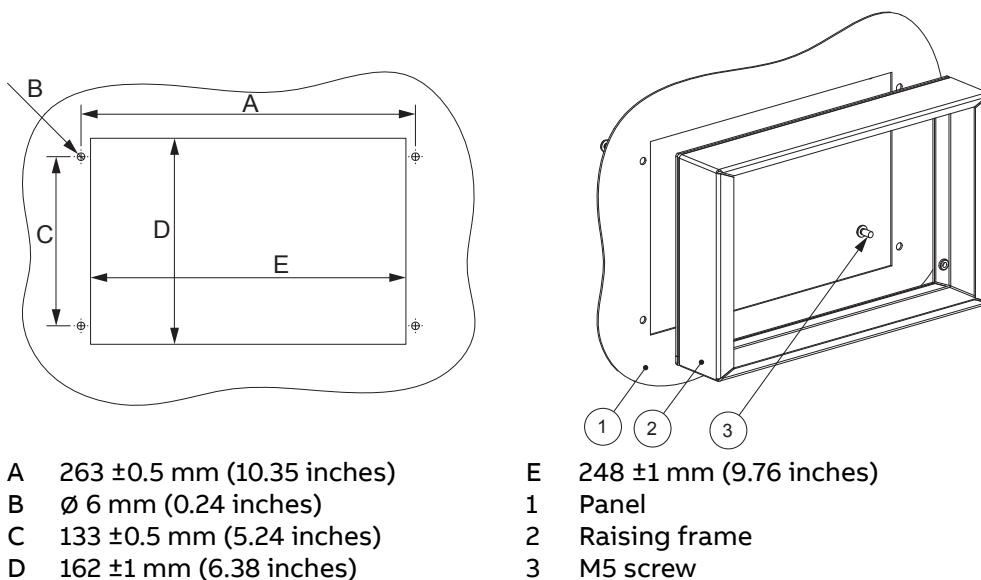


Figure 19: Mounting the wide size raising frame

2. Loosen the four M5 fixing screws in the case to fit the case to the raising frame.
3. Remove the protective film temporarily from the top side of the case.

4. Mount the case to the raising frame.

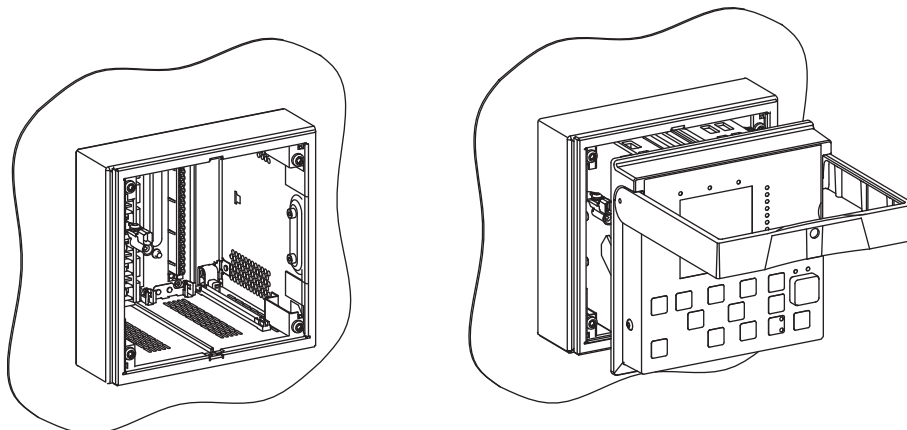


Figure 20: Mounting the case

5. Tighten the M5 screws.



The allowed range for the fixing screws' tightening torque is 0.7...1 Nm (0.52...0.74 foot-pounds).

6. Attach the protective film back on the top side of the case.
7. Install the plug-in unit into the case.

The purpose of the protective film is to prevent debris falling inside the unit while installing electrical wiring. Remove the protective film before energizing the protection relay.

8. Seal or secure the handle.

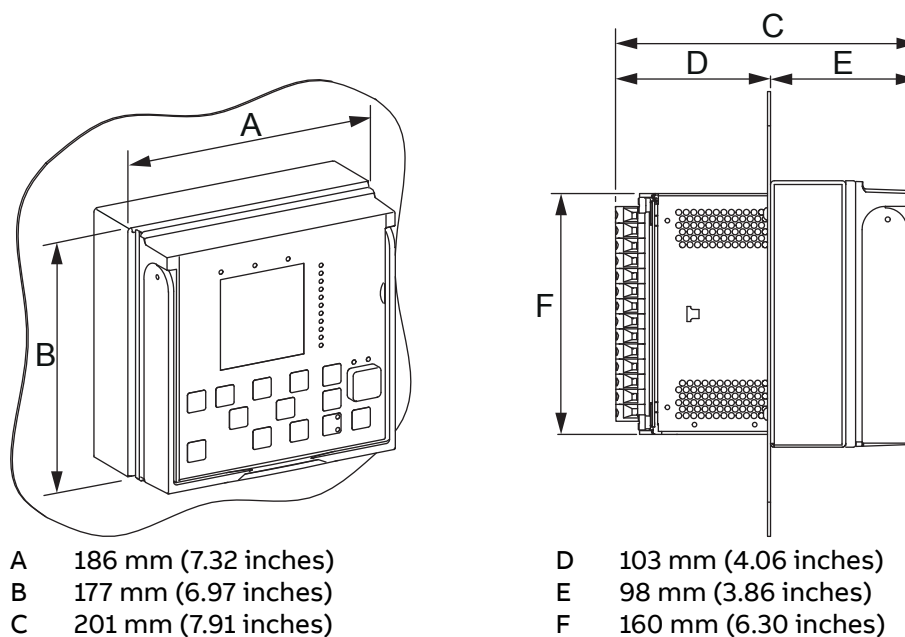


Figure 21: Semi-flush mounted standard size relay

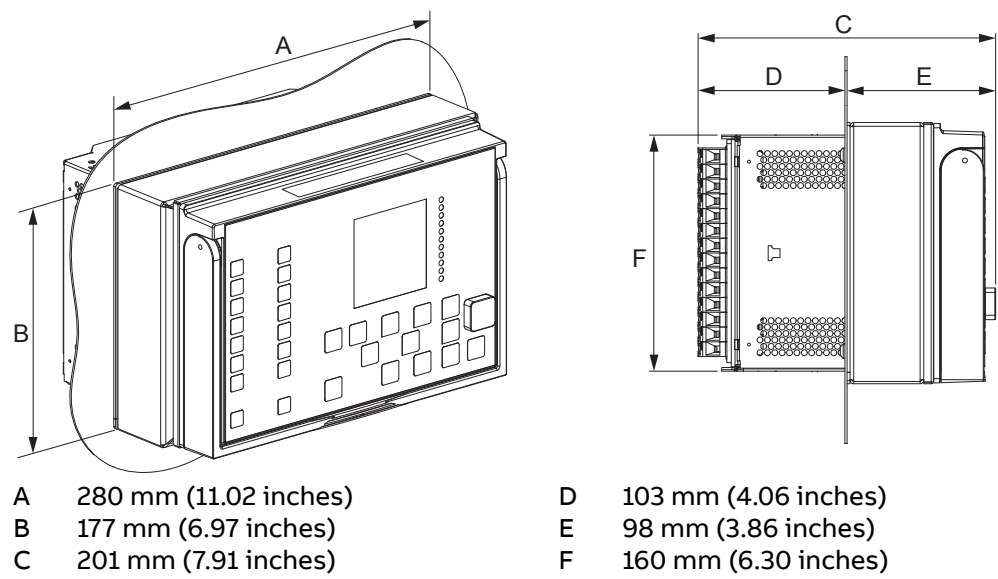


Figure 22: Semi-flush mounted wide size relay

4.4.4 Semi-flush mounting protection relay inclined

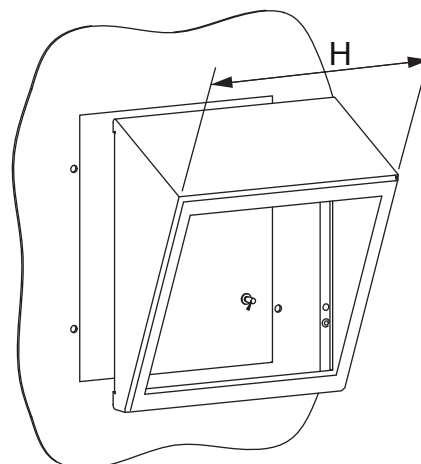
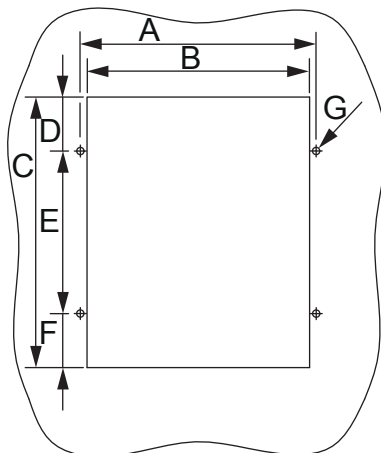
A mounting kit is needed for semi-flush mounting the protection relay inclined. In addition to the detailed mounting instructions, the mounting kit includes:

- Angle frame
- Gasket
- Screws

Requirements for installation

- Panel cut-out of 167 × 203 mm (6.57 × 7.99 inches) standard case with mounting holes
- Panel cut-out of 248 × 203 mm (9.76 × 7.99 inches) wide case with mounting holes
- Depth behind the panel 107 mm (4.21 inches)
- When IP 54 degree of protection (according to IEC 60529) is required for the front side, a gasket has to be used in the installation.

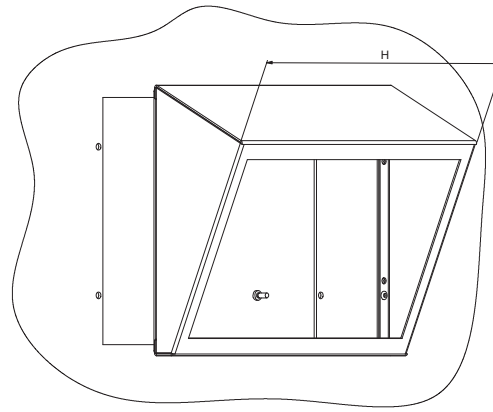
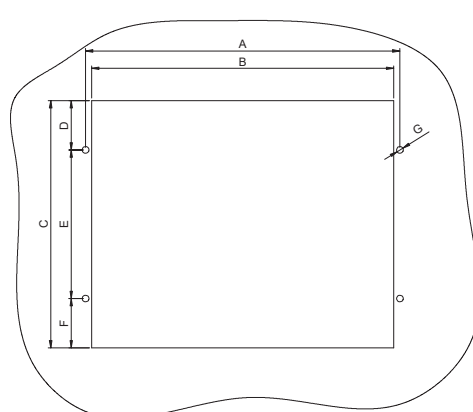
1. Mount the angle frame into the panel cut-out with four M4 screws.



A 177 ±0.3 mm (6.97 inches)
 B 167 ±1 mm (6.57 inches)
 C 203 ±1 mm (7.99 inches)
 D 40.5 mm (1.59 inches)

E 122 mm ±0.3 mm (4.80 inches)
 F 40.5 mm (1.59 inches)
 G ø 5.5 mm (0.22 inches)
 H 190 mm (7.48 inches)

Figure 23: Mounting the standard size angled frame



A 258 ±0.3 mm (10.16 inches)
 B 248 ±1 mm (9.76 inches)
 C 203 ±1 mm (7.99 inches)
 D 40.5 mm (1.59 inches)

E 122 mm ±0.3 mm (4.80 inches)
 F 40.5 mm (1.59 inches)
 G ø 5.5 mm (0.22 inches)
 H 272 mm (10.71 inches)

Figure 24: Mounting the wide size angled frame

2. Loosen the four M5 fixing screws in the case to fit the case into the angle frame.
3. Mount the case to the angle frame.
 With the angled frame, the protection relay can be mounted inclined downward to a 25° angle.
4. Tighten the screws.



The allowed range for the fixing screws' tightening torque is 0.7...1 Nm (0.52...0.74 foot-pounds).

5. Install the plug-in unit into the case.
6. Seal or secure the handle.

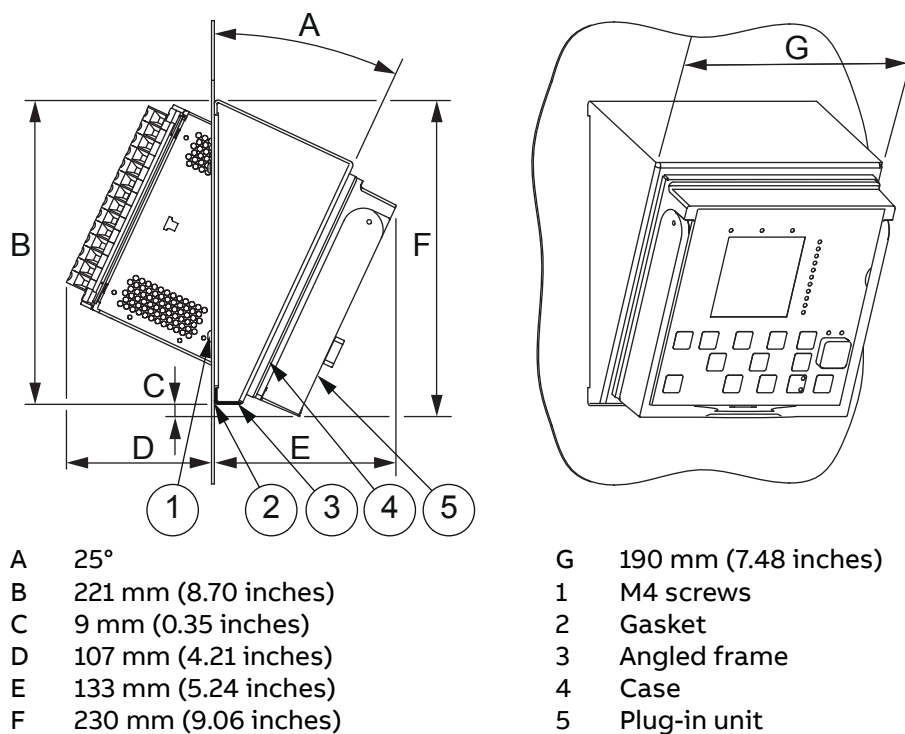


Figure 25: Standard size relay semi-flush mounted inclined

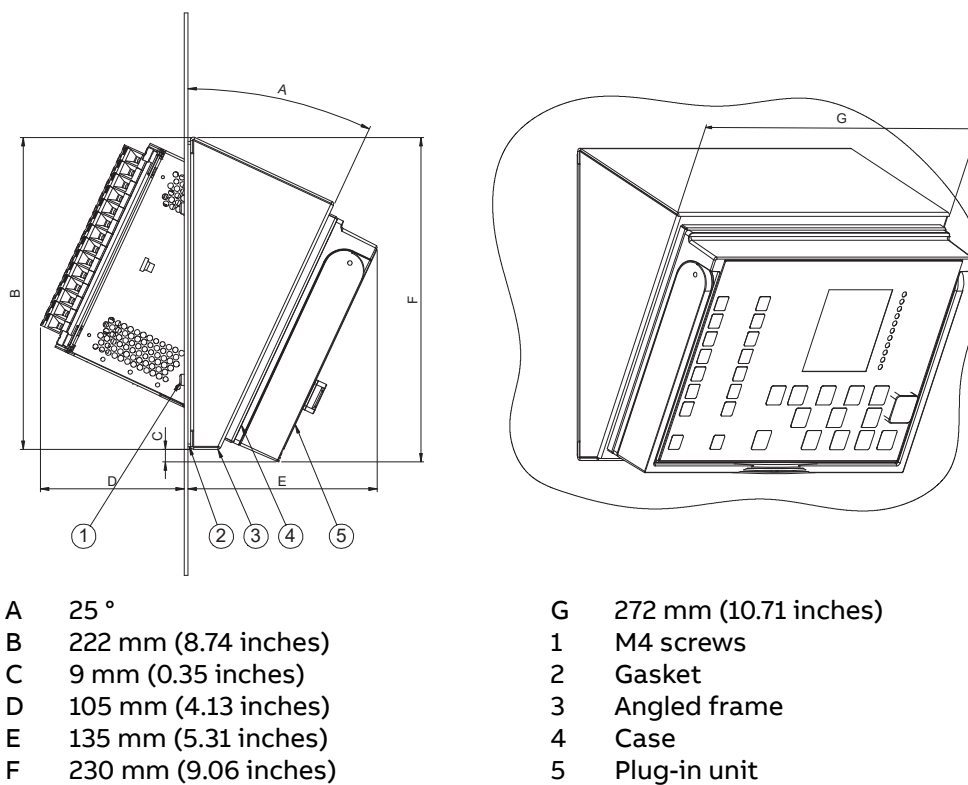


Figure 26: Wide size relay semi-flush mounted inclined

4.4.5 Rack mounting protection relay

A mounting kit is needed for rack mounting the protection relay. In addition to the detailed mounting instructions, the 19" rack mounting kit includes:

- Mounting panel; the type of the mounting panel depends on the number of mounted devices
 - Screws
1. Mount the mounting panel to a 19" rack.
 2. Loosen the four M5 fixing screws in the case to fit the case into the panel cut-out.

3. Mount the case to the panel cut-out.

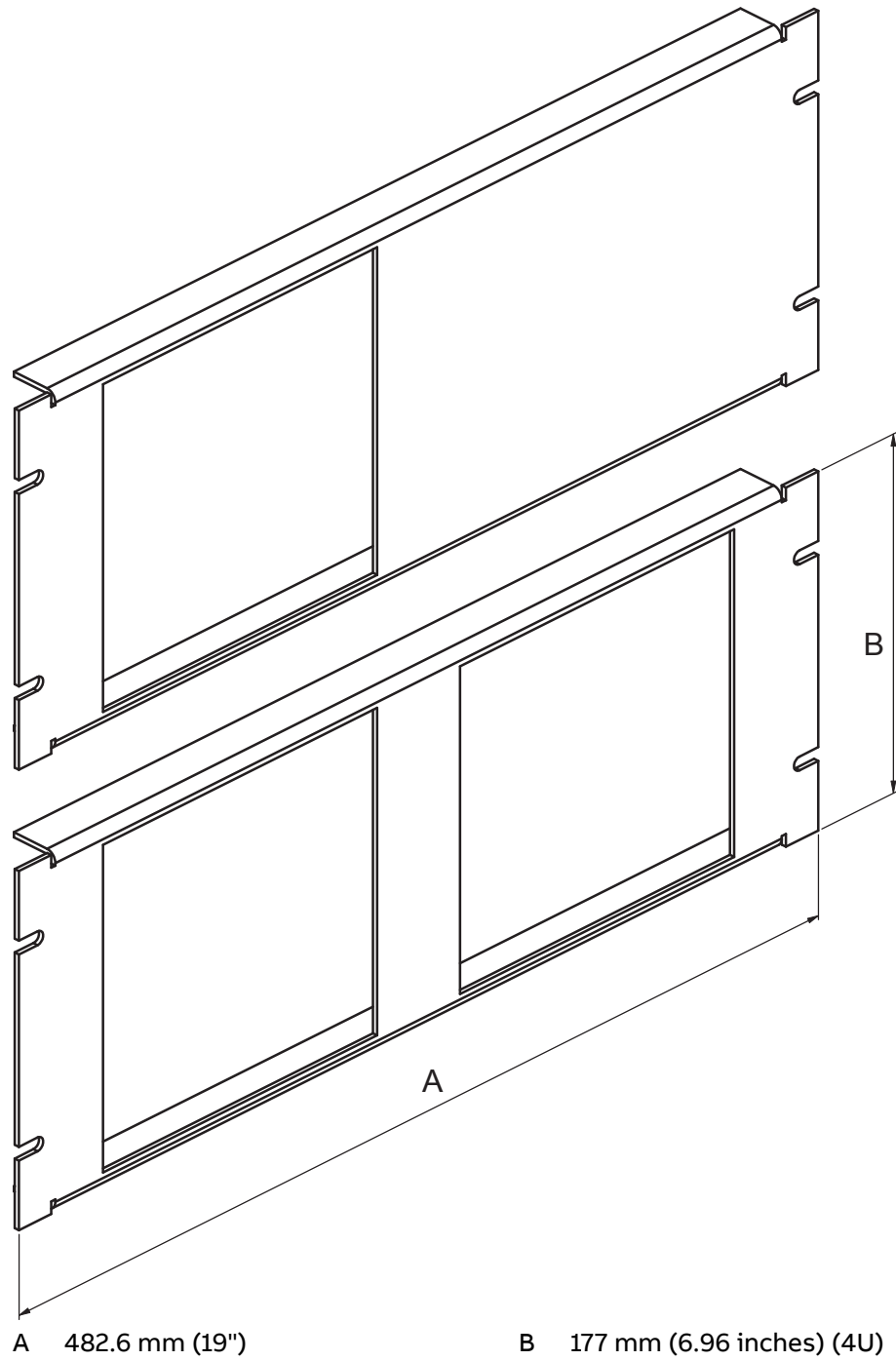
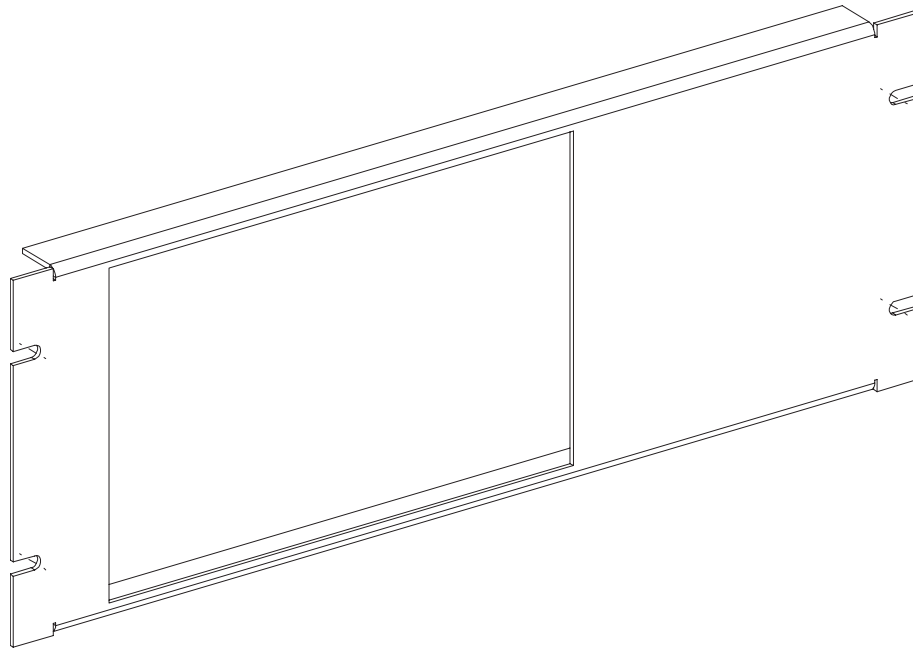


Figure 27: 19" rack mounting panels for standard size relay



A 482.6 mm (19")

B 177 mm (6.96 inches) (4U)

Figure 28: 19" rack mounting panel for wide size relay

4. Tighten the screws.



The allowed range for the fixing screws' tightening torque is 0.7...1 Nm (0.52...0.74 foot-pounds).

5. Install the plug-in unit into the case.
6. Seal or secure the handle.

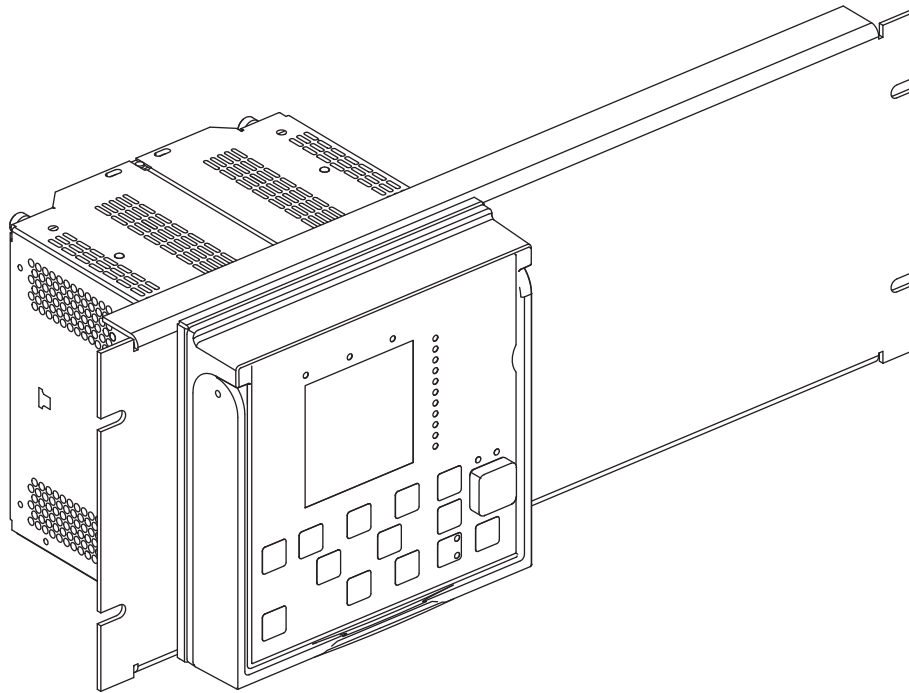


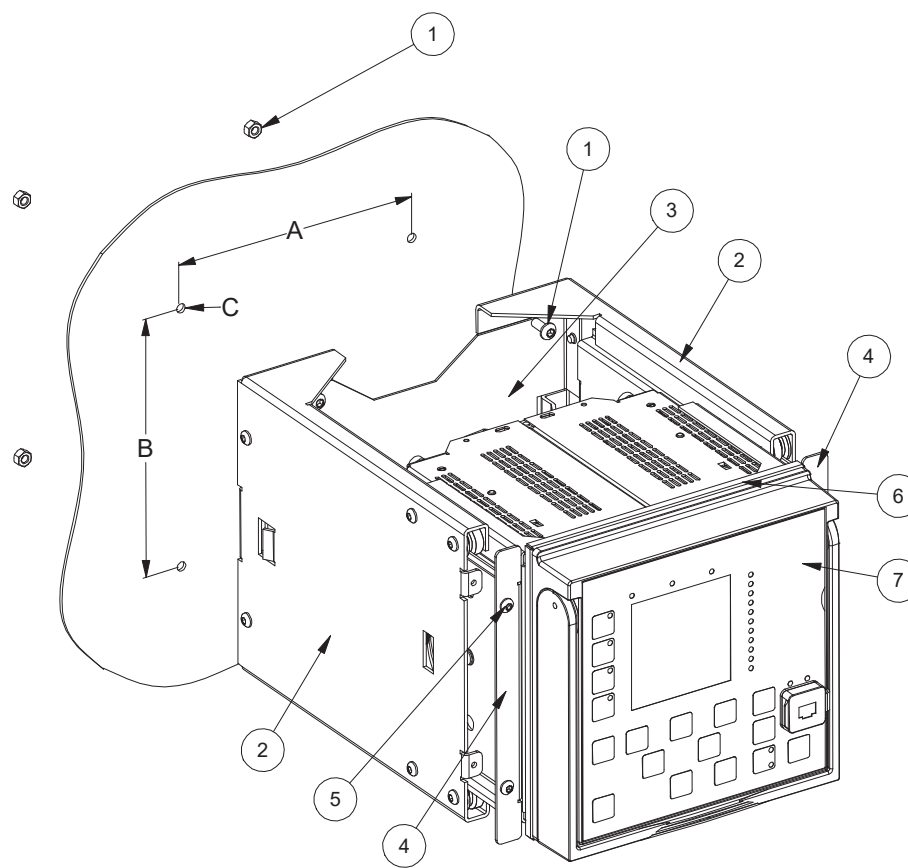
Figure 29: Rack mounted protection relay

4.4.6 Wall mounting protection relay

4.4.6.1 Wall mounting standard size relay

A mounting kit is needed for wall mounting the protection relay. In addition to the detailed mounting instructions, the wall mounting kit includes:

- Side parts and rails
- Back plate
- Screws and nuts
- Dimensions for screw holes



- | | | | |
|---|-----------------------------------|---|--------------|
| A | 149 mm (5.87 inches) | 3 | Back plate |
| B | 156 mm (6.14 inches) | 4 | Side part |
| C | Ø 6 mm (0.24 inches) | 5 | Screws M4x6 |
| 1 | Mounting screws M5x12 and nuts M5 | 6 | Case |
| 2 | Rail | 7 | Plug-in unit |

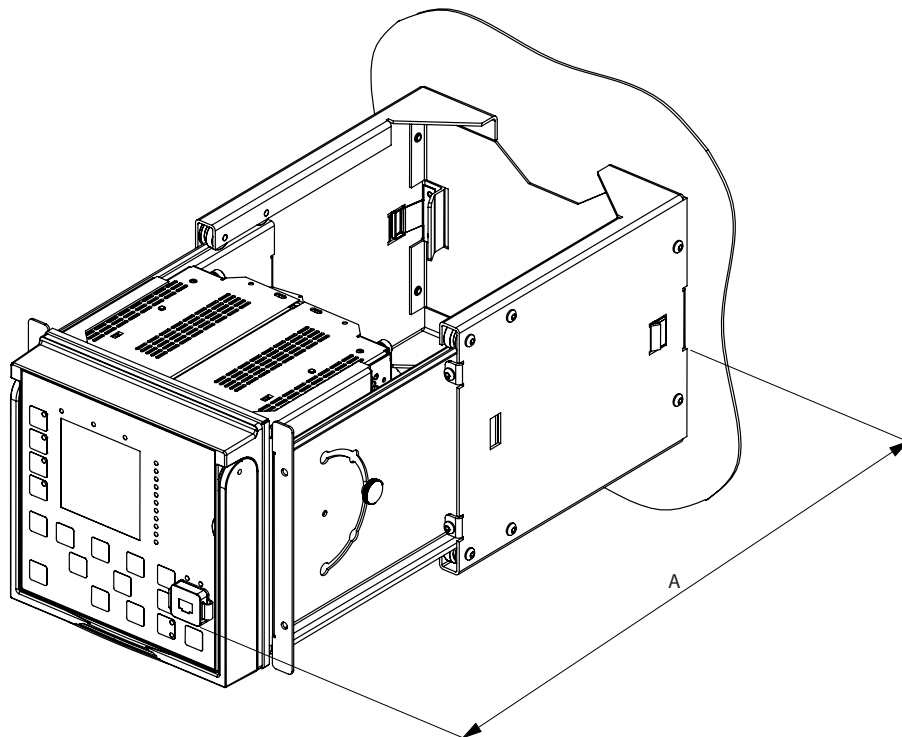
Figure 30: Wall-mounting kit for standard size relay

1. Open M4x6 screws and remove the side parts from the rails.
2. Drill screw holes according to the dimensional drawing.
3. Install the frame (=back plate+rails) to the wall with M5x12 screws and nuts.
4. Remove the gasket from the case.
5. Loosen the four M5 fixing screws in the case to fit the case to the side parts.
6. Tighten the fixing screws.



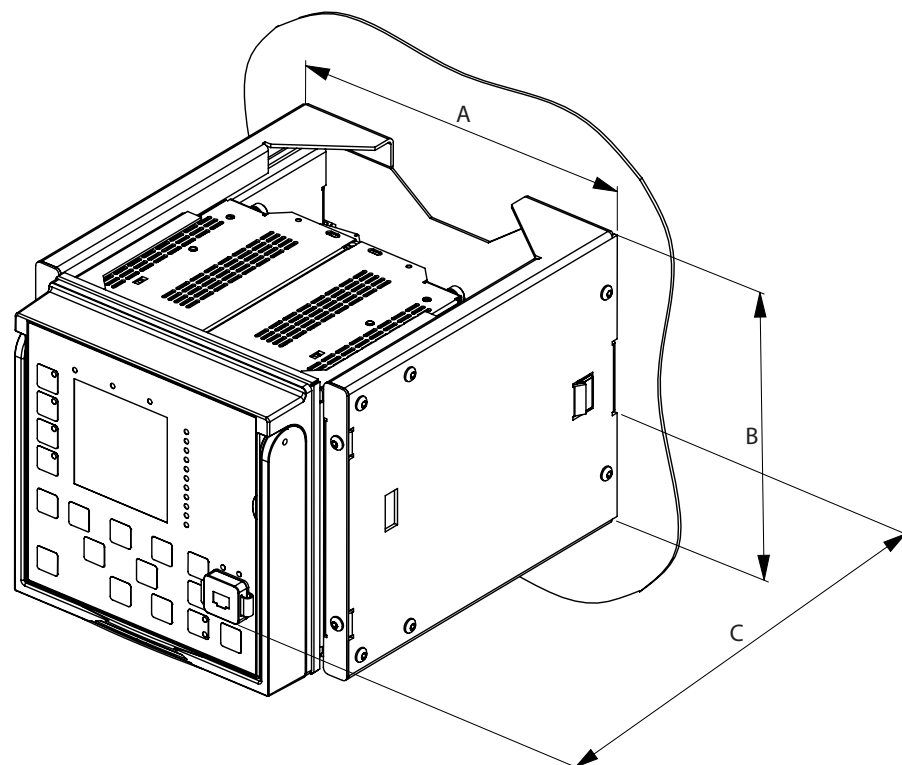
The allowed range for the fixing screws' tightening torque is 0.7...1 Nm (0.52...0.74 foot-pounds).

7. Mount the case with the side parts between the rails.
8. Install the plug-in unit into the case.
9. Seal or secure the handle.
10. Lock the relay with M4x6 screws.



A 430 mm (16.93 inches)

Figure 31: Wall mounting the protection relay



A 214 mm (8.43 inches)

B 177 mm (6.97 inches)

C 270 mm (10.63 inches)

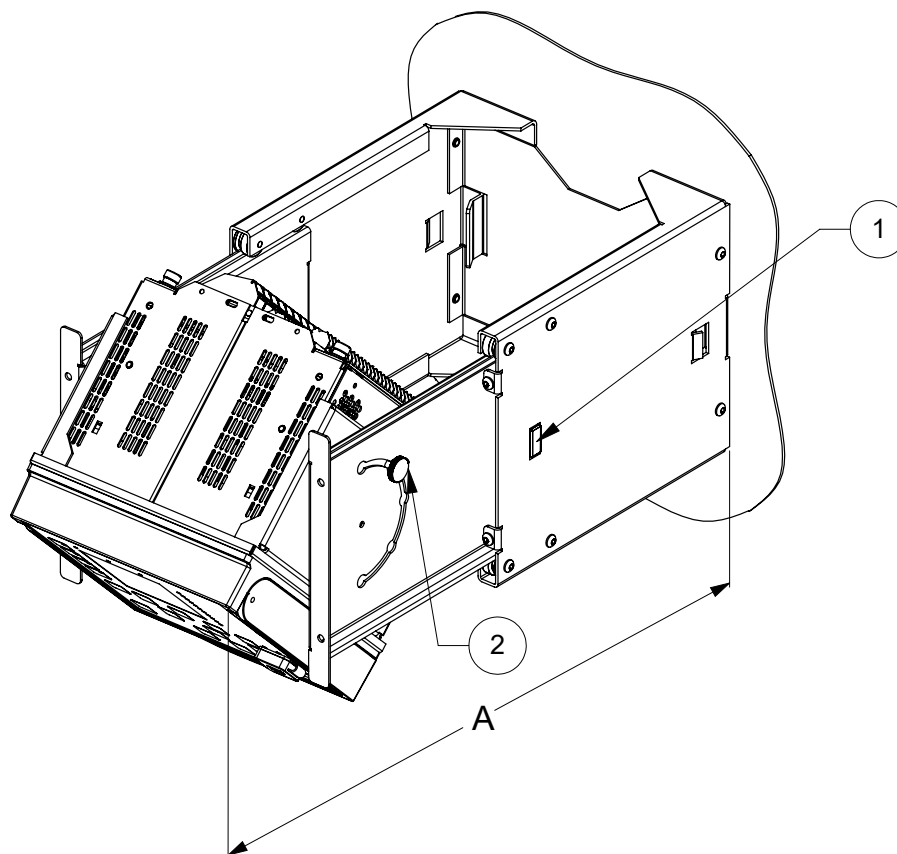
Figure 32: Wall mounted protection relay



Minimum of 50 mm (1.97 inches) space is needed between two kits side by side.

When connecting the wires, a wall-mounted protection relay can be pulled out and turned 45° (or 90°) degrees downwards or upwards.

- To release the protection relay for pulling it out, push the locks beside the mounting frame.
- To rotate the protection relay, loosen the knurled-head screws in the rails.



A 455 mm (17.91 inches)

2 Knurled-head screw

1 Mounting frame lock

Figure 33: Wall mounted protection relay in rotated position



Minimum of 50 mm (1.97 inches) space is needed above and below the frame for rotating.

4.4.6.2

Wall mounting wide size relay

A mounting kit is needed for wall mounting the protection relay. In addition to the detailed mounting instructions, the wall mounting kit includes:

- Wall mounting frame
- Screws
- Dimensions for screw holes

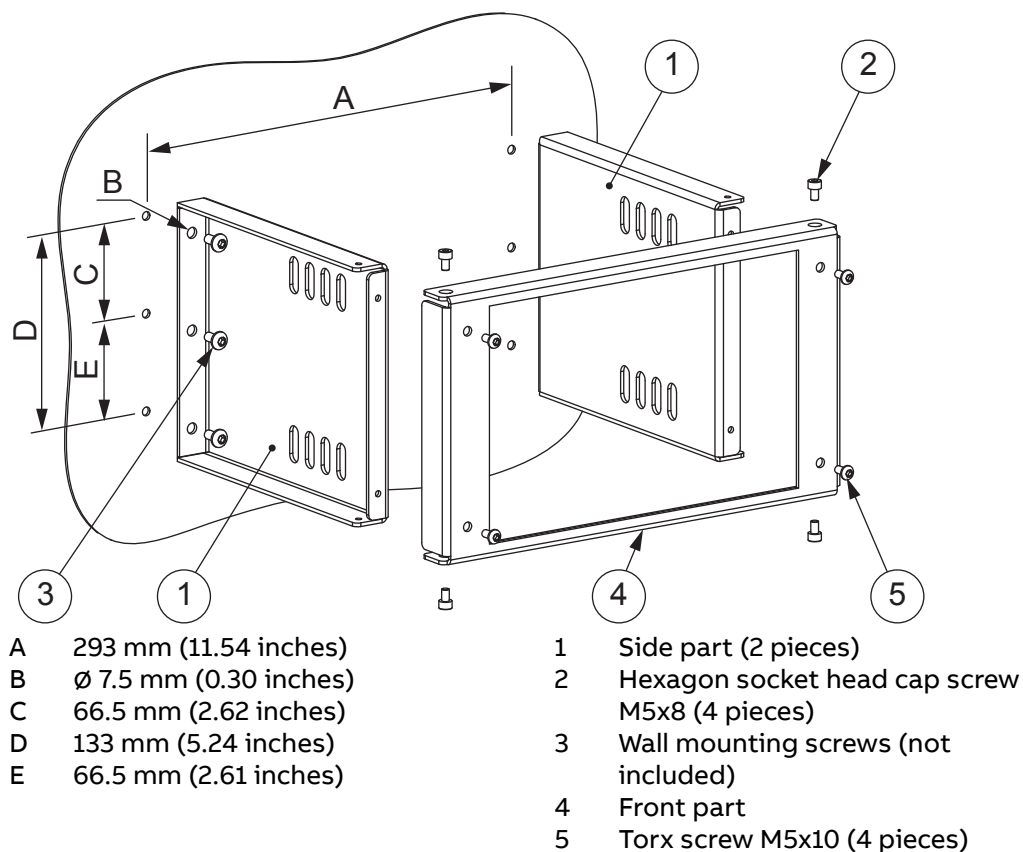


Figure 34: Wall-mounting kit for wide size relay

1. Drill screw holes according to the dimensional drawing.
2. Mount the side parts with M6 screws (screws not included).
3. Mount the front part with the included M5 screws.

4. Detach the plug-in unit from the case and mount the case to the front part by tightening the integrated mounting screws 0.7...1.0 Nm (0.52...0.74 foot-pounds).

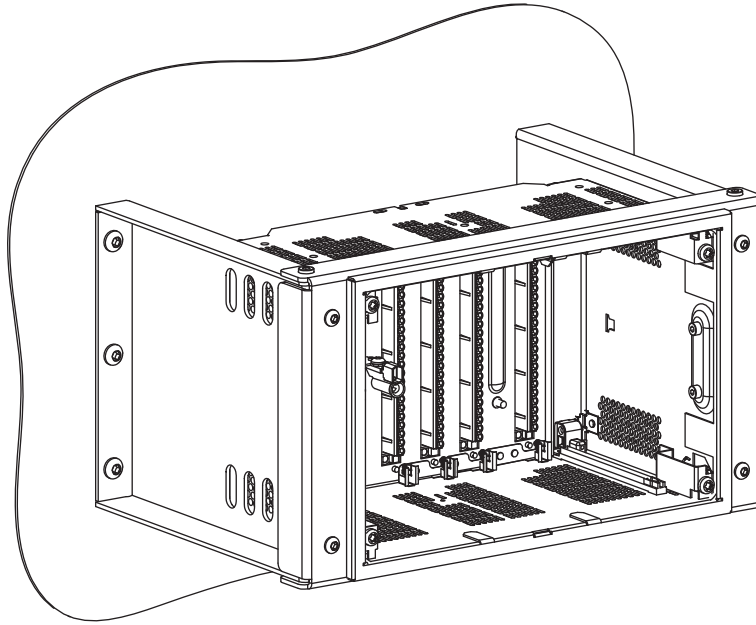
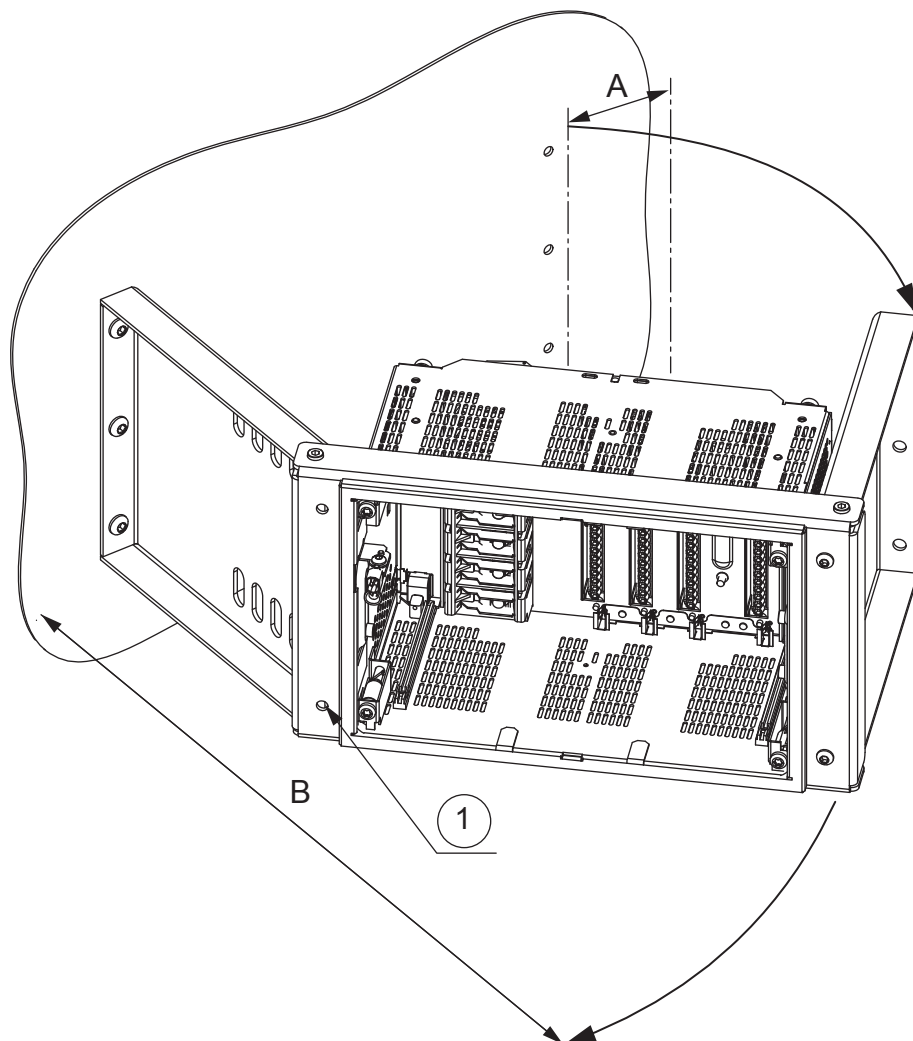


Figure 35: Mounting the case

5. The frame can be swung out to access terminals for wiring. Remove the screws as seen to swing out the frame.



A 57 mm (2.24 inches)
B 503 mm (19.80 inches)

1 Remove the screws before
swinging out the frame

Figure 36: Swing mechanism

6. After finished wiring, turn the frame to the wall again and insert all removed screws.
7. Install the removed plug-in unit.
8. Seal or secure the handle.

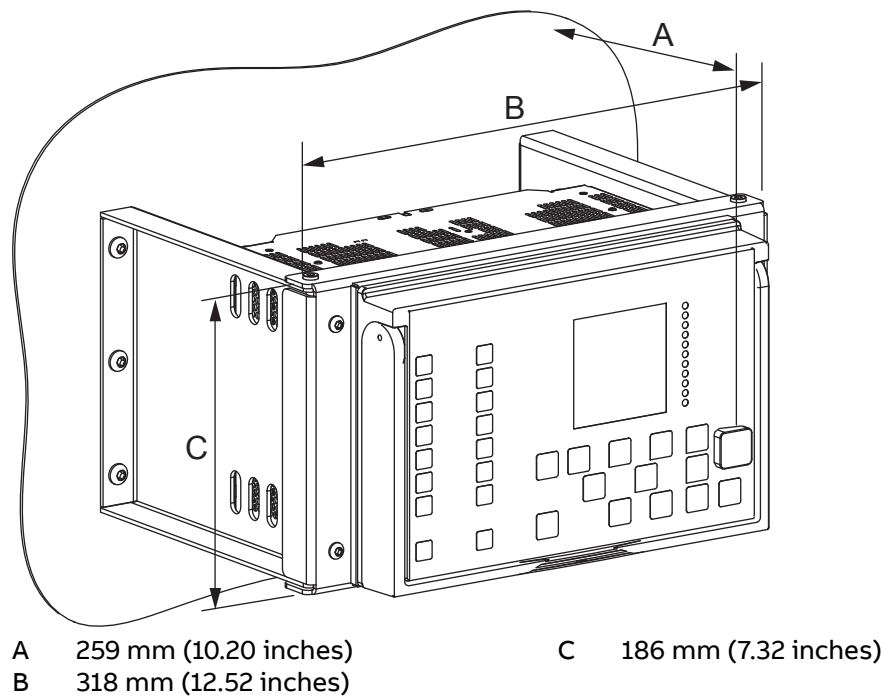


Figure 37: Wall-mounted protection relay



Leave enough slack to wires and cables to allow installation movement.

4.4.7

Rack mounting protection relay and test switch RTXP into 19" equipment frame

A mounting kit is needed for rack mounting the protection relay into a 19" equipment frame. In addition to the detailed mounting instructions, the mounting kit includes:

- Mounting panel
- Metallic frame for mounting the RTXP 18 (standard only) or RTXP 24 (standard and wide) test switch to the panel
- Test switch is not included in this kit.



A protection relay equipped with optical connections requires a minimum depth of 180 mm (7.09 inches). The allowed minimum bending radius has to be checked from the optical cable manufacturer.

1. Mount the mounting panel into the 19" rack.
2. Loosen the four M5 fixing screws in the case to fit the case into the panel cut-out.
3. Mount the case to the panel cut-out.
4. Install the optional metallic frame to mount the RTXP 18 or 24 test switch to the panel.
5. Seal or secure the handle.

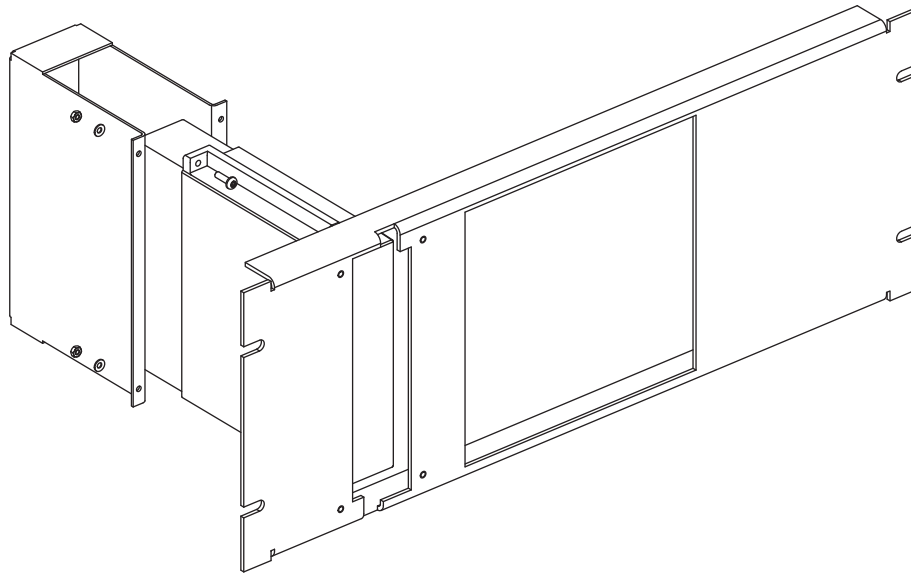


Figure 38: Mounting of the metallic frame for an RTXP 18 test switch

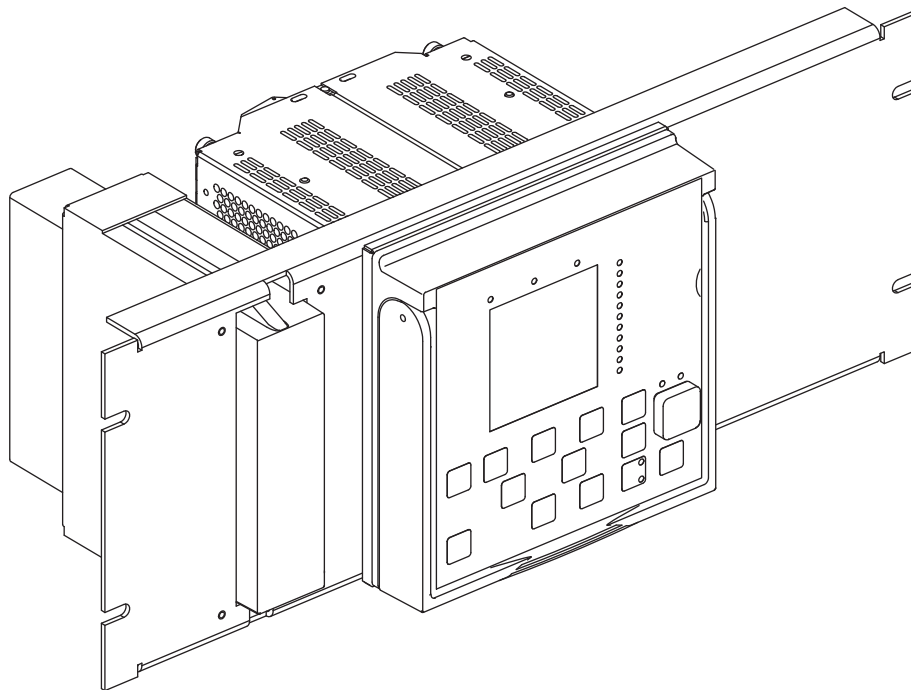


Figure 39: Protection relay and the test switch RTXP 18 mounted into a 19" equipment panel

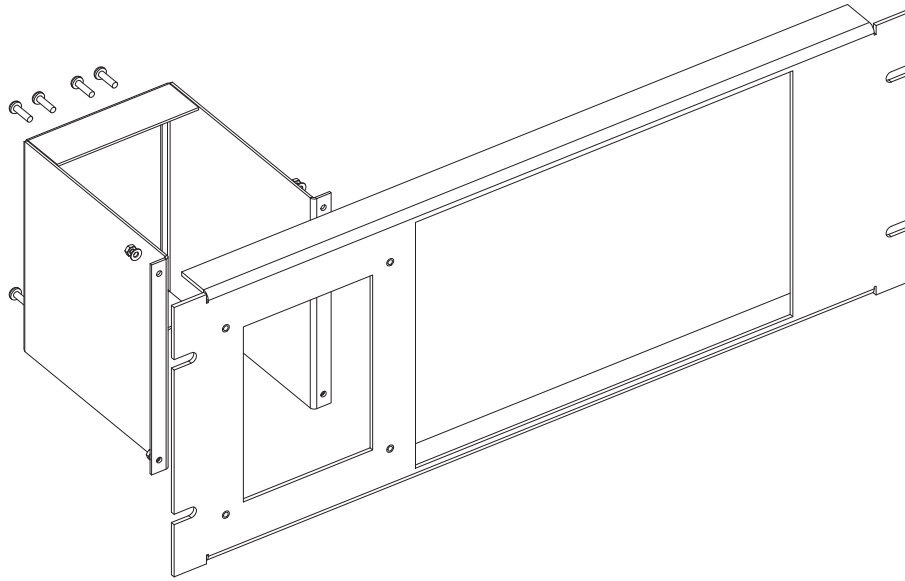


Figure 40: Mounting of the metallic frame for an RTXP 24 test switch

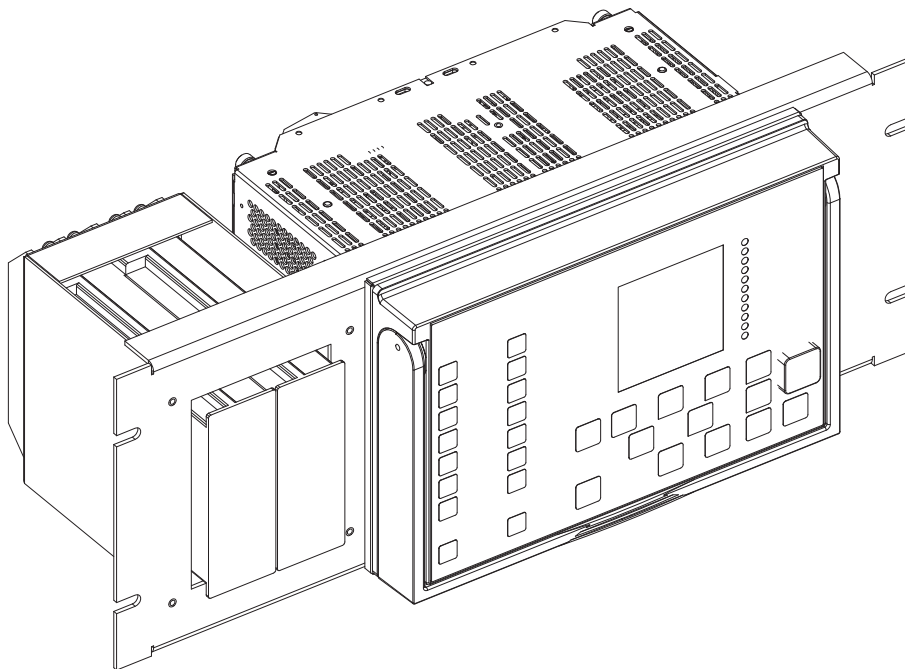


Figure 41: Protection relay and the test switch RTXP 24 mounted into a 19" equipment panel

4.4.8 Rack mounting protection relay into combiflex 19" equipment frame (Type RHGT 19" 4U variant C)

A mounting bracket is needed for rack mounting the protection relay into a combiflex 19" equipment frame, a different bracket is in installation with test switch. In addition to the detailed mounting instructions, the mounting kit includes:

- Mounting bracket for the case and optionally with RTXP test switch
1. Install the mounting bracket into the combiflex equipment frame by using tapping screws ST3.5x13 for standard size and M4x10 for wide size, from rear side.

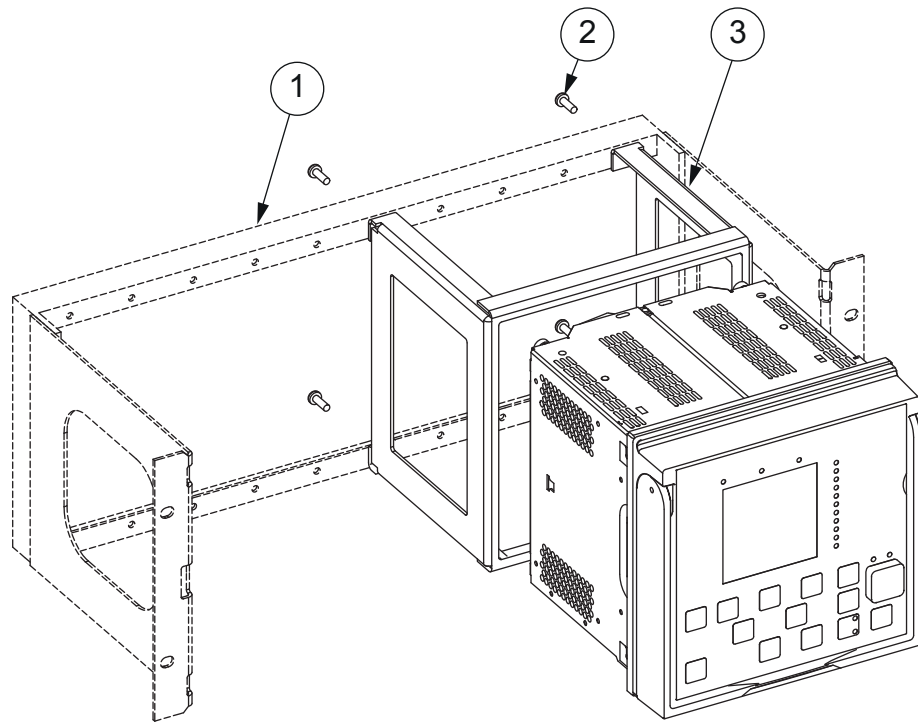


The type of mounting bracket to be used depends on whether the protection relay is installed into the frame on its own or with a test switch.

2. Loosen the four M5 fixing screws in the case to fit the case into the mounting bracket.
3. Mount the case to the mounting bracket.
4. Tighten the M5 screws.

The allowed range for the fixing screws' tightening torque is 0.7...1 Nm (0.52...0.74 foot-pounds).

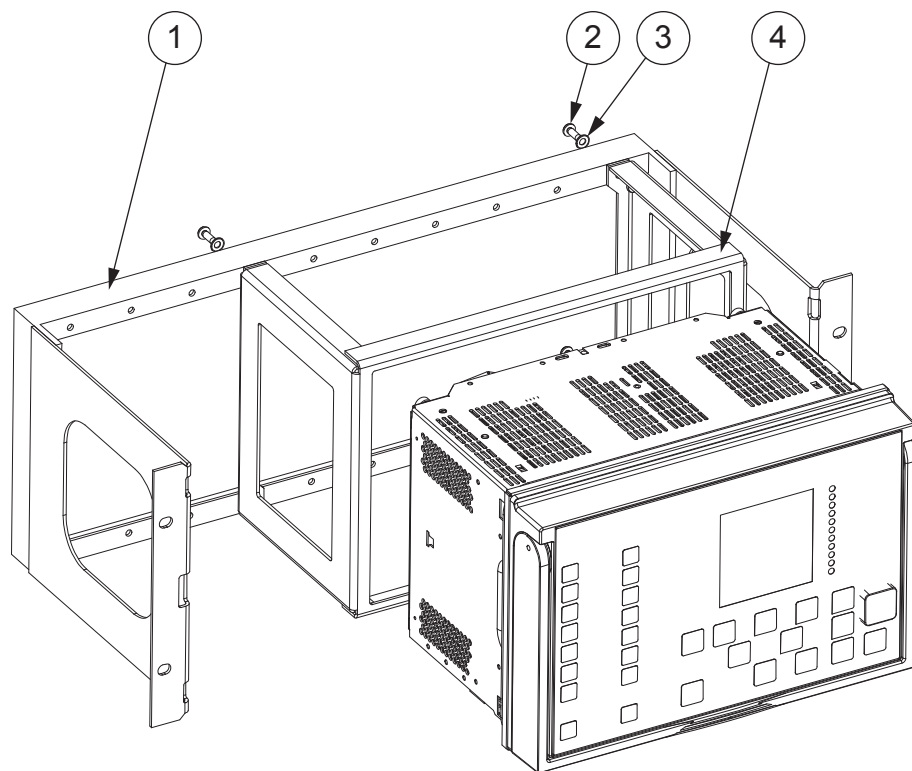
5. Install the plug-in unit into the case.



- 1 RHGT 19" 4U equipment frame,
variant C, with support frame
2 Tapping screw ST3.5x13

- 3 Mounting bracket

Figure 42: Mounting the standard size relay into a 19" combiflex equipment frame



- 1 RHGT 19" 4U equipment frame,
variant C, with support frame
- 2 Screw M4x10

- 3 Washer
- 4 Mounting bracket

Figure 43: Mounting the wide size relay into a 19" combiflex equipment frame

- 6. Seal or secure the handle.

7. Install the optional RTXP 18 or 24 test switch.

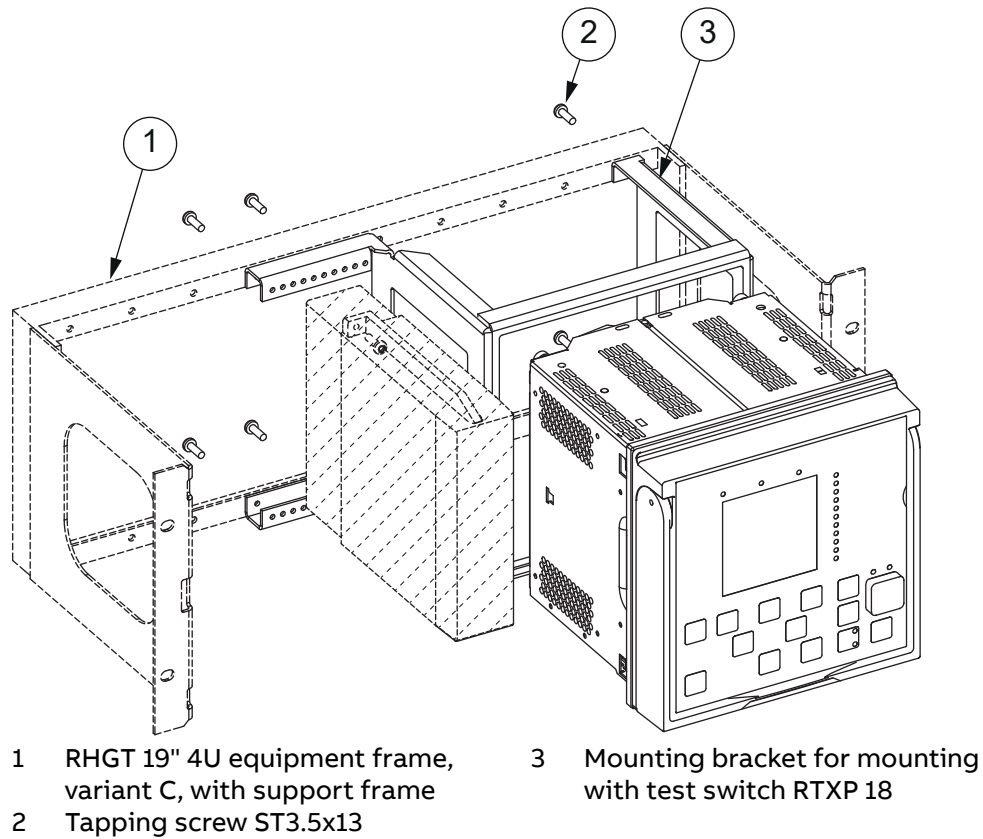
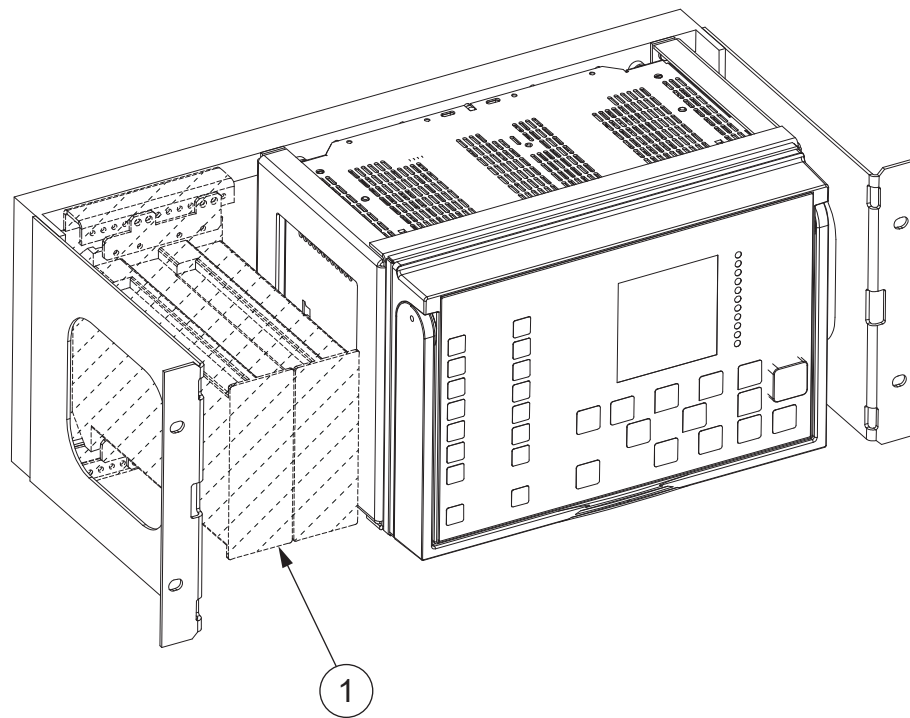


Figure 44: Protection relay and the test switch



1 Test switch RTXP 24

Figure 45: Protection relay and the test switch RTXP 24

4.4.9 Replacing a Strömberg SP_J40 series relay by a standard size REX615

1. Drill new screw holes (Ø4.5 mm (0.18 inches)) to the panel according to the dimensional drawing.

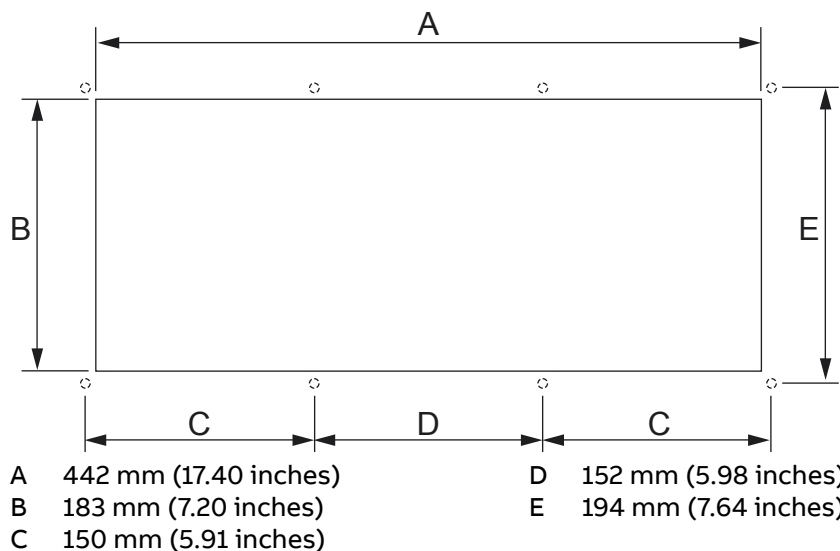


Figure 46: Screw hole and existing panel cut-out dimensions for replacing a SP_J40 series IED

2. Mount the installation plate to the existing panel cut-out with M5 thread forming screws.

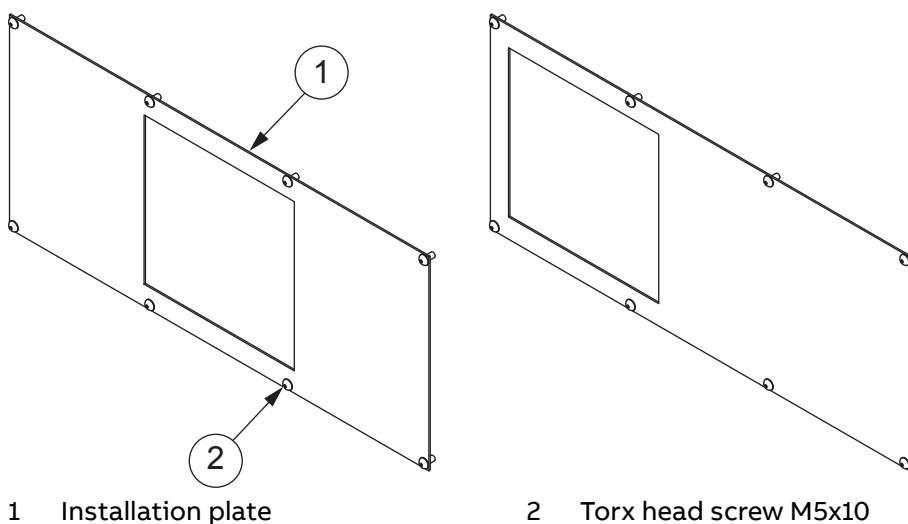


Figure 47: Installation plates for replacing SP_J40 relay

3. Tighten the screws.
4. Mount the new IED.



Available only for standard size REX615.

4.4.10 Replacing two Strömberg SP_J3 series relays by a standard size REX615

1. Make a new panel cut-out to the existing mounting panel according to the dimensional drawing.

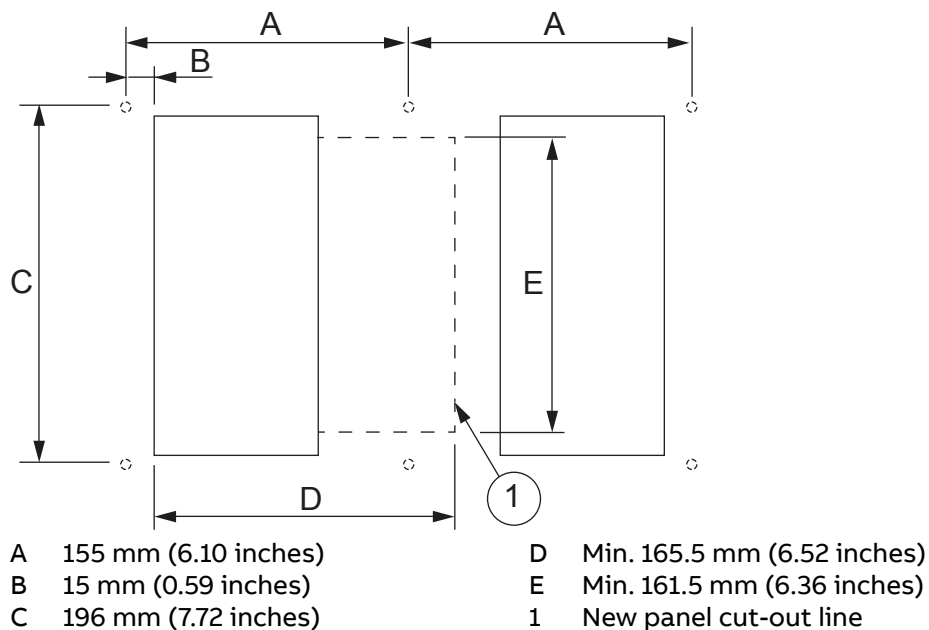
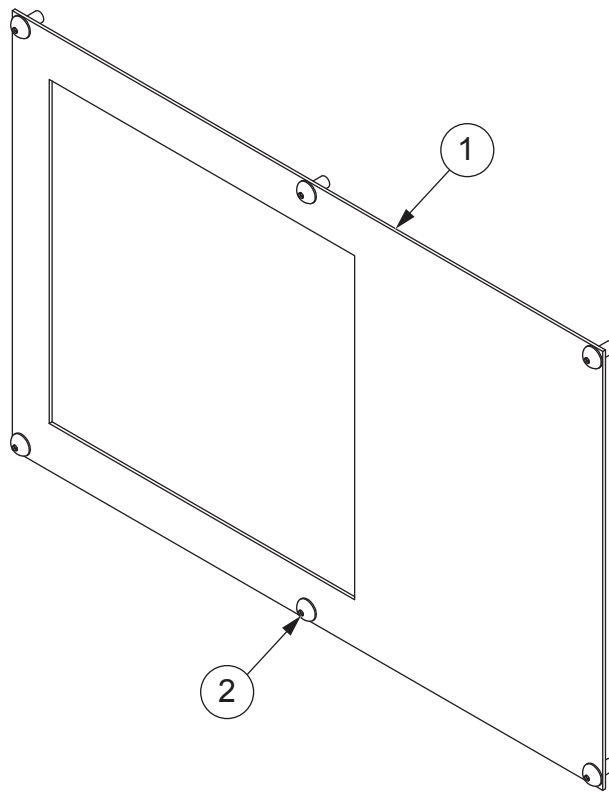


Figure 48: New panel cut-out and screw hole dimensions for replacing two SP_J3 series relays

2. Drill new screw holes (Ø4.5 mm (0.18 inches)) to the panel according to the dimensional drawing.

3. Mount the installation plate to the panel cut-out with M5 thread forming screws.



1 Installation plate

2 Torx head screw M5x10

Figure 49: Installation plate for replacing two SP_J3 relays

4. Tighten the screws.
5. Mount the new IED.
6. Seal or secure the handle.



For ordering numbers, see the ordering number table.

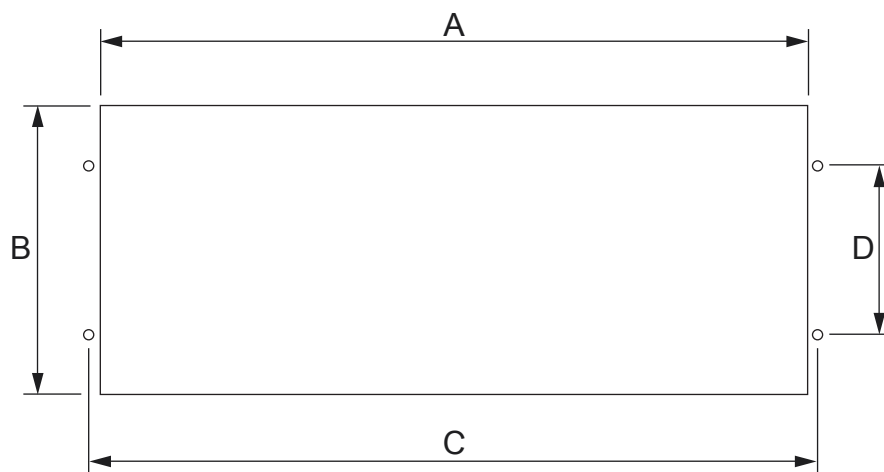


Available only for standard size REX615.

4.4.11 Replacing a Strömberg SP_J3/J6 series relay in 19" rack by a standard size REX615

1. Mount the installation plate to the existing panel cut-out with M6 screws and nuts.

Use the existing screw holes.



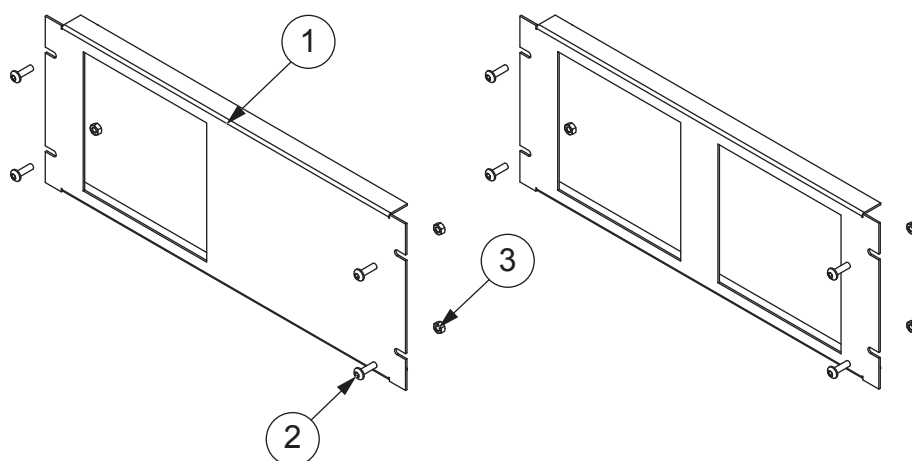
A 446 mm (17.56 inches)

B 196 mm (7.72 inches)

C 460 mm (18.11 inches)

D 114.5 mm (4.51 inches)

Figure 50: Screw hole and existing panel cut-out dimensions for replacing a SP_J3/J6 series relay



1 Installation plate

2 Torx head screw M6x20

3 Hexagon nut M6

Figure 51: Installation plates for replacing SP_J3/J6 relay in 19" rack

2. Tighten the screws.
3. Mount the new IED with the mounting frame to the installation plate.
4. Seal or secure the handle.



For ordering numbers, see the ordering number table.



Available only for standard size REX615.

4.4.12 Replacing a Strömberg SP_J6 series relay by a standard size REX615

1. Drill new screw holes (ø4.5 mm (0.18 inches)) to the panel according to the dimensional drawing.

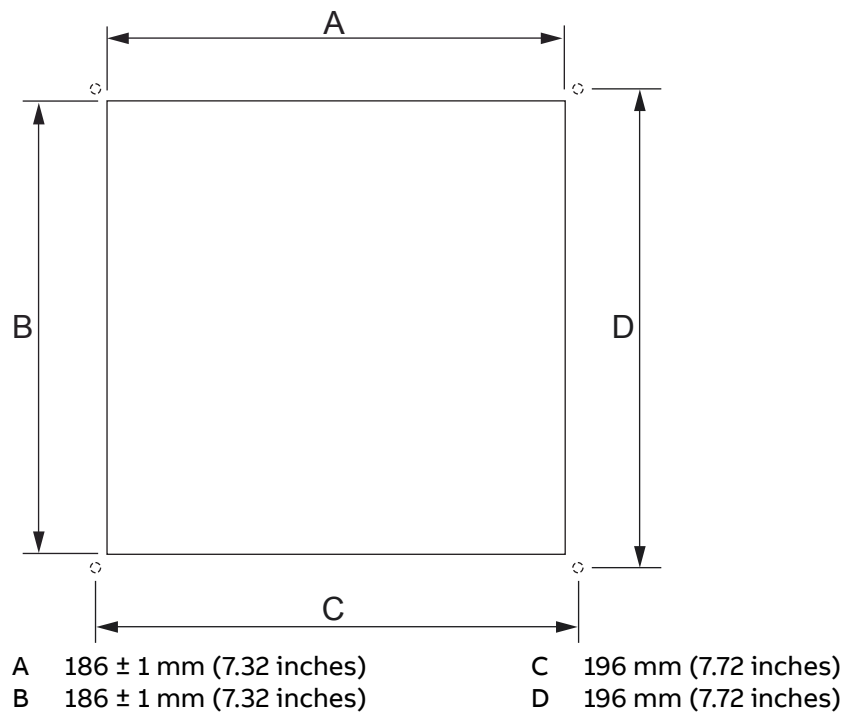
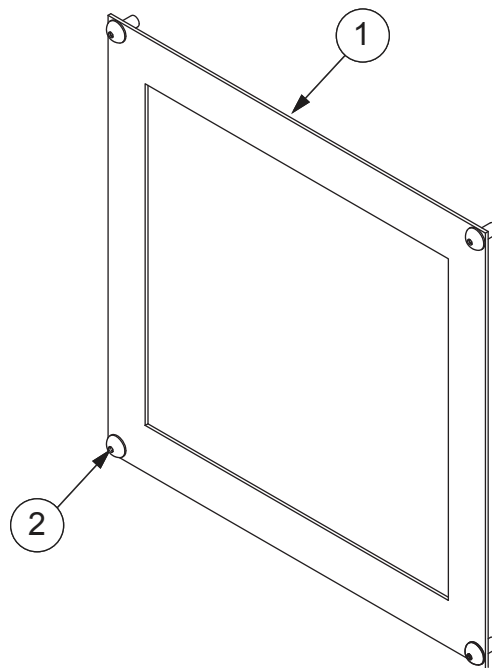


Figure 52: Screw hole and existing panel cut-out dimensions for a SP_J6 series relay

2. Mount the installation plate to the existing panel cut-out with M5 thread forming screws.



1 Installation plate

2 Torx head screw M5x10

Figure 53: Installation plate for replacing SP_J6 relay

3. Tighten the screws.
4. Mount the new IED.
5. Seal or secure the handle.



For ordering numbers, see the ordering number table.



Available only for standard size REX615.

4.4.13 Replacing three BBC S_ series relays by a standard size REX615

1. Make a new panel cut-out to the existing mounting panel according to the dimensional drawing.

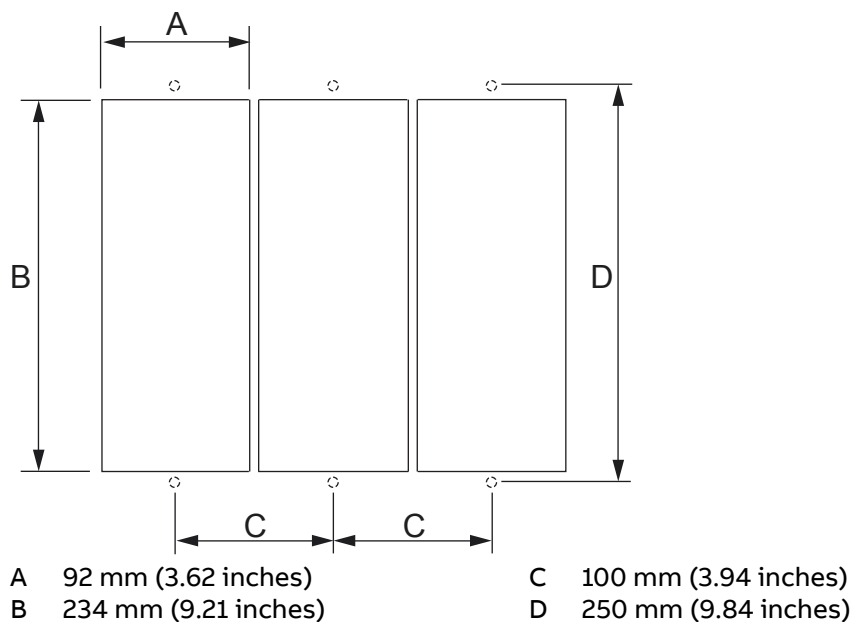
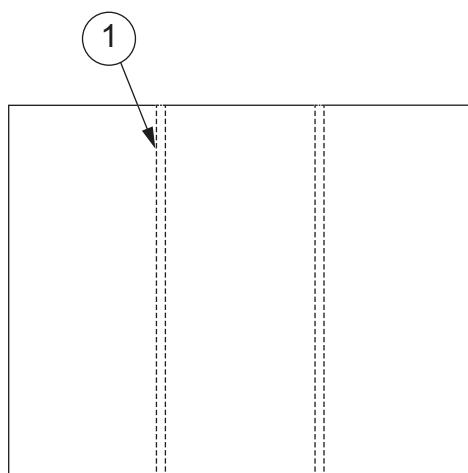


Figure 54: Existing panel cut-out and screw hole dimensions for three BBC S_ series relays



1 New panel cut-out line

Figure 55: New panel cut-out for replacing BBC S_ series relays

2. Mount the installation plate to the panel cut-out with M5 screws, nuts and washers.

Use the existing screw holes.

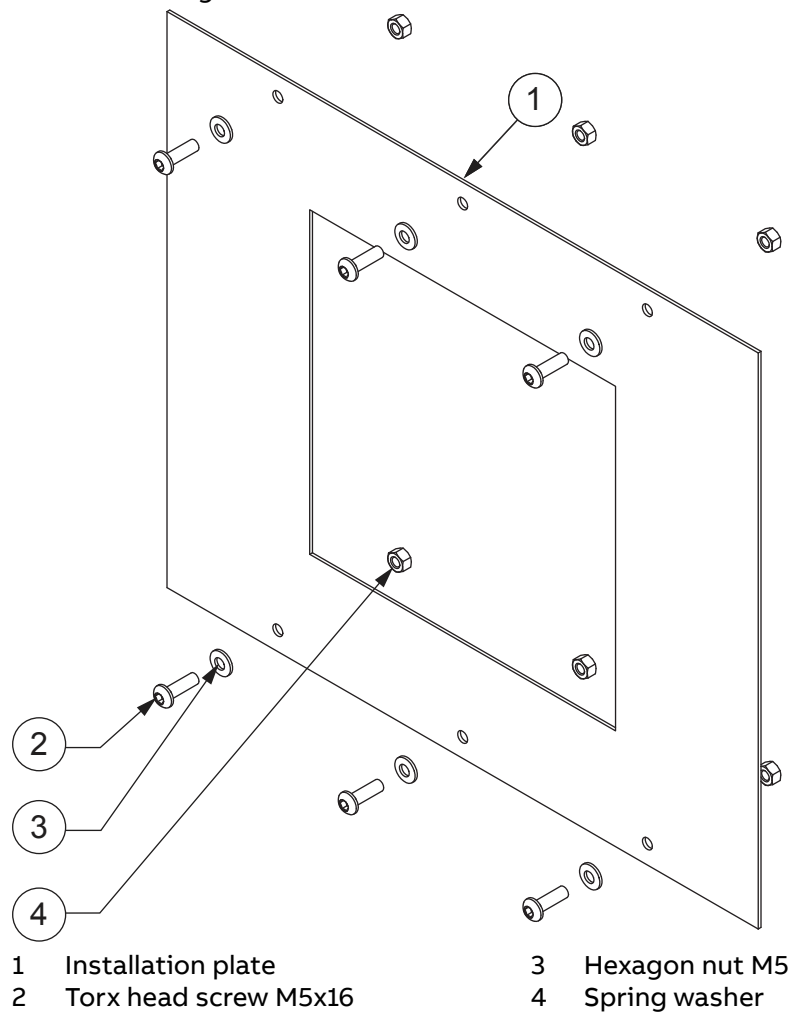


Figure 56: Installation plate for replacing three BBC S₁ relays

3. Tighten the screws.
4. Mount the new IED.



For ordering numbers, see the ordering number table.



Available only for standard size REX615.

4.4.14 Replacing a SPA 300 series relay by a standard size REX615

1. Make a new panel cut-out to the existing mounting panel according to the dimensional drawing.

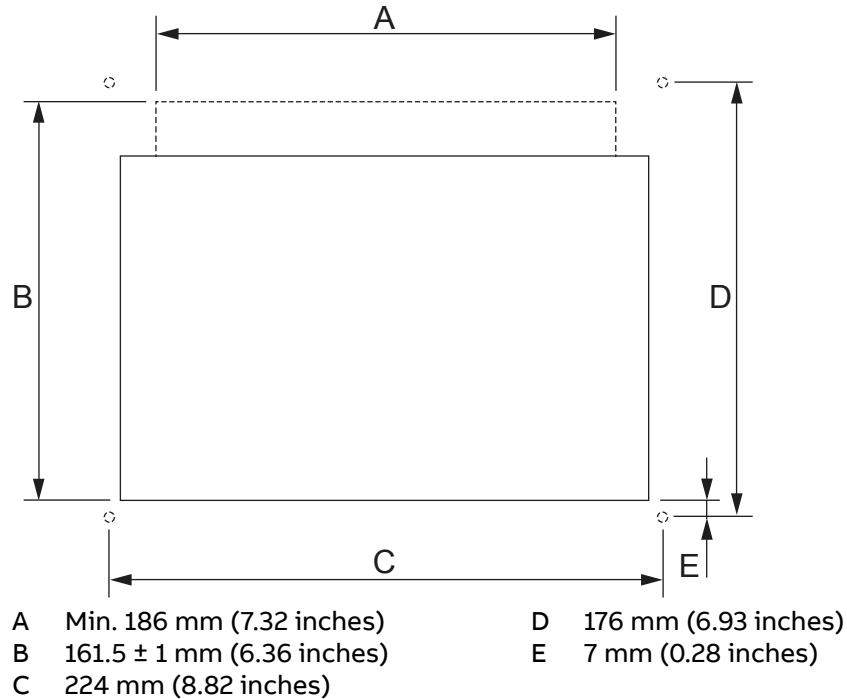
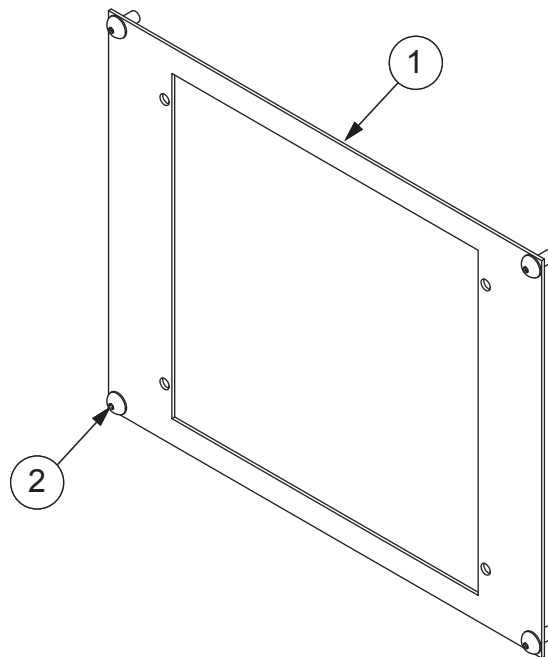


Figure 57: New panel cut-out and screw hole dimensions for replacing a SPA 300 series relay

2. Drill new screw holes (ø4.5 mm (0.18 inches)) to the panel according to the dimensional drawing.

3. Mount the installation plate to the panel cut-out with M5 thread forming screws.



1 Installation plate

2 Torx head screw M5x10

Figure 58: Installation plate for replacing SPA 300 relay

4. Tighten the screws.
5. Mount the new IED.
Use a raising frame if needed.
6. Seal or secure the handle.



For ordering numbers, see the ordering number table.



Available only for standard size REX615.

4.4.15

Replacing a SPA 300 series relay in 19" rack by a standard size REX615

- Mount the IED with the mounting frame to the panel cut-out according to the rack mounting instructions in [Chapter 4.4.5 Rack mounting protection relay](#).
No modifications are needed to the panel.



For replacement kit ordering numbers, see the ordering number table.

4.4.16 Flush mounting the standard size relay with protection cover

A mounting kit is needed for flush mounting the protection relay with a protection cover. The mounting kit includes detailed mounting instructions and the protection cover.

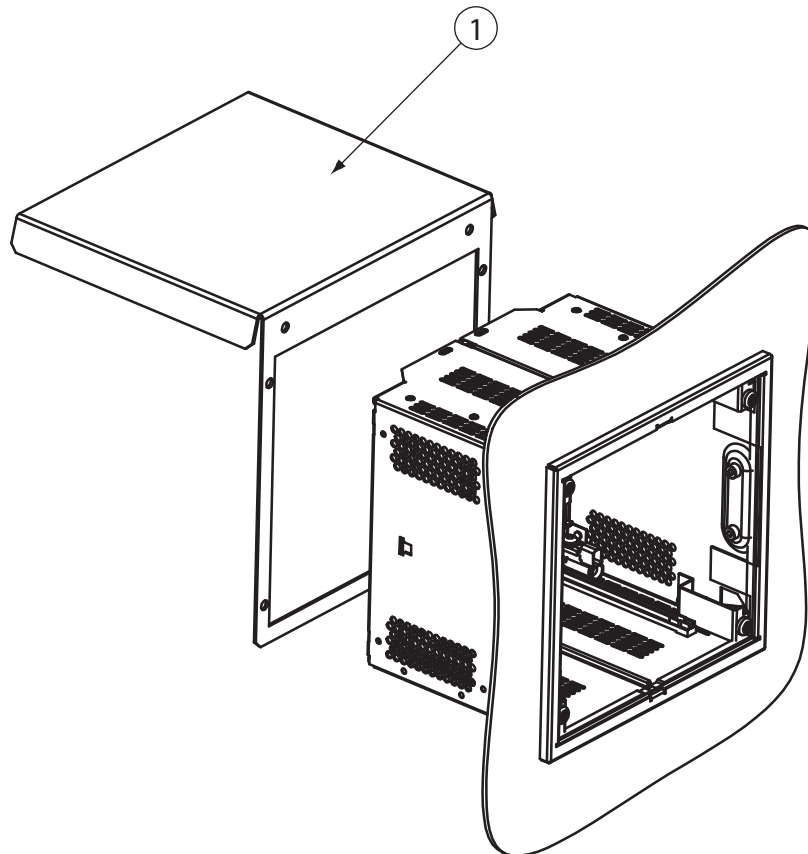
Installation requirements:

- Panel cut-out of 165.5 × 161.5 mm (6.52 × 6.36 inches)
- Depth behind the panel 170 mm (6.70 inches)



A protection relay equipped with optic connections requires a minimum depth of 180 mm (7.09 inches). Check the allowed minimum bending radius from the optical cable manufacturer.

1. Loosen the four M5 fixing screws in the case to fit the case into the panel cut-out.
2. Mount the case to the panel cut-out from the front.
3. Mount the protection cover from the rear.



- 1 Protection cover

Figure 59: Mounting the protection cover

4. Tighten the M5 (T25) screws.



The allowed fixing screws' tightening torque is 0.7...1 Nm (0.52...0.74 foot-pounds).

5. Install the plug-in unit into the case.

There is a protective film on the top side of the protection relay. Its purpose is to prevent debris falling inside the unit while installing electrical wiring. Remove the protective film before energizing the protection relay.

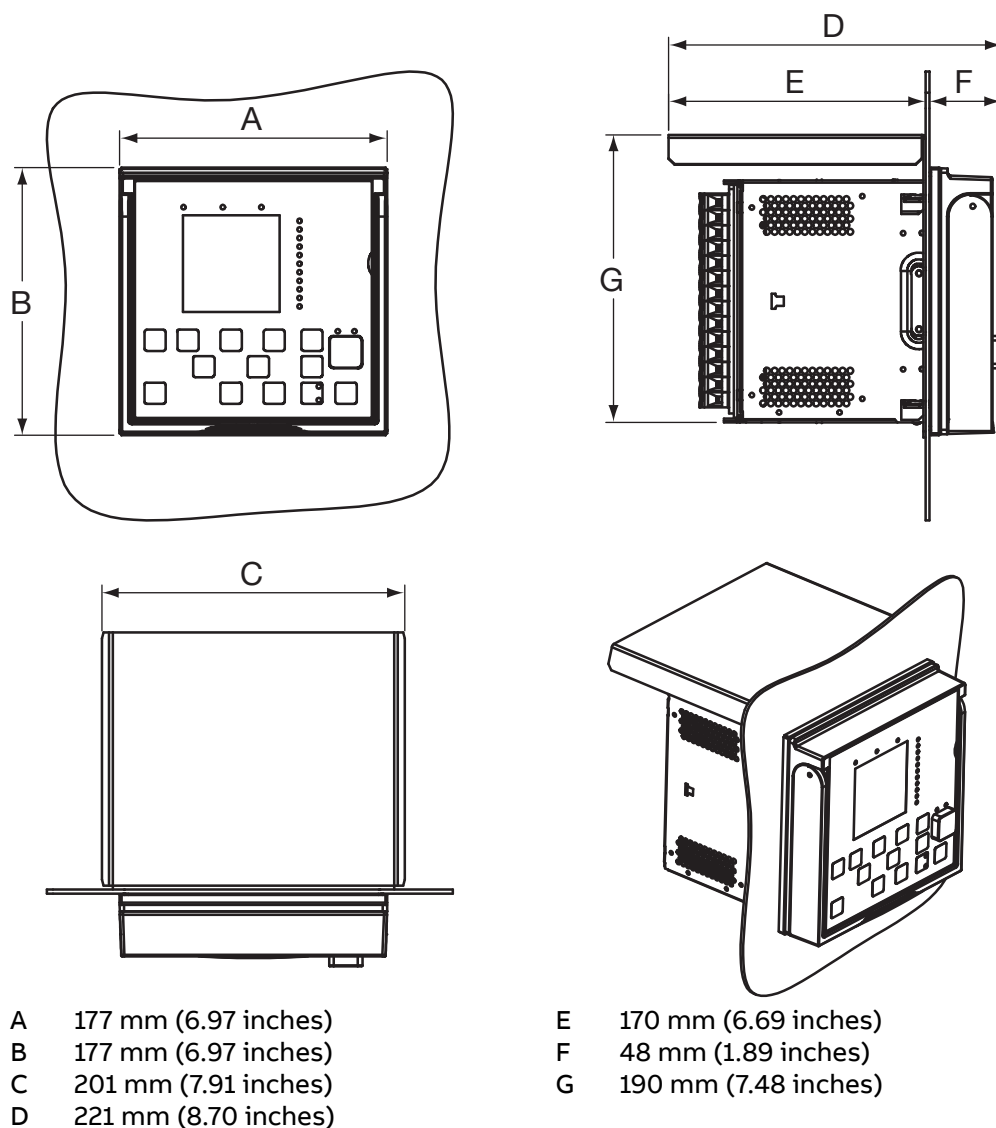


Figure 60: Flush mounted case and a plug-in unit with the protection cover

4.4.17 Flush mounting the wide size relay with protection cover

A mounting kit is needed for flush mounting the IED with a protection cover. The mounting kit includes detailed mounting instructions and the protection cover.

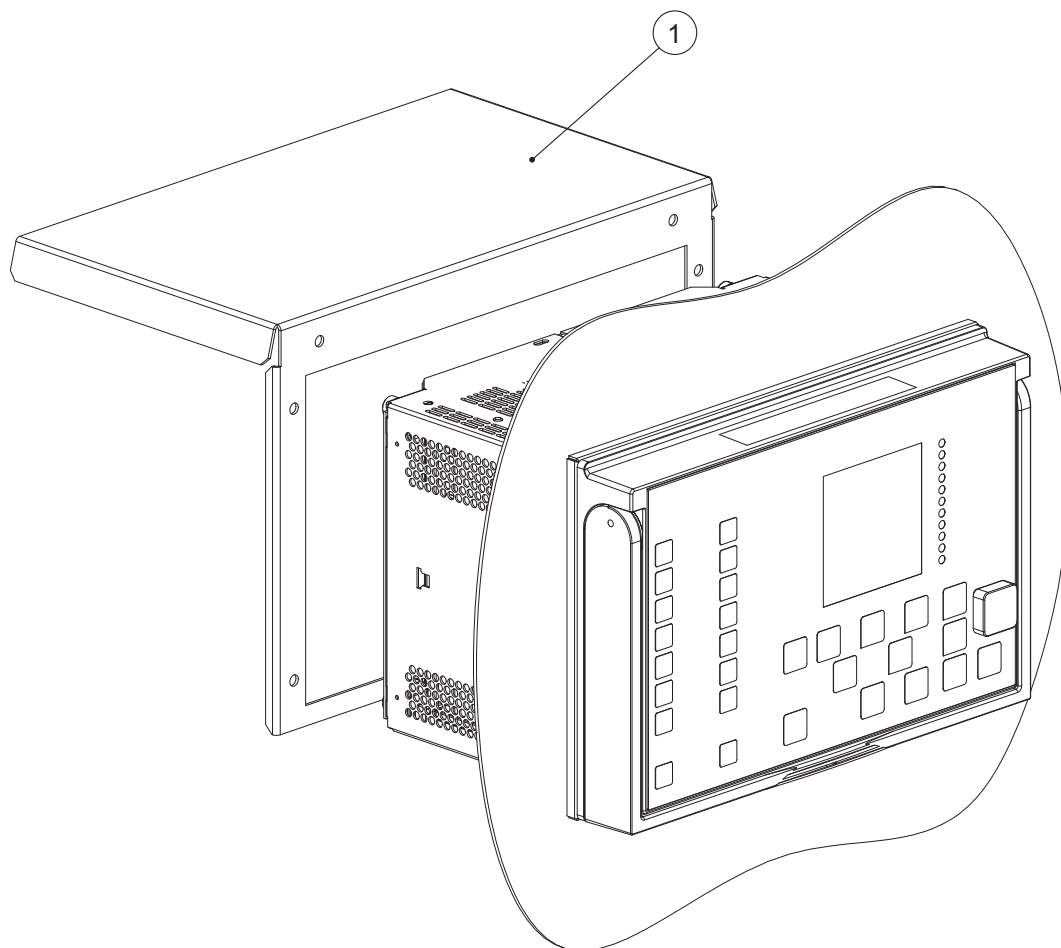
Installation requirements:

- Panel cut-out of 248 x 162 mm (9.76 x 6.38 inches)
- Depth behind the panel 170 mm (6.69 inches)



An IED equipped with optic connections requires a minimum depth of 180 mm (7.09 inches). Check the allowed minimum bending radius from the optical cable manufacturer.

1. Loosen the four M5 fixing screws in the case to fit the case into the panel cut-out.
2. Mount the case to the panel cut-out from the front.
3. Mount the protection cover from the rear.



- 1 Protection cover

Figure 61: Mounting the protection cover

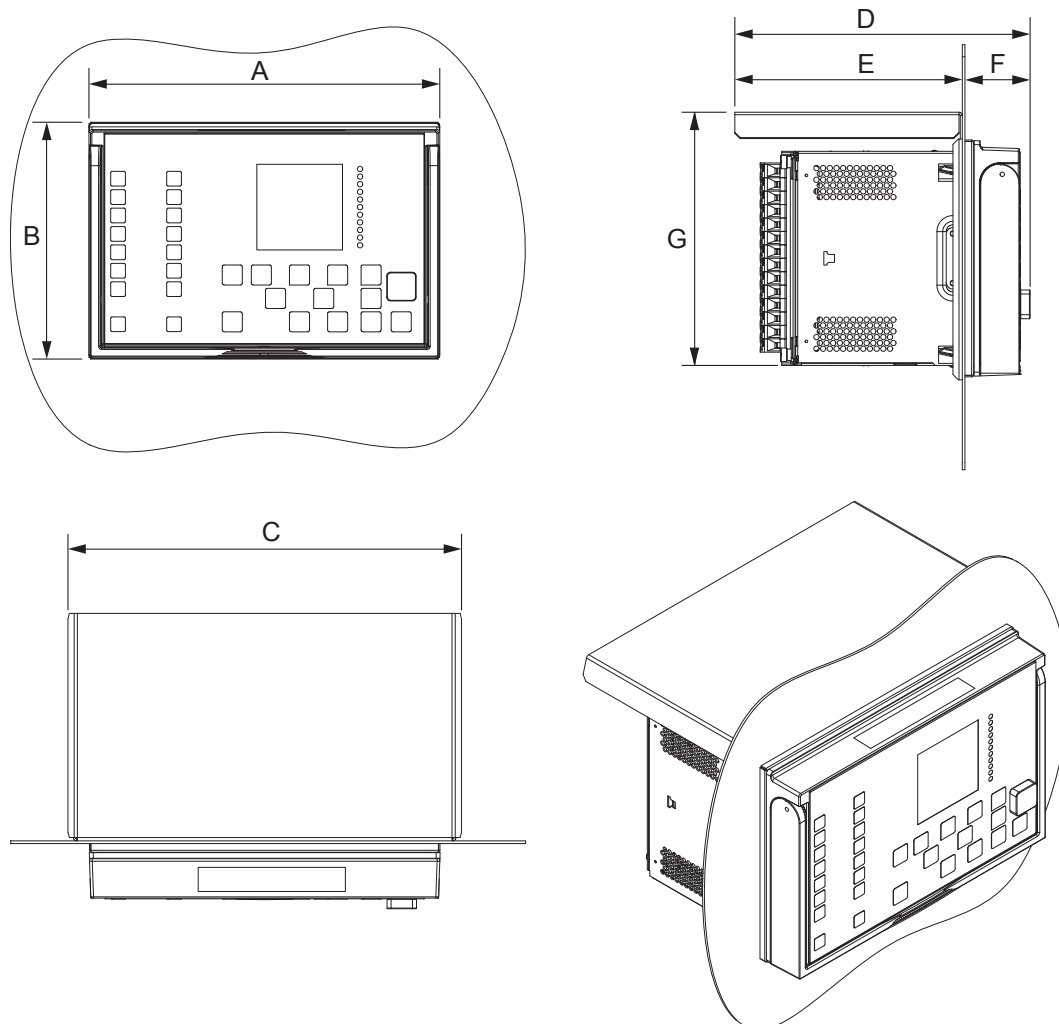
4. Tighten the M5 (T25) screws.



The allowed fixing screws' tightening torque is 0.7...1 Nm (0.52...0.74 foot-pounds).

5. Install the plug-in unit into the case.

There is a protective film on the top side of the IED. Its purpose is to prevent debris falling inside the unit while installing electrical wiring. Remove the protective film before energizing the IED.



A 262.2 mm (10.32 inches)

B 177 mm (6.97 inches)

C 294.1 mm (11.58 inches)

D 221 mm (8.70 inches)

E 170 mm (6.69 inches)

F 48 mm (1.89 inches)

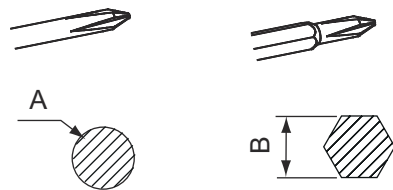
G 190 mm (7.48 inches)

Figure 62: Flush mounted case and a plug-in unit with the protection cover

5 Connecting

5.1 Required tools

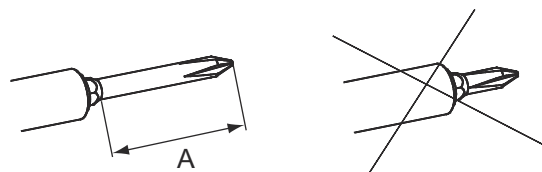
Only use a screwdriver and insert bits for Phillips (PH 1) cross-recessed head screws (M3.5) when handling CT/ VT terminals of screw-compression type.



A Max. $\varnothing 5.5$ mm (0.22 inches)

B Max. $\varnothing 5$ mm (0.20 inches)

Figure 63: Screwdriver for CT/VT terminals of screw-compression type



A Min. 15 mm (0.59 inches)

Figure 64: Insert bits for CT/VT terminals of screw-compression type

5.1.1 Tools for connector installation

Table 1: Tools for connector installation

Connector type	Tool for anchoring connector		Tool for connecting wire	
	Slotted head (DIN 5264) (mm)	ISO 8764/2	Slotted head (DIN 5264) (mm)	ISO 8764/2
CT/VT connectors (X115...X120)				
Compression	–	–	–	PH 1
Ring lug	–	–	1.2x8.0	PH 2
Signal connectors (X100...X115)				
Compression	–	–	0.6x3.5	–
Ring lug	–	–	1.2x8.0	PH 2
VT / signal connectors on X130				
Compression	0.6x3.5	PH1	0.6x3.5 ¹	PH1

Table continues on the next page

¹ For SIM1005 terminal 0.8x4.0

Connector type	Tool for anchoring connector		Tool for connecting wire	
	Slotted head (DIN 5264) (mm)	ISO 8764/2	Slotted head (DIN 5264) (mm)	ISO 8764/2
Ring lug	0.5x3.0	–	1.2x8.0	PH 2
IRIG-B connector (X000)				
IRIG-B	0.4x2.5	–	0.4x2.5	–

5.2 Connecting wires

All connections are made on the rear of the case. No soldering is needed.

- Open the screw-compression type terminals before inserting any wires. By default the terminals are closed at the time of delivery.
- Use fine wire in door mounting.



See the application manual for product-specific connection diagrams.

5.2.1 Connecting wires to screw-compression type connectors

Terminal blocks of screw-compression type are used for electrical connections.

1. Open the screw terminal before inserting a wire into it for the first time. To open the screw terminal, turn the fixing screw anti-clockwise until the terminal hole is wide open (the inside of the terminal hole is surrounded by metal).
2. Insert the wire and turn the fixing screw clockwise until the wire is firmly fixed.

Table 2: Wire specifications for screw-compression type connectors

Connector type	Wire size		Stripping length ¹	Contact screw tightening torque	Connector fixing screw tightening torque
	Stranded	Crimp ferrule			
CT/VT connectors	0,5...6,0 mm ² ^{2,3,4}	0,5...4 mm ² DIN 46228 pt1 ⁵	9 mm (0.354 in)	0,8 Nm ^{3,4} (0.5900 ft-lb)	NA (Assembled on the factory) ⁶

Table continues on the next page

¹ Depending of used ferrule

² For two wires of max. 2,5 mm² (AWG 14)

³ VTs terminal on X130: One 0,5...2,5 mm² (20...14 AWG) wire. Contact screw tightening torque: 0,4...0,5 Nm (0.2950...0.3688 ft-lb)

⁴ SIM0005 terminal on X130: One 0.5...2.5 mm² (20...14 AWG) wire. Contact screw tightening torque: 0,6 Nm...0,8 Nm (0.4425...0.5900 ft-lb)

⁵ In case of insulated crimp ferrules, TWIN ferrules need to be used with daisy chain requirement 2 × 2.5 mm² (AWG 14), two conductors without crimp ferrule with the same cross section 0,14...1,5 mm² (AWG 26-16)

⁶ Terminal on X130: 0,2...0,25 Nm (0.1475...0.1844 ft-lb)

Connector type	Wire size		Stripping length ¹	Contact screw tightening torque	Connector fixing screw tightening torque	
	Stranded	Crimp ferrule				
	(AWG 20...10)	(AWG 20...12)				
Signal connectors	0,5...2,5 mm ² ⁷ (AWG 20...14)	0,5...2,5 mm ² DIN 46228 pt1 (pt4) ^{7,8} (AWG 20...14)	6...8 mm (0.236 ...0.315 in)	0,4...0,5 Nm (0.2950...0.3688 ft-lb)	NA (Assembled on the factory) ⁶	
Serial EIA-485 / IRIG-B connector	0,2...1,5 mm ² (AWG 24...16)	0,2...1,5 mm ² - DIN 46228 pt1 (pt4) ⁸ (AWG 24...16)	6...8 mm (0.236 ...0.315 in)	0,2...0,25 Nm (0.1475...0.1844 ft-lb)	0,15...0,2 Nm (0.1106...0.1475 ft-lb)	

5.2.2 Connecting ring-lug type wires

Ring-lug type insulated terminal can be used for signal connector X120 (and X115 if similar connector available). The maximum outside diameter for the M4 ring-lug type terminals is 9 mm (0.354 inches).

Table 3: Wire specifications for ring-lug type connectors

Connector type	Wire size		Contact screw tightening torque	Connector fixing screw tightening torque	
	Stranded	Ring tongue terminal			
CT/VT connectors	0,5...6,0 mm ² ^{1,2} (AWG 20...10)	M4 max. Ø 9 mm (0.350 in)	0.8...1,0 Nm ² (0.5900...0.7376 ft-lb)	NA (Assembled on the factory) ³	
Signal connectors	0,5...2,5 mm ² ⁴ (AWG 20...14)	M3 max. Ø 8 mm (0.310 in)	1,13 Nm (0.8333 ft-lb)	NA (Assembled on the factory) ³	

5.3 Connecting protective earthing



The earth lead must be at least 6.0 mm² (9 AWG). If the earth lead is long, the cross section of the wire must be increased.



Use fine copper wire as the earth lead.

¹ Depending of used ferrule

⁷ For two wires 0,5...1,0 mm² (AWG 20...17). Crimp ferrule 2 x 0,5...1,0 mm² (20...17 AWG)

⁸ Plastic collar ferrule type: DIN 46228 pt 4

¹ For two wires of max. 2,5 mm² (AWG 14)

² VTs terminal on X130: One 0,5...2,5 mm² (20...14 AWG) wire. Contact screw tightening torque: 1,13 Nm (0.8333 ft-lb)

³ Terminal on X130: 0,2...0,25 Nm (0.1475...0.1844 ft-lb)

⁴ For two wires 0,5...1,0 mm² (AWG 20...17)

1. Loosen the protective earth screw (T20) to connect a separate earth protection lead.

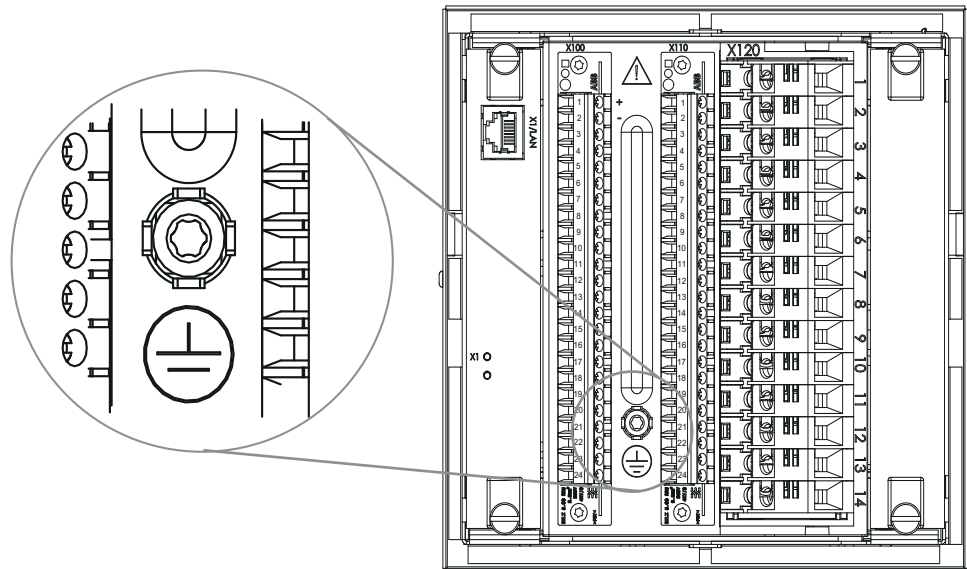


Figure 65: The protective earth screw is located between the connectors X100 and X110 (or between X100 and X105)



The earth lead should be as short as possible but extra length is required for door mounting.



Each protection relay must have its own earth lead connected to the earth circuit connector.

2. Connect the earth lead to the earth bar.
Use either stripped wire screwed between a washer cup and the protective earth screw or a ring lug.



Select a suitable ring lug to fit under the M4 screw.

3. Tighten the protective earth screw.
4. Support the earth lead so that it cannot break or weaken.
Be aware of the mechanical, chemical and electrochemical environment.

5.4 Connecting analog signals

A connection diagram is needed to connect the analog signals.

When using the ring-lug type for CT/VT terminals, follow these steps.

1. Open or remove the lid that covers the ring-lug fixing screw with the tip of a screwdriver. Every fixing screw has its own lid.

2. Remove the fixing screw.
3. Slide the screw through the terminal lug and screw it back on.

5.4.1 Connecting current and voltage inputs

- Connect the wires from the CTs/VTs to the correct device according to the phase order and the connection diagram.
- Each terminal X115/X120 for CTs/VTs is dimensioned for one 0.5...6.0 mm² (20...10 AWG) wire or for two wires of maximum 2.5 mm² (14 AWG).
- Each terminal X130 for VTs is dimensioned for one 0.5...2.5 mm² (20...14 AWG) wire.



SIM0005 terminal X130 is dimensioned for one 0.5...2.5 mm² (20...14 AWG) wire.



See the application manual for configuration specific current and voltage inputs.

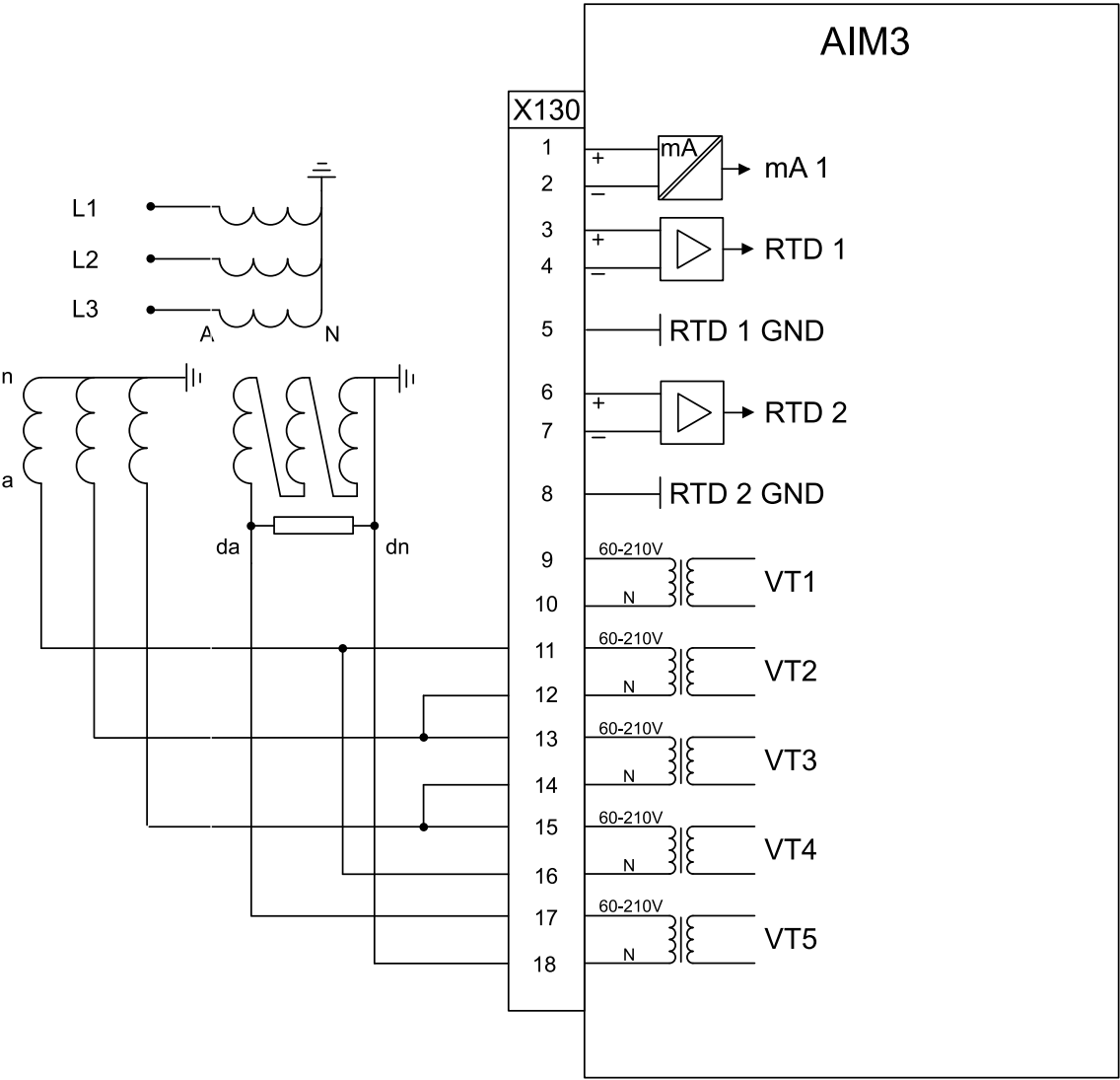


Figure 66: Example of AIM3 card variant (5 U, 2 RTD + 1 mA channel)

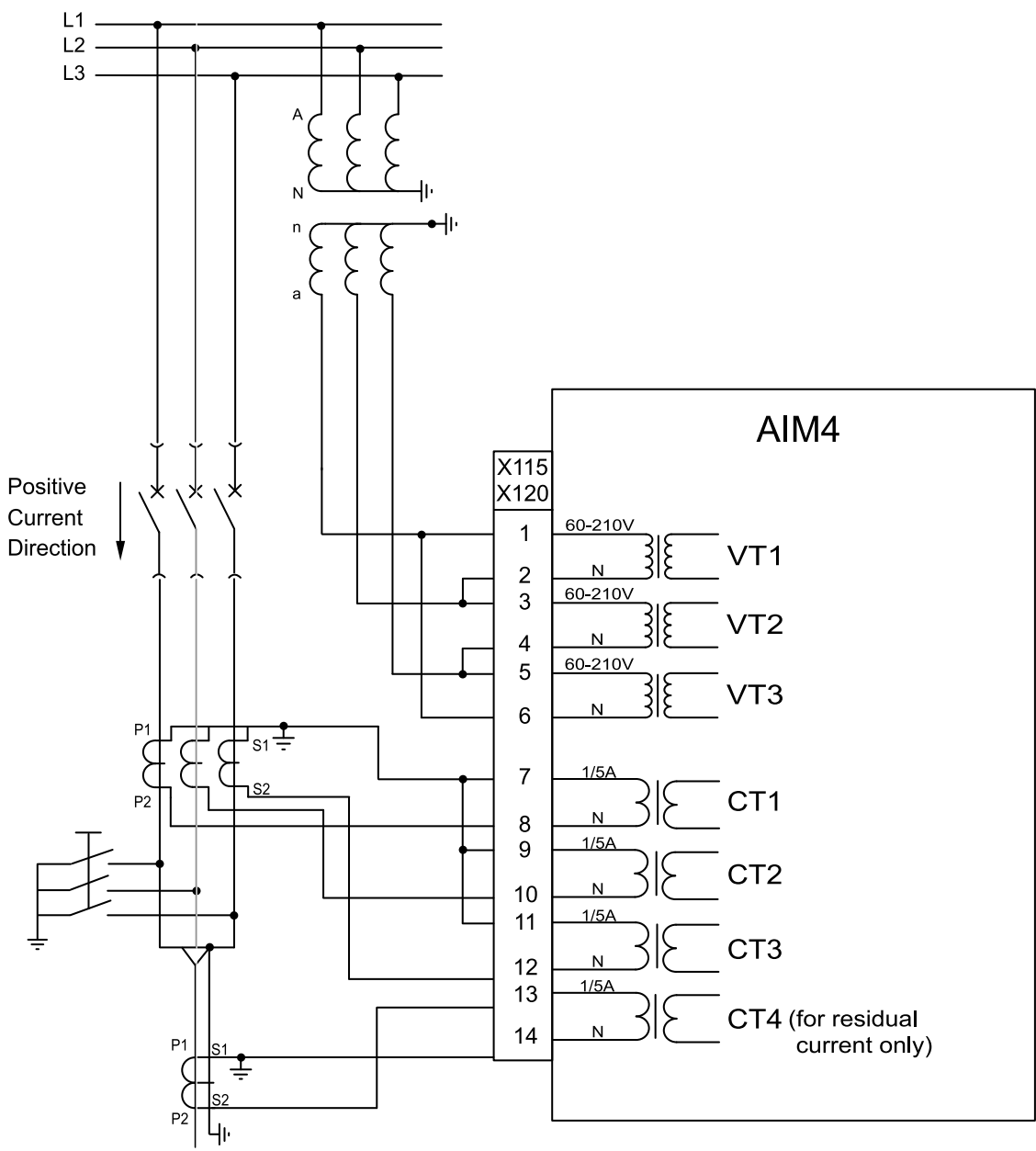


Figure 67: Example of AIM4 card variant (4 I + 3 U with 1/5 A Io channel)

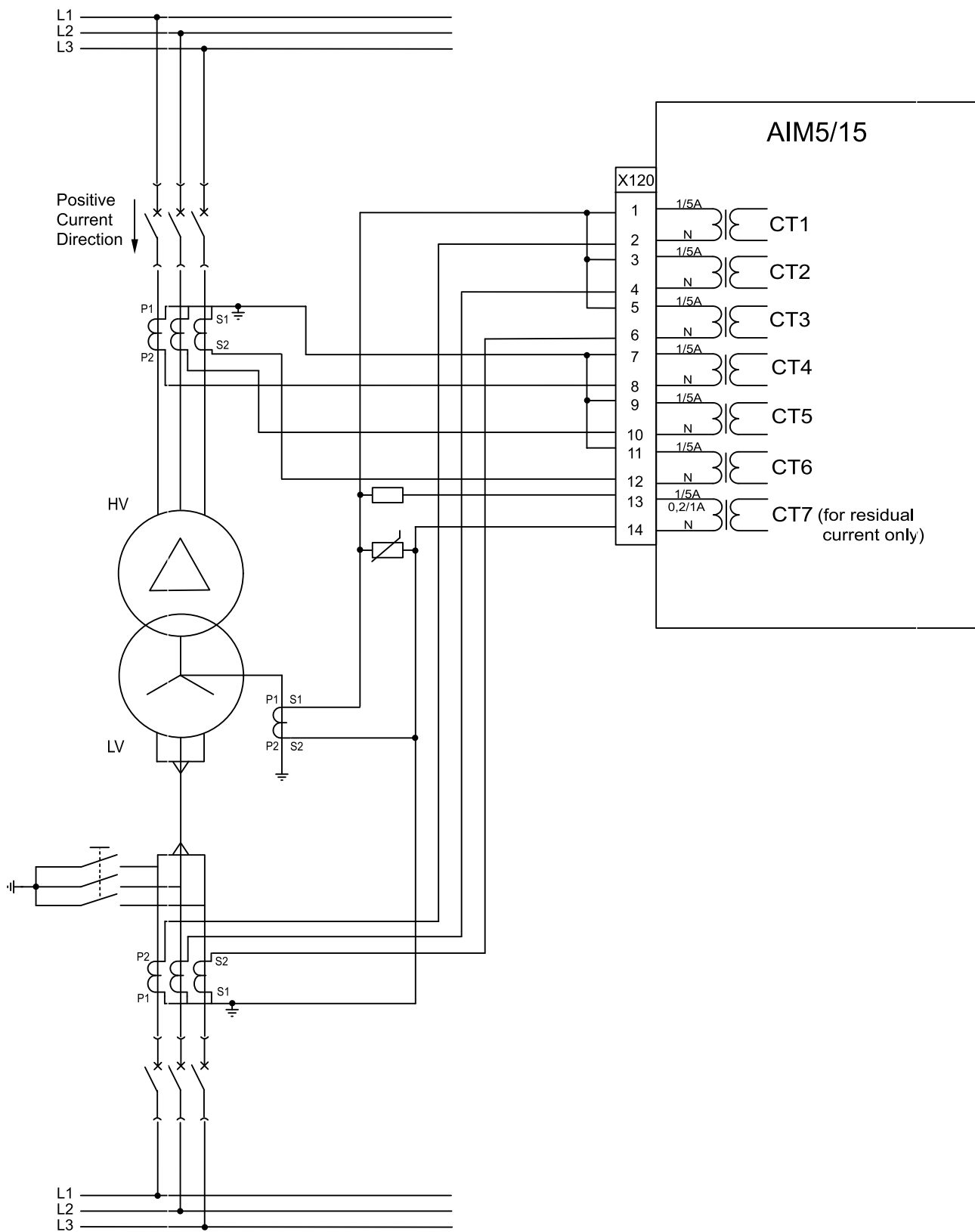


Figure 68: Example of AIM5/15 high-impedance restricted earth-fault card variant (7I with 1/5 A I_o channel)

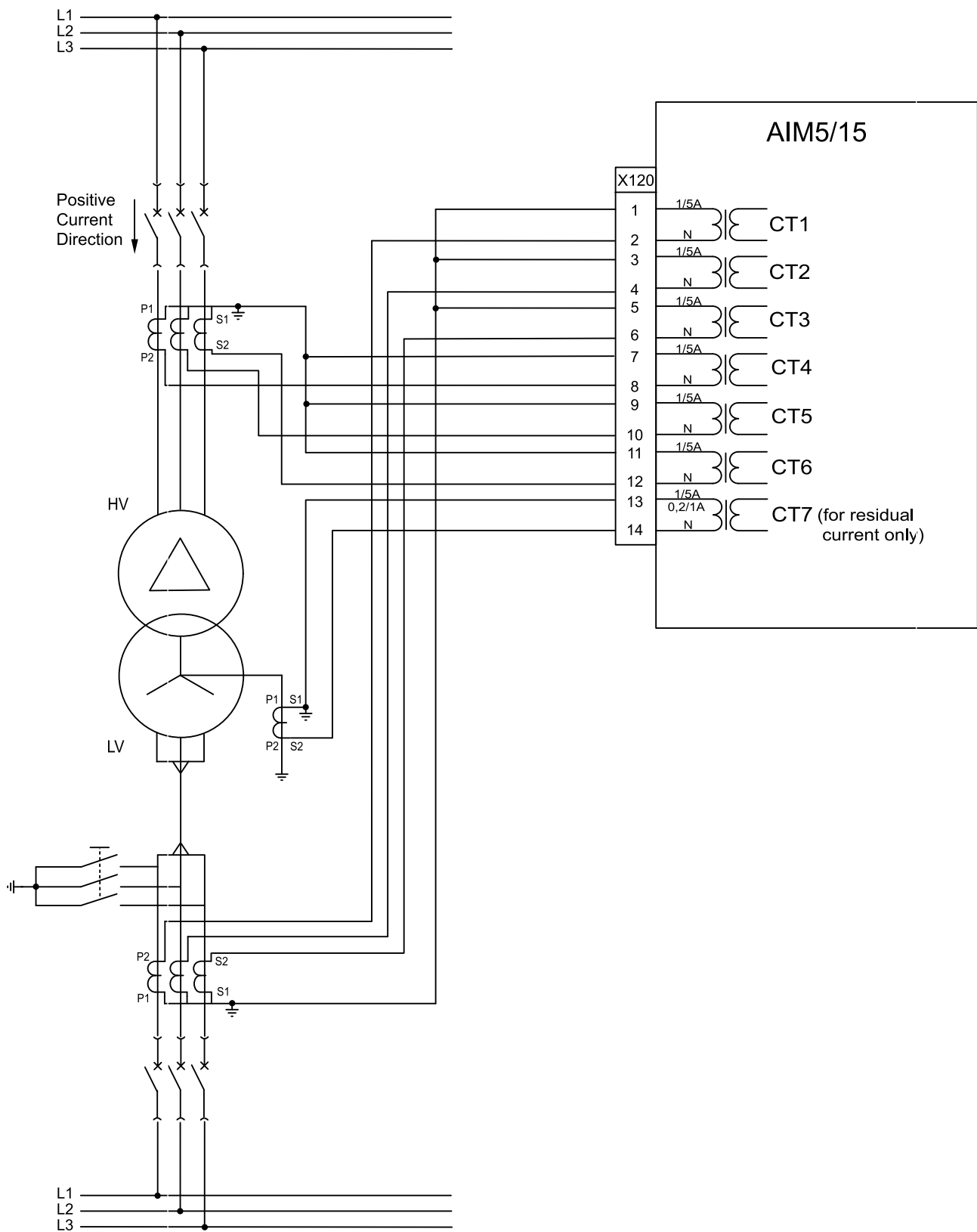


Figure 69: Example of AIM5/15 low-impedance restricted earth-fault card variant (7 I with 1/5 A I_o channel)

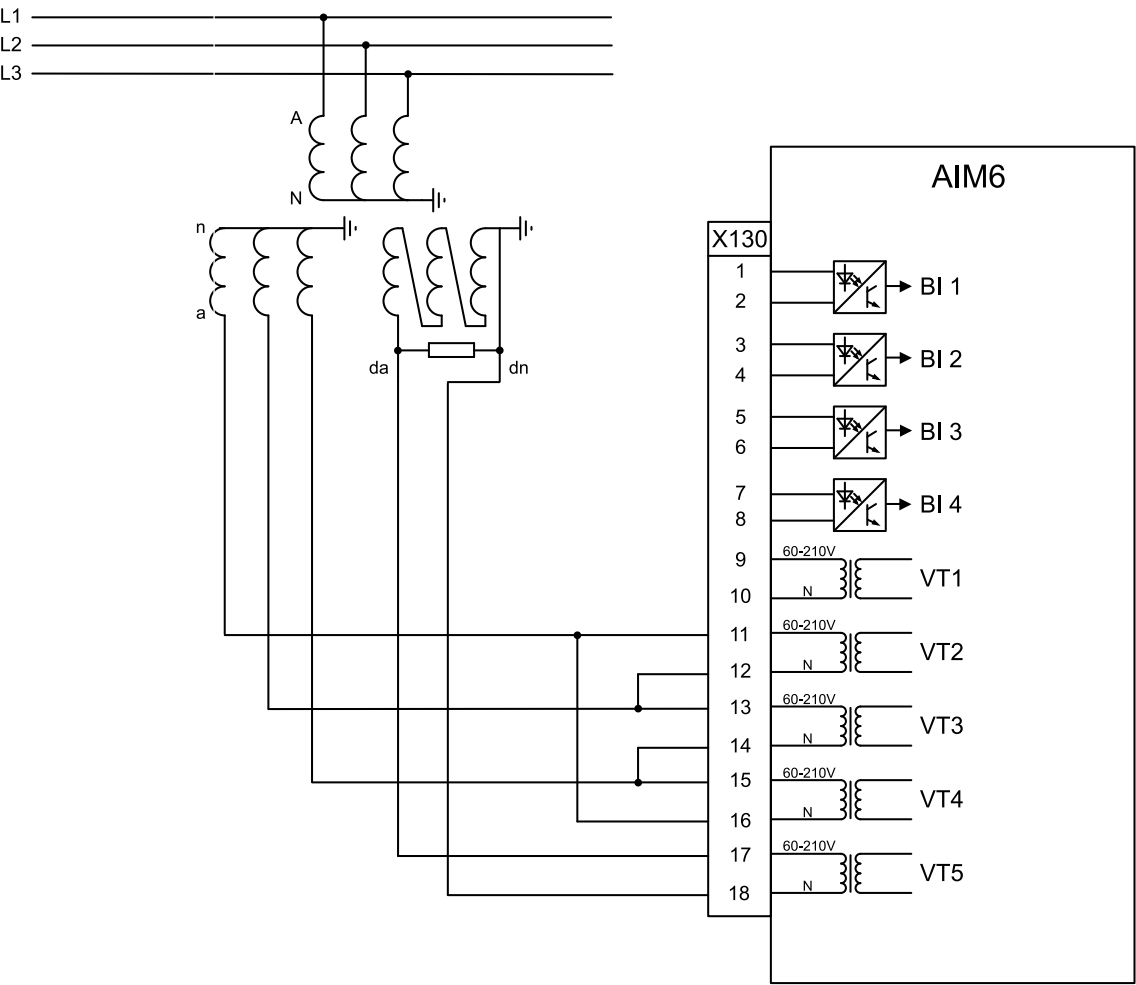


Figure 70: Example of AIM6 card variant (5 U)

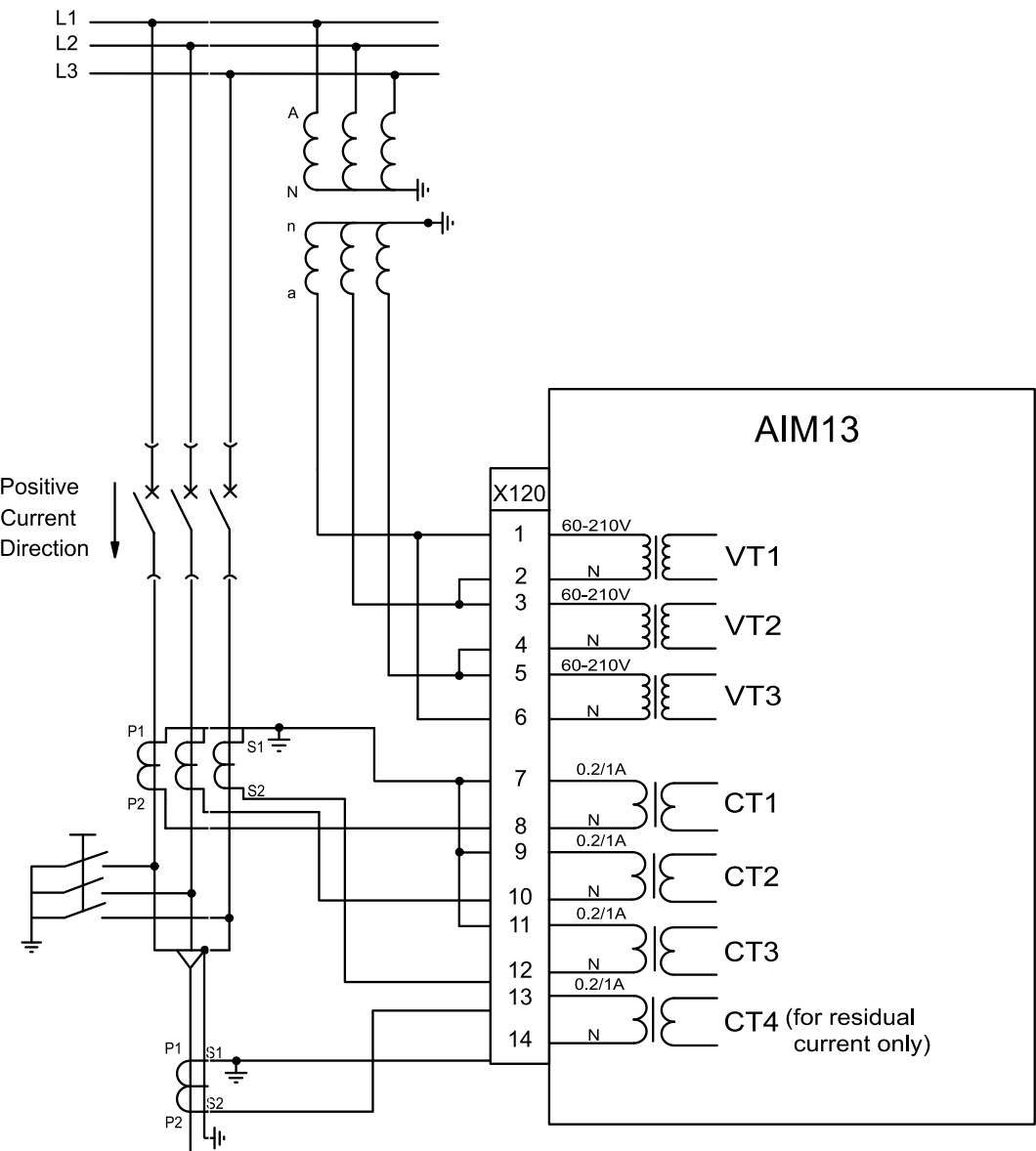


Figure 71: Example of AIM13 card variant (4 I + 3 U with 0.2/1A Io channel)

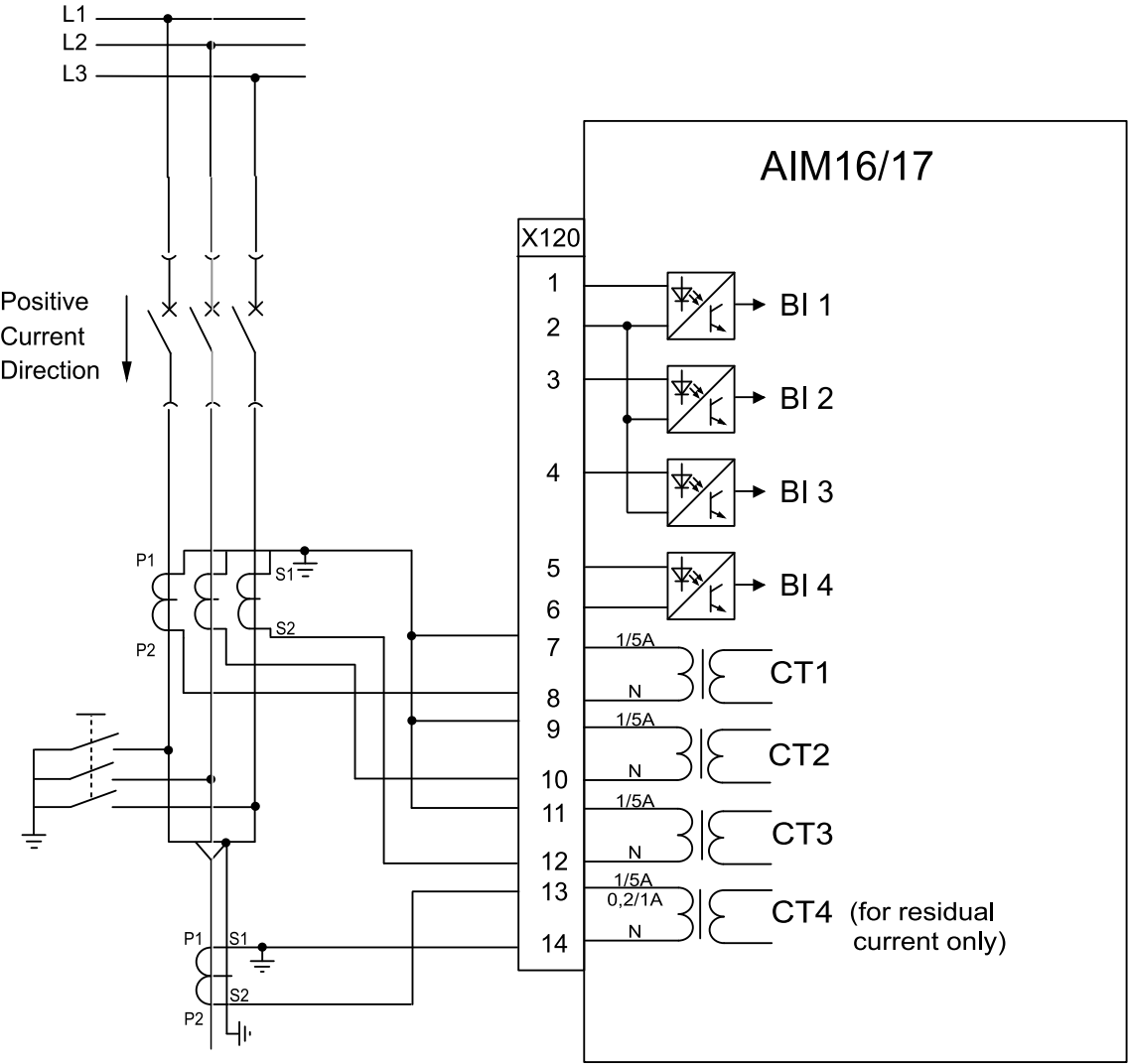


Figure 72: Example of AIM16/17 card variant (4 I with 1/5 A Io channel)

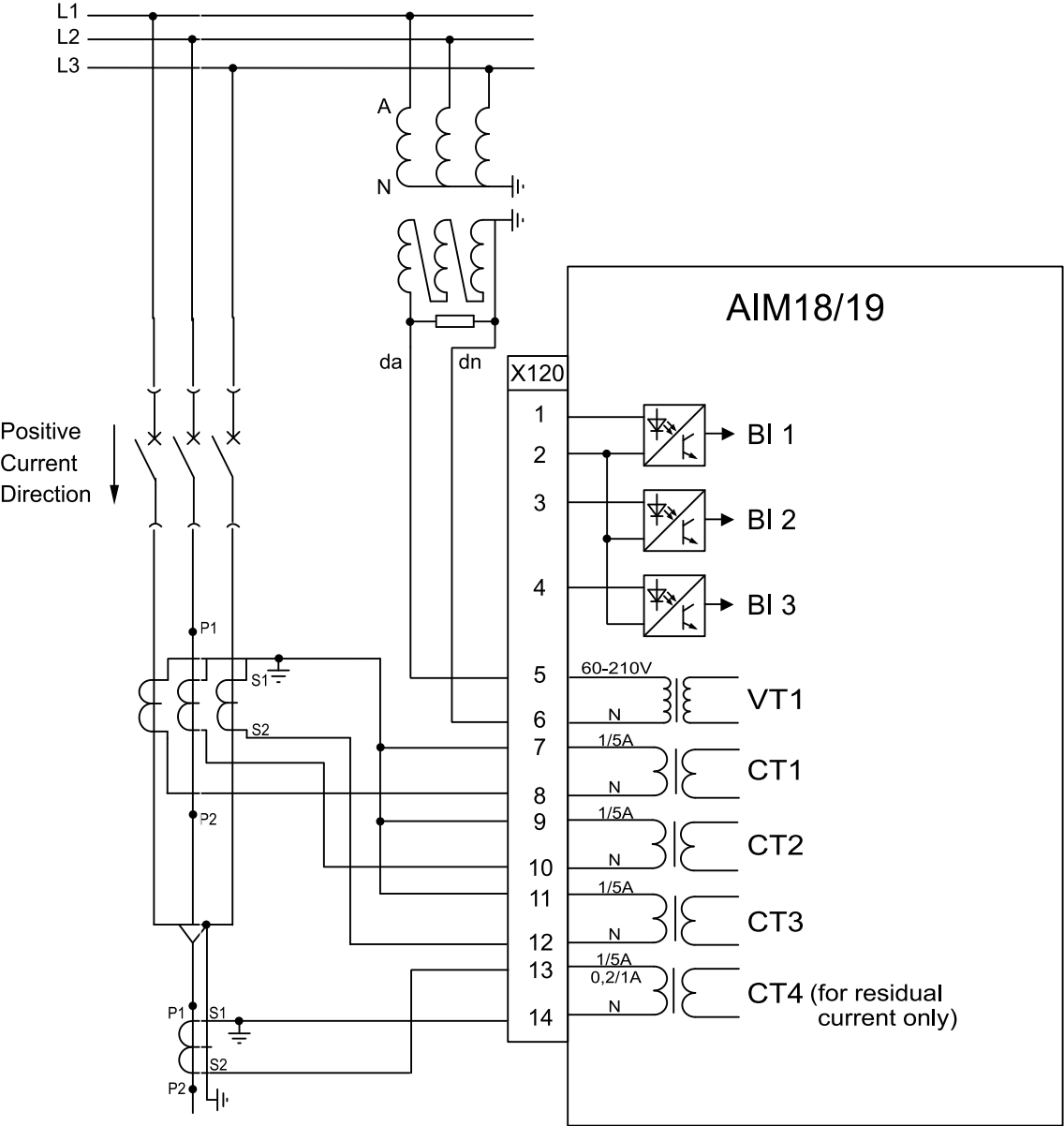


Figure 73: Example of AIM18/19 card variant (4 I + Uo with 1/5 A Io channel)

5.4.2 Connecting sensors (non-conventional instrument transformers)

Current measurement is based on Rogowski coil and voltage measurement is based on voltage divider. Both current and voltage measurements interface the relay via one RJ45 connector per phase. REX615 has one type of sensor input module (SIM0005) available which supports IEC 61869 standard.



SIM0005 has different RJ45 connector pin lay-out compared to SIM0002 module used in previous 615 series and 620 series relays. For better visual recognition RJ45 connectors on SIM0005 module are marked with red color.

There are a number of sensor interface adapters available which can be used to connect the sensors to the relay, supporting different scenarios both in green field and retrofit cases. More information can be found in Sensor accessories .

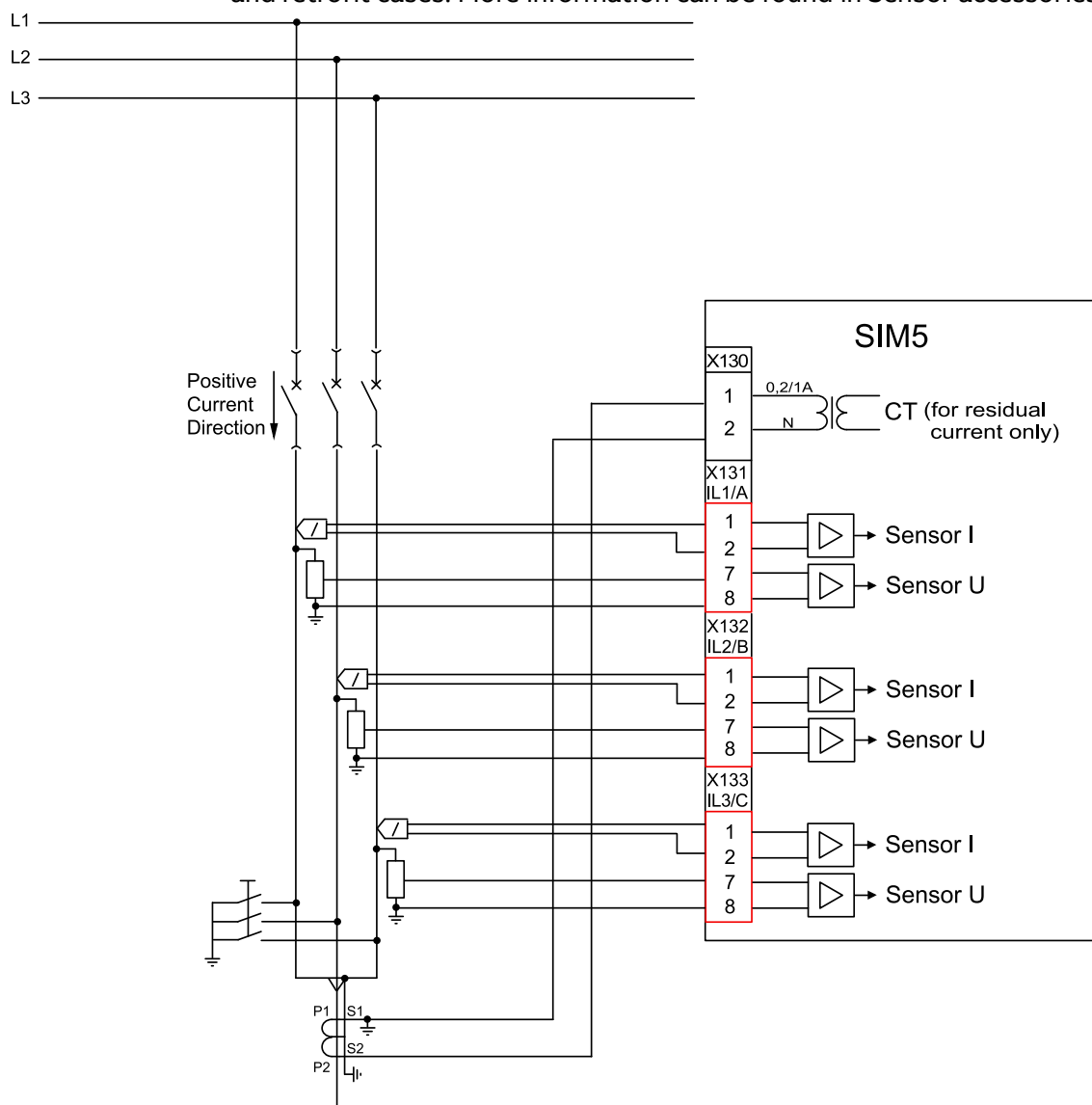


Figure 74: Example of SIM5 card variant (4I+3U with 0.2/1 A Io channel). Including red sensor connectors.

5.4.3 Connecting RTD and mA inputs

- Connect the wires from the transducers to the correct device according to the connection diagram.

Each terminal for RTDs and mA inputs is dimensioned for one 0.5...2.5 mm² (20...13 AWG) wire or for two 0.5...1.0 mm² (20...17 AWG) wires.



In addition to the specific RTD card, RTD/mA inputs are available with the AIM0003 card. See the application manual for standard-configuration specific RTD and mA inputs.

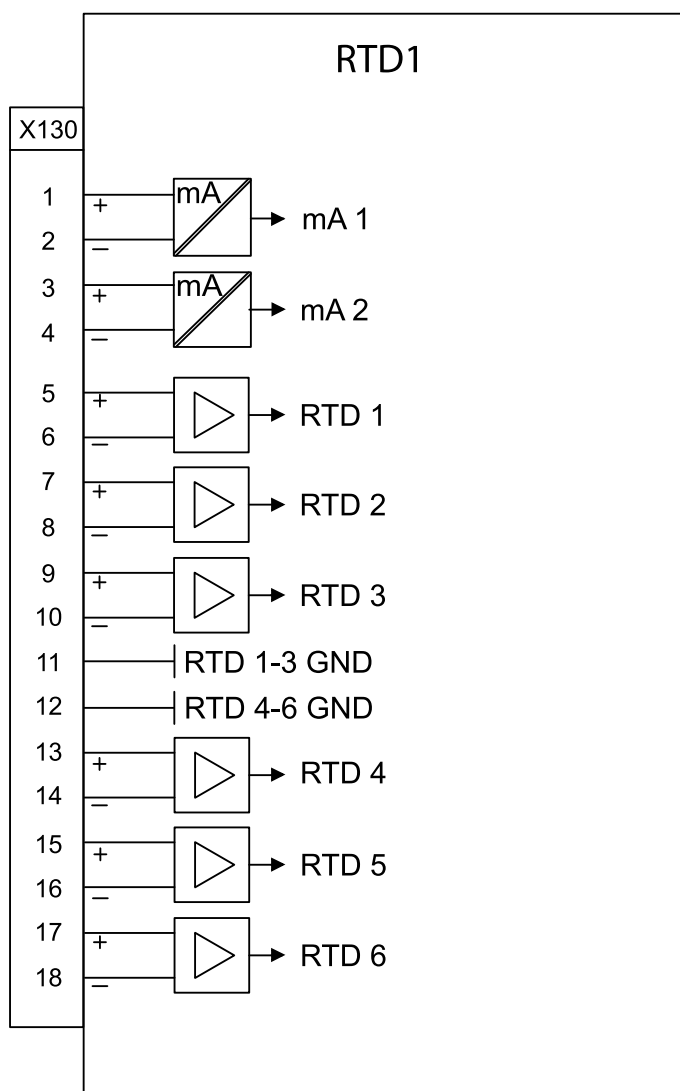


Figure 75: Example of RTD1 card variant (6 RTD + 2 mA channel)

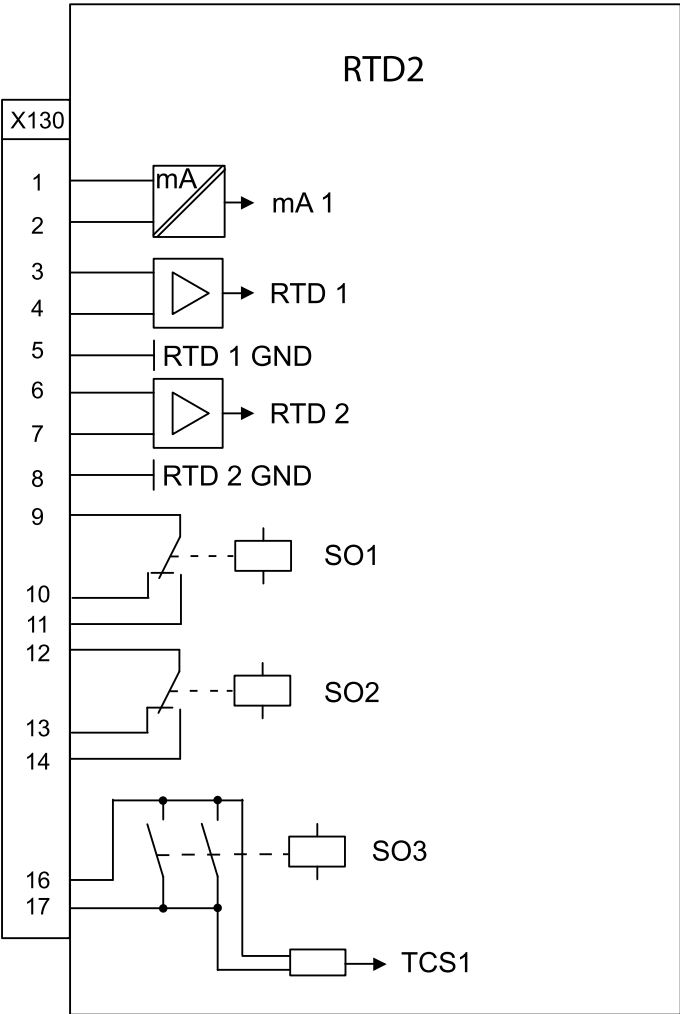


Figure 76: Example of RTD2 card variant, (2 RTD + 1 mA + 3 SO + TCS)

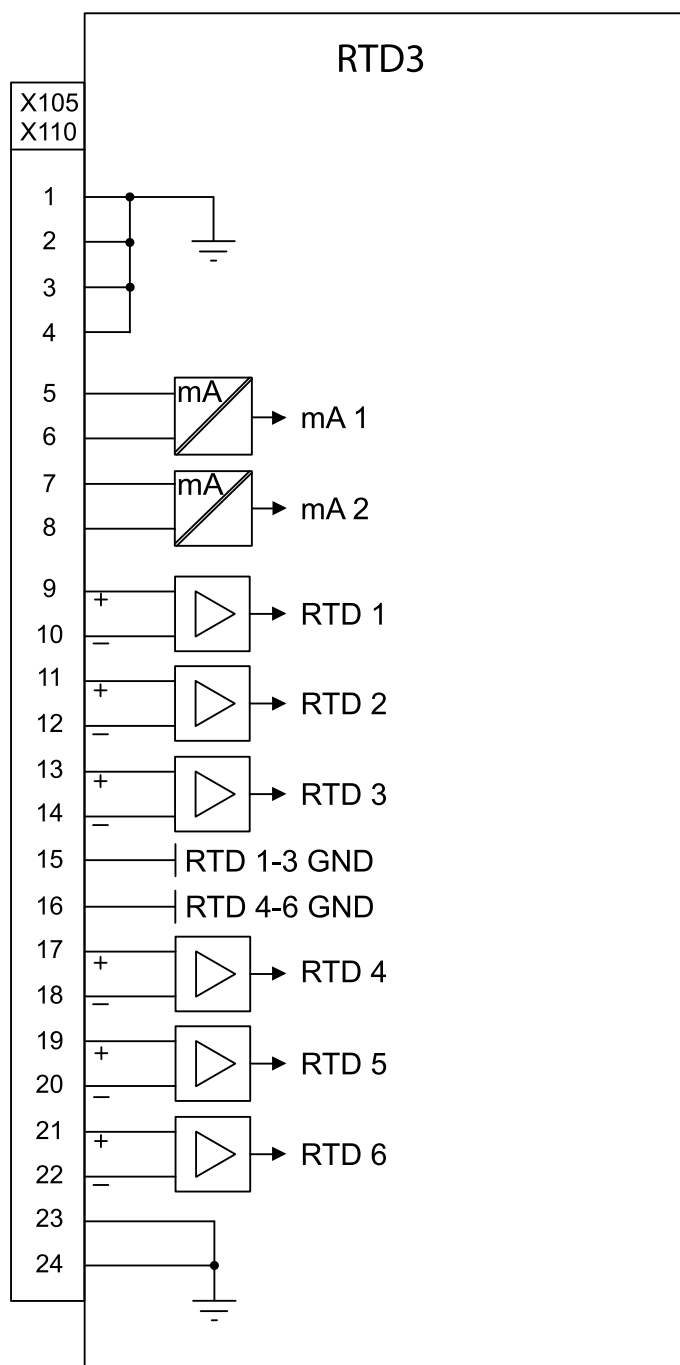


Figure 77: Example of RTD3 card variant, (6 RTD + 2 mA)

5.4.4 Connecting protection relay with a test switch

- When the protection relay is used with a test switch, connect the current and voltage transformers directly to the switch.

5.4.5 Shielding RTD/mA instrument cables

The RTD/mA inputs are connected by screened cables (1 mm²/17 AWG), for example, Unitronic 300S (formerly Unitronic 300CY), manufactured by LAPP Group.



Other manufacturers' cables with similar technical features can also be used. When selecting a suitable sensor cable, observe also other possible customer installation requirements.

1. Prepare screened cable with a drain wire.
 - a) Select a drain wire (0.75 mm²/18 AWG) as short as possible.

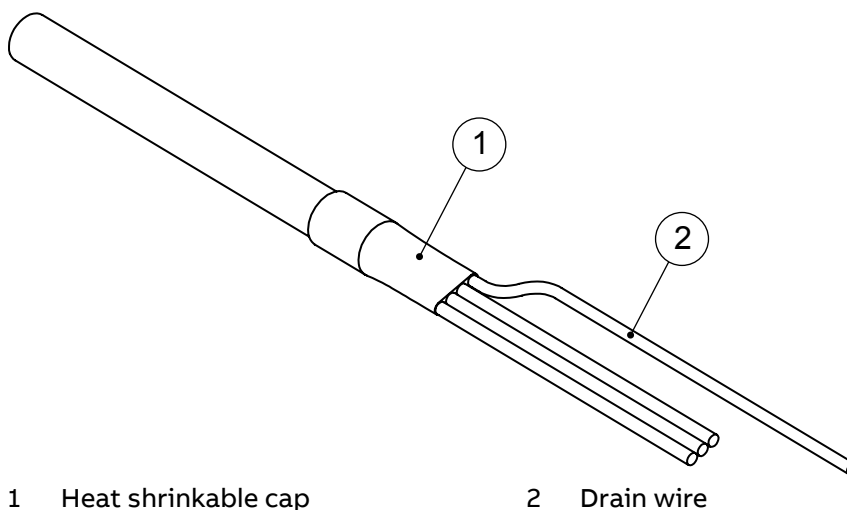


The wire combination green/yellow should not be used as the drain wire has no protective function.

- b) Install the drain wire.
The wire can be installed, for example, with a Shield-Kon one-piece connector (Thomas & Betts). For detailed instructions, see the manufacturer's manual.
 - c) Protect the cable end with a heat shrinkable cap.

Table 4: Drain wire lengths

Wire from connector	Length
X105	~200 mm (7.87 inches)
X110	~200 mm (7.87 inches)
X130	~300 mm (11.81 inches)

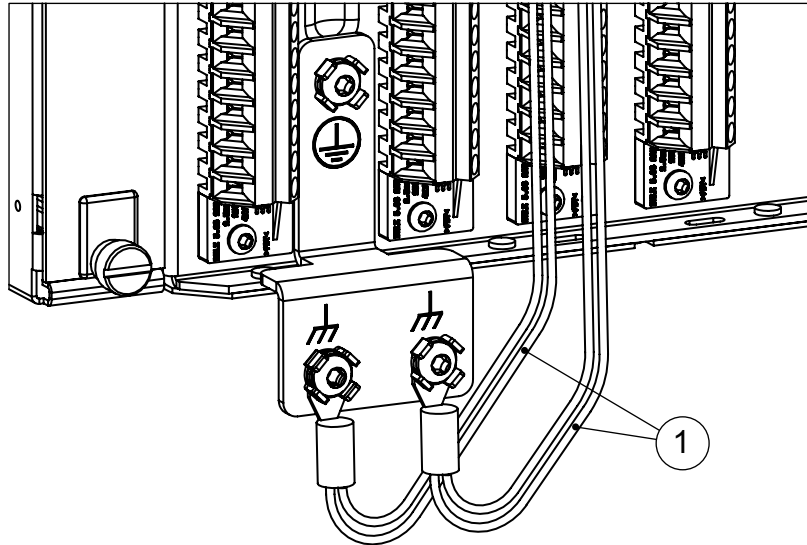


1 Heat shrinkable cap

2 Drain wire

Figure 78: Screened cable

2. Connect the functional earthing wires (drain wires) to the protection relay's body via a connection flange (2RCA036978A0001) to separate them from the protective earthing conductor.



1 Drain wires

Figure 79: Connection of drain wires

¹ Cannot be used when the protection relay is mounted with the Combiflex 19" equipment frame (2RCA032826A0001)

5.5 Connecting binary signals

- Connect the wires for the binary signals to the correct device according to the connection diagram. Each terminal for binary input and output signal is dimensioned for one 0.5...2.5 mm² (20...13 AWG) wire or for two 0.5...1.0 mm² (20...17 AWG) wires.



In addition to specific BIO cards, BI/O signals are available with some AIM and PSM cards. See the application manual for standard-configuration specific binary signal options.



Use only DC power for the binary inputs. Use of AC power, including full- or half-wave rectified AC power, may cause damage to the binary input modules.

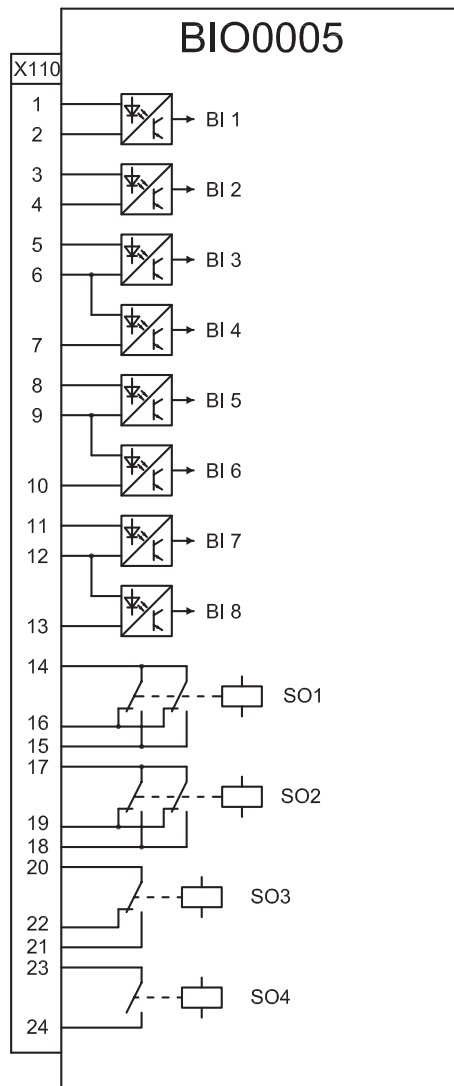


Figure 80: Example of BIO0005 card variant (8 BI + 4 BO)

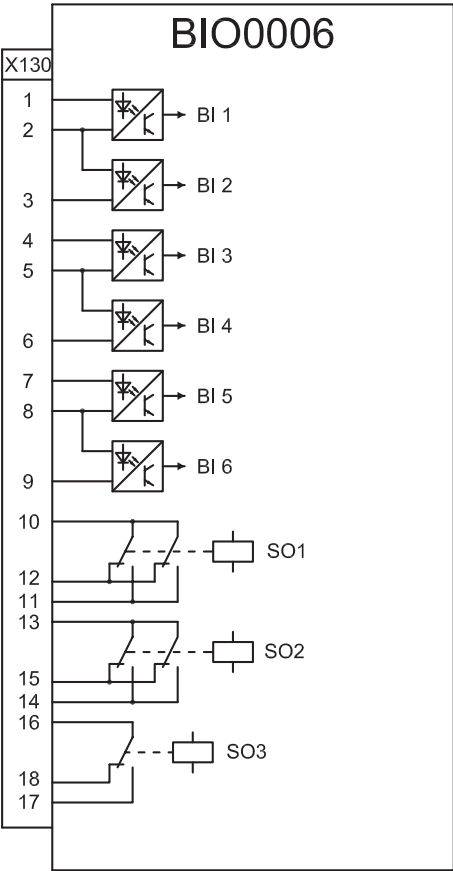


Figure 81: Example of BIO0006 card variant (6 BI + 3 BO)

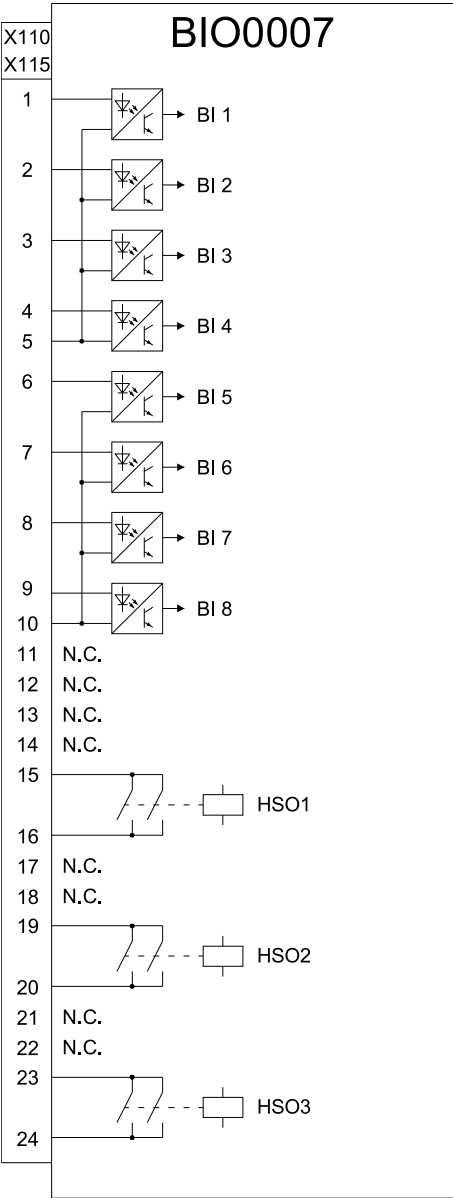


Figure 82: Example of BIO0007 card variant (8 BI + 3 BO)

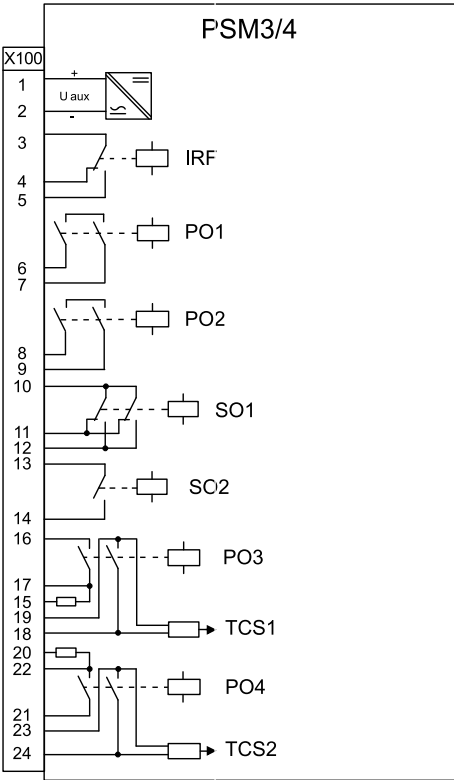


Figure 83: Example of PSM3 and PSM4 card variant (depending on power supply nominal auxiliary voltage)

5.6 Connecting power supply

The permitted auxiliary voltage range of the protection relay is marked on top of the protection relay's LHMI.

- Connect the protection relay's auxiliary voltage to terminals X100-1 and X100-2.
- Connect the positive lead to terminal X100-1.

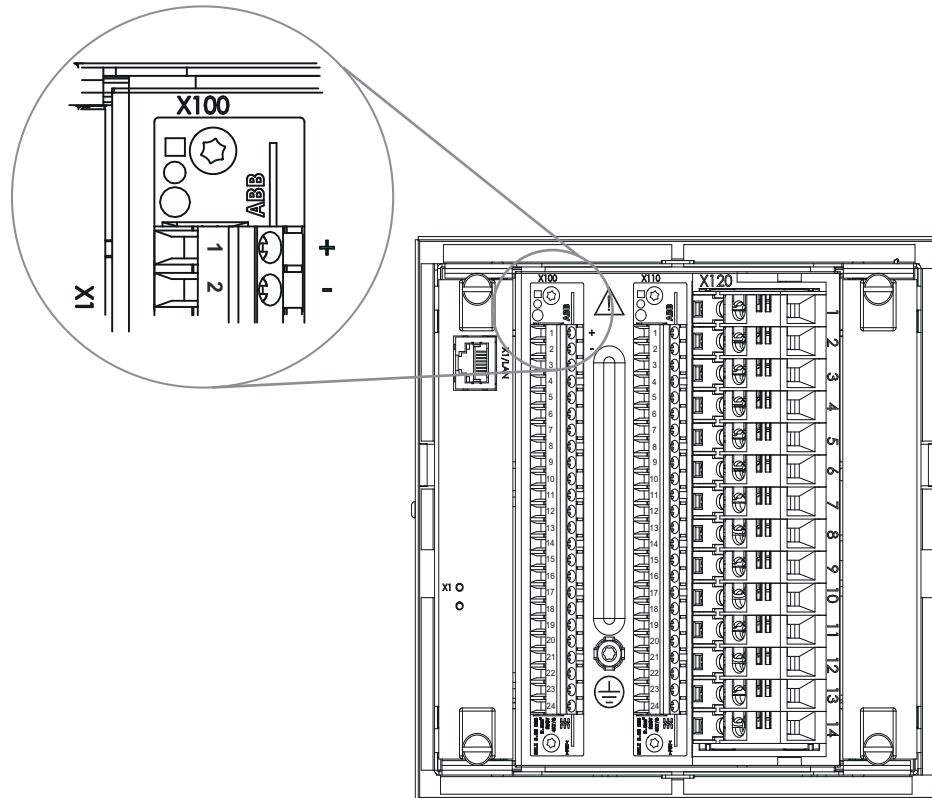


Figure 84: Connecting auxiliary voltage

5.7 Connecting communication

- Before connecting communication, check that the HW module has the correct communication interfaces.

The communication module is located on the left side of the protection relay when viewing the case from the rear.



See the technical manual for product-specific communication interfaces.

5.8 Mounting lens sensors for an arc protection system

Arc protection is used to detect arc situations in air insulated metal-clad switchgear. The arc protection system determines where in the switchgear cubicle the optional lens sensors are installed.

1. Drill a hole ($\varnothing$ 10 mm (0.39 inches)) in the wall of the supervised space.

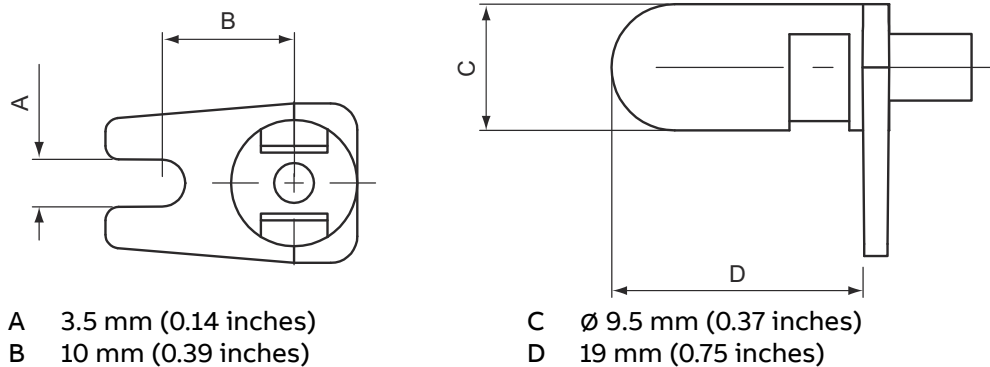


Figure 85: Dimensions of the lens sensor

2. Fit the lens sensor into the hole and fasten it with a self-tapping M3 screw. Alternatively, the lens sensor can be fastened with a cable tie. To do this, secure the cable tie to a suitable point of attachment on the cubicle wall and wrap the cable tie tightly around the sensor.

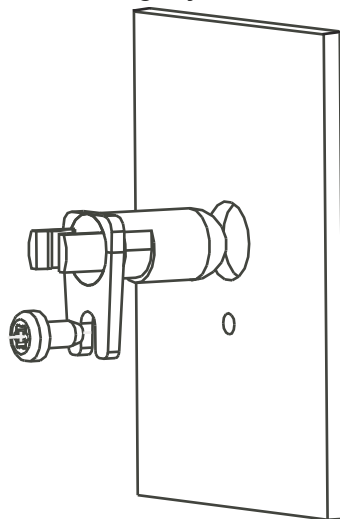


Figure 86: Mounting the lens sensor

3. Make sure that the cable tie lies in the groove of the sensor to prevent it from blocking the light.

5.9 Energizing protection relay



Before connecting the auxiliary power, check that the protective film is removed from top of the protection relay.

- Before connecting the auxiliary power, check that the terminal strip is wired and placed correctly.
- Check that all fixing screws are tight and that all cables are connected.
- Remove the protective film from the top side of the unit. Check that there is no debris visible in the ventilation holes.

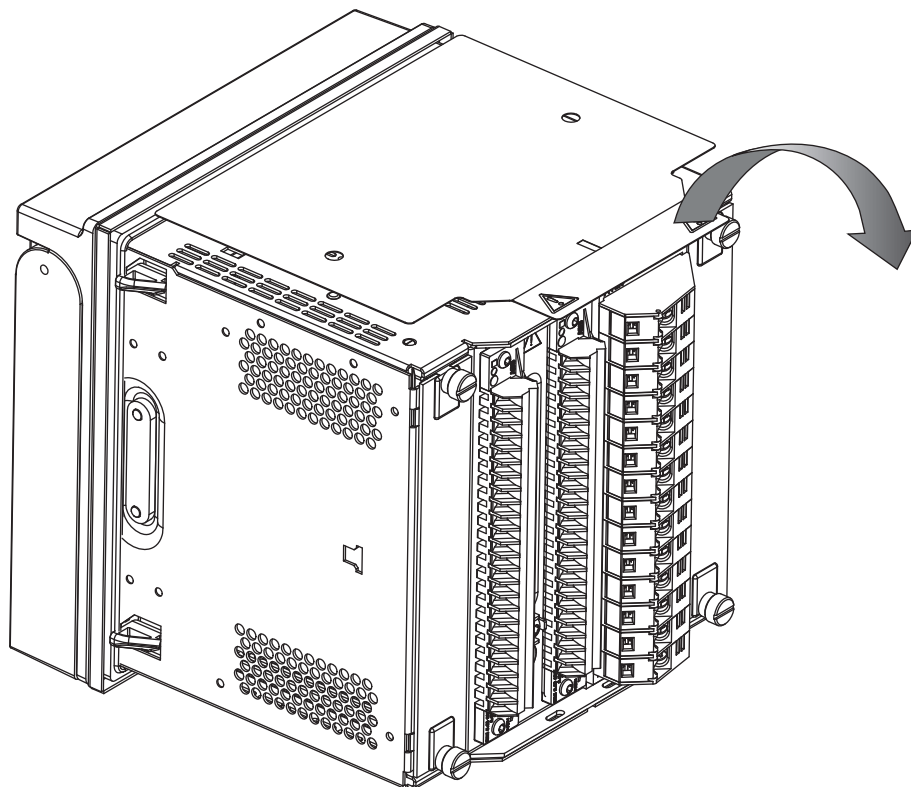


Figure 87: Removing the protective film

During the start-up, indications and self test procedures are shown in a certain order.

1. Green Ready LED starts to flash.
2. LCD lights up and ABB logo is displayed.
3. LCD test patterns are displayed and all the LEDs are lit for a short period.
4. The measurements view is displayed. A steady green Ready LED indicates a successful start-up.

If the protection relay detects a diagnostic error during start-up, the green Ready LED flashes and the internal fault code is displayed on the LCD.

6 Removing, repairing and exchanging

6.1 Product life cycle

At some point of the product life cycle, the protection relay is upgraded to a next generation unit. When selecting the original product, already consider the upgrading and extension possibilities that the specific product offers for its whole life cycle.

Protection relay specific options can be found in Retrofit Solutions Database on the Internet www.abb.com by following the links within ABB Service Guide or via ABB Product Guide from the product-specific Service & Support sheet.

6.2 Checking protection relay information

The protection relay information includes detailed information about the device, such as version and serial number. The protection relay information is shown on the display for a few seconds when the device starts up. The same information is found also in the protection relay menu.

1. Select **Main Menu > Information.**

2. Select a submenu with  and .

3. Enter the selected submenu with .

4. Browse the information with  and .

The **Product identifiers** submenu contains product related information like product type, serial number, order number, production date, configuration name, SW version and SW date.

The **Site identifiers** submenu contains information about the site where the protection relay has been installed.

The **System identifiers** submenu contains the Technical key and IEC 61850 version. The Technical key is unique and cannot be changed.

The **HW modules** submenu contains information about the HW modules.

6.3 Removing protection relay

1. Turn off the power.
2. Detach the plug-in unit from the case.
3. Disconnect the wiring.
4. Loosen the four M5 fixing screws.

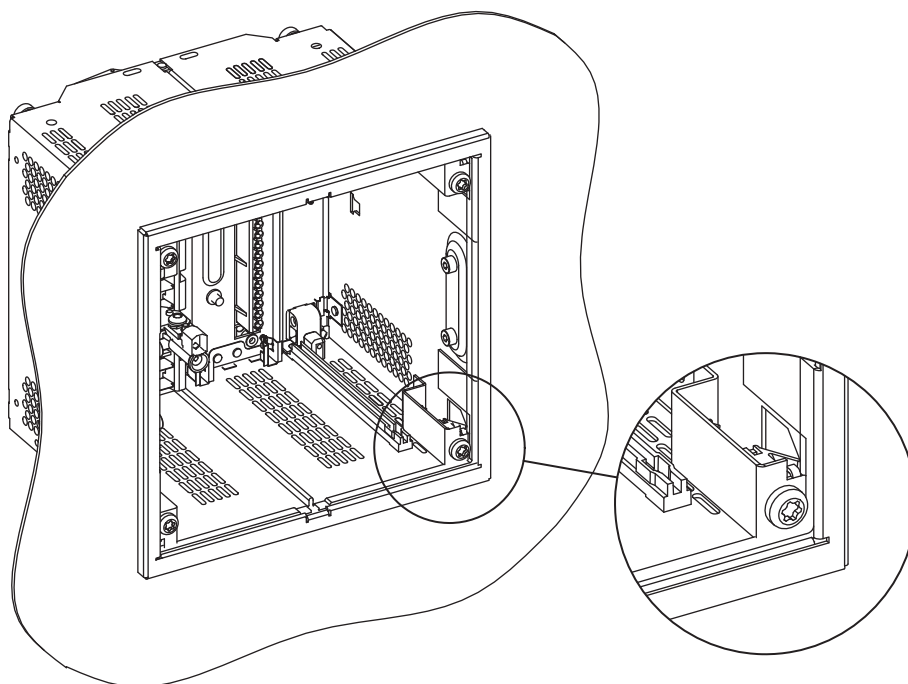


Figure 88: Loosening the M5 screws

5. Detach the case from the panel cut-out.

6.4 Sending protection relay for repair

- In case of product problems, contact the nearest ABB office or representative for consultation and instructions.

6.5 Exchanging protection relay

- To exchange the protection relay with another identical unit, remove the protection relay and install the new one.

The exchangeable units can be found from the PartsOnLine system, see www.abb.com/partsonline. Use of PartsOnLine requires user registration.

- To exchange a protection relay to a different unit, change the case and connect the wires.
- When replacing only the plug-in unit for maximum system availability, check that the order code of the case and the spare plug-in unit are equal to ensure the type compatibility.

The serial numbers of the original case and the spare plug-in unit are temporarily mismatched. For matching the serial numbers, replace also the case of the spare unit as soon as possible, for example, during the next scheduled maintenance break.

7 Technical data

7.1 Case and LHMI display variants

7.1.1 Front side of the protection relay



Figure 89: Standard size LHMI (IEC/ANSI variant)

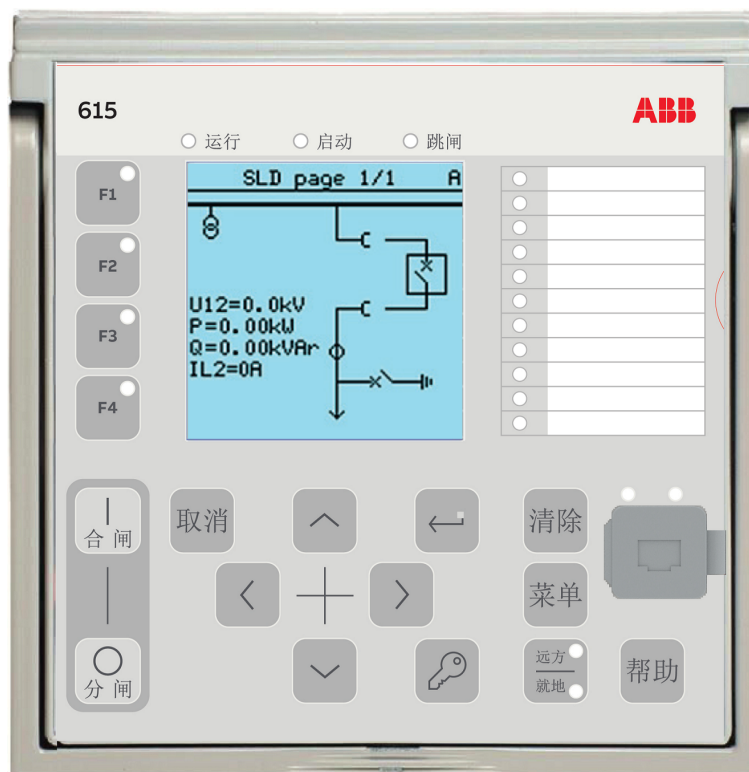


Figure 90: Standard size LHMI (CN variant)

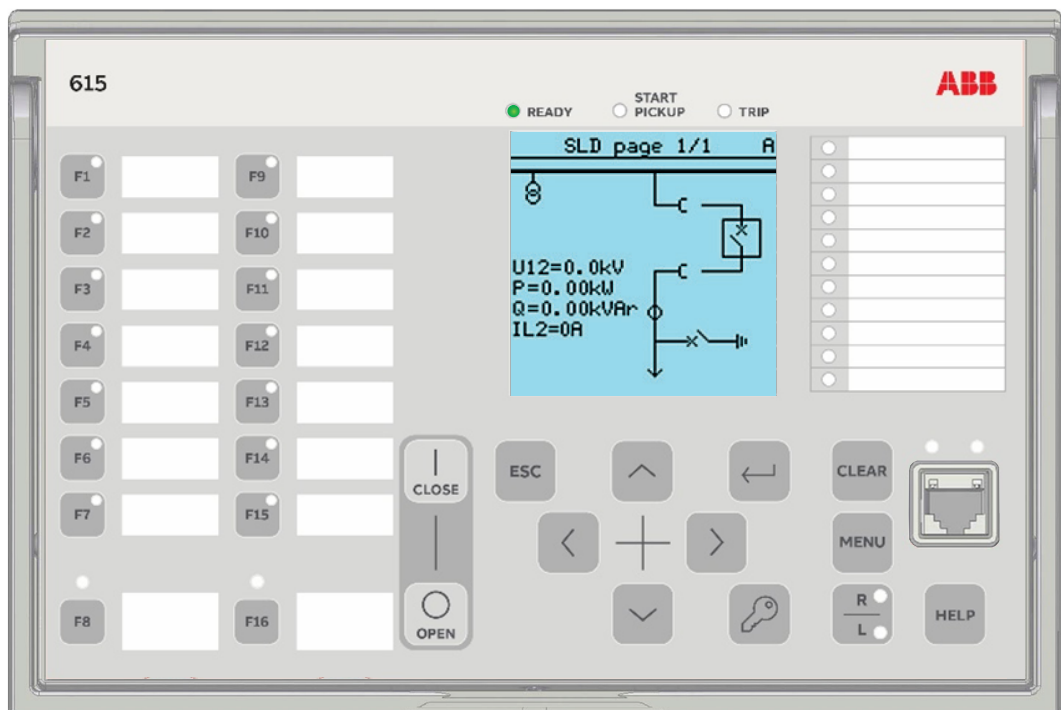


Figure 91: Wide size LHMI (IEC/ANSI variant)

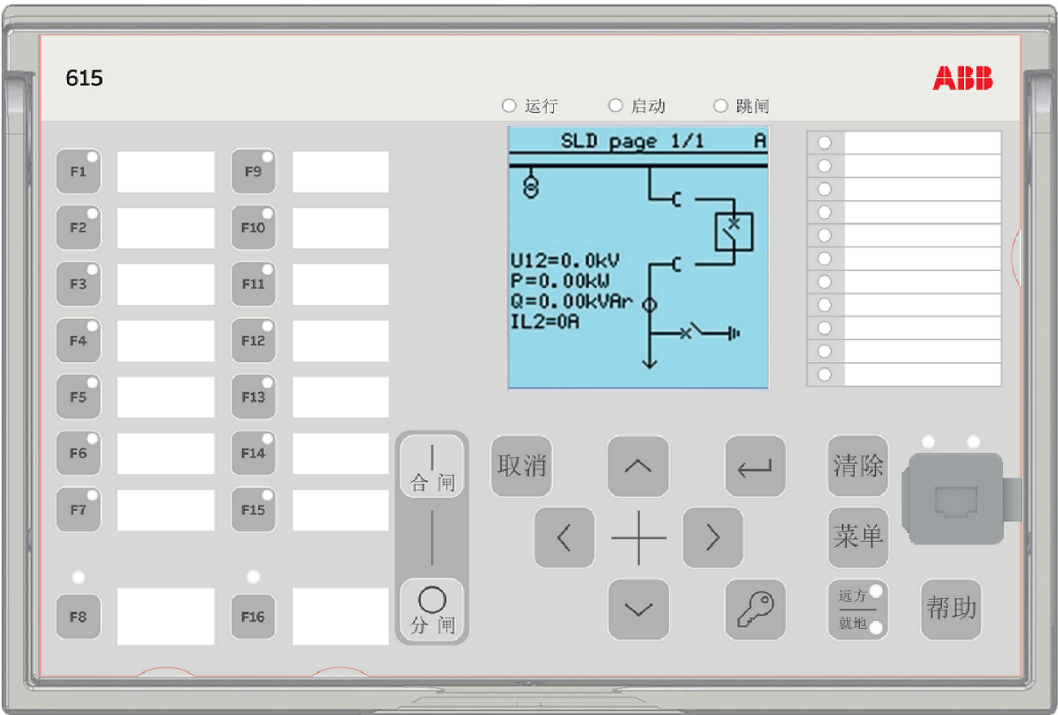


Figure 92: Wide size LHMI (CN variant)

Table 5: Display characters

Character size ²	Rows in the view	Characters per row
Small, mono-spaced (6 × 12 pixels)	10	20
Large, variable width (13 × 14 pixels)	7	8 or more

² Depending on the selected language

7.1.2 Rear side of the protection relay

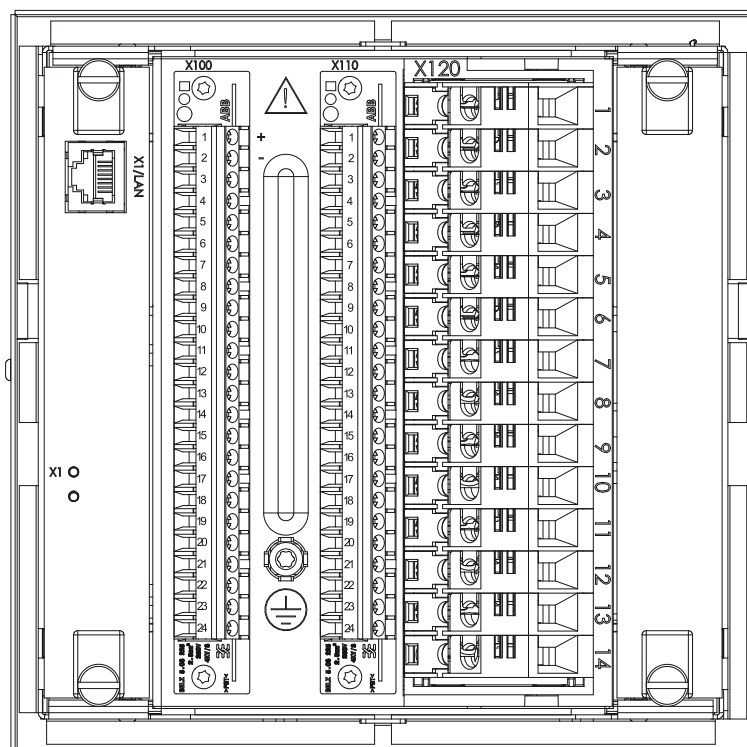


Figure 93: Rear view of a REX615 standard series protection relay with communication module

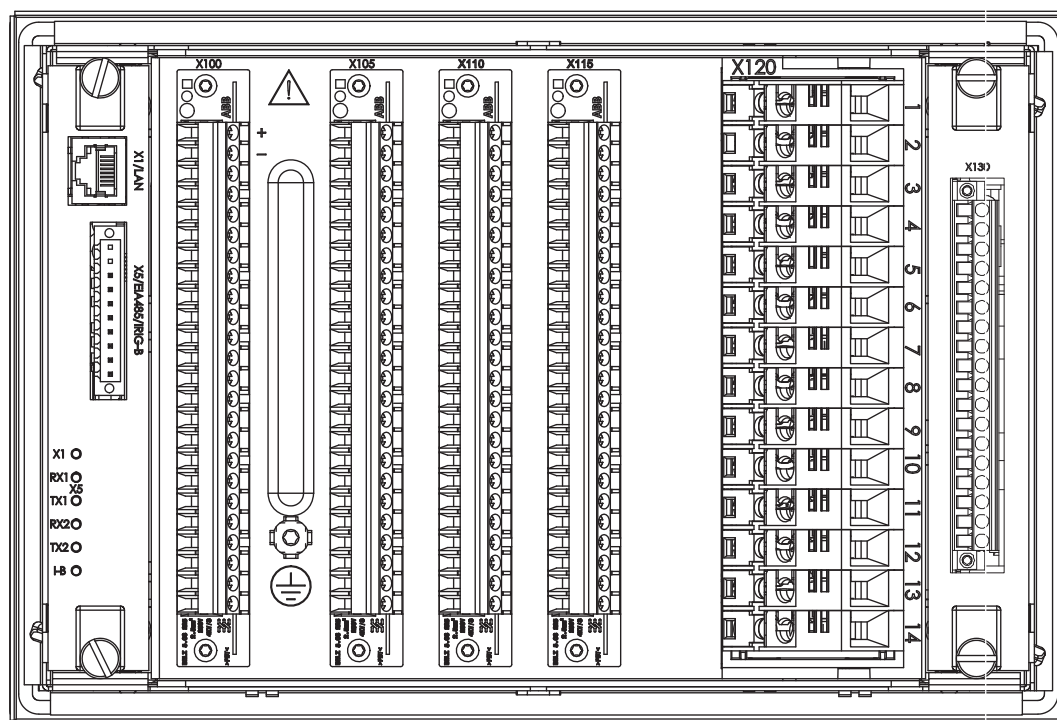


Figure 94: Rear view of REX615 wide with one wide analog input module (X120)

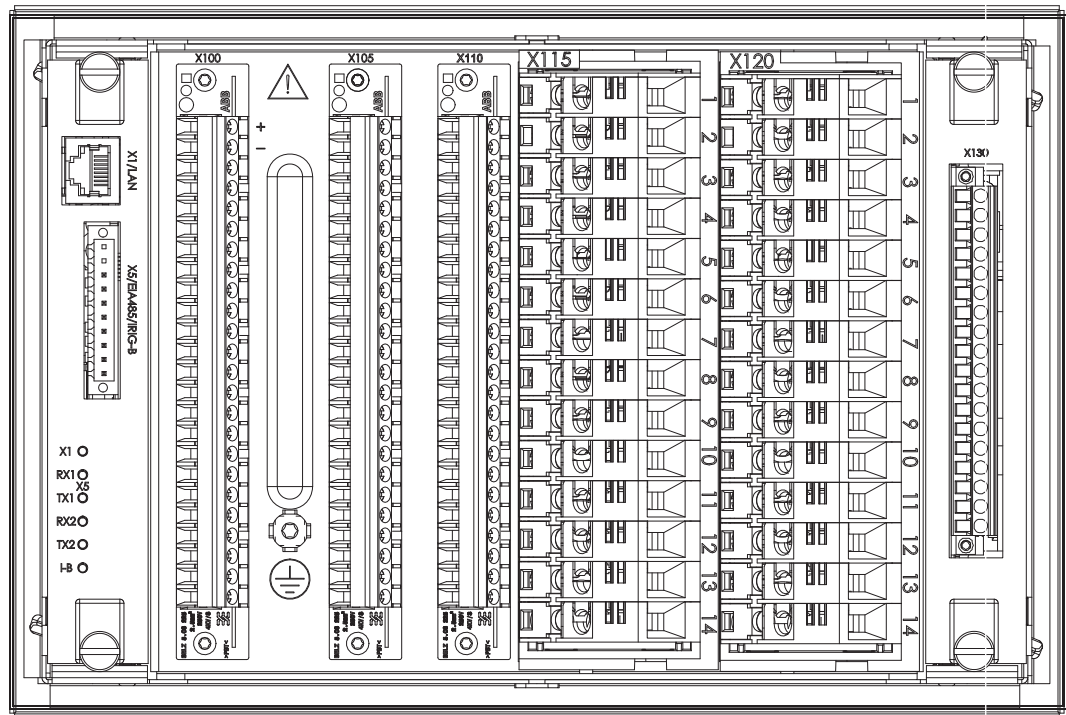
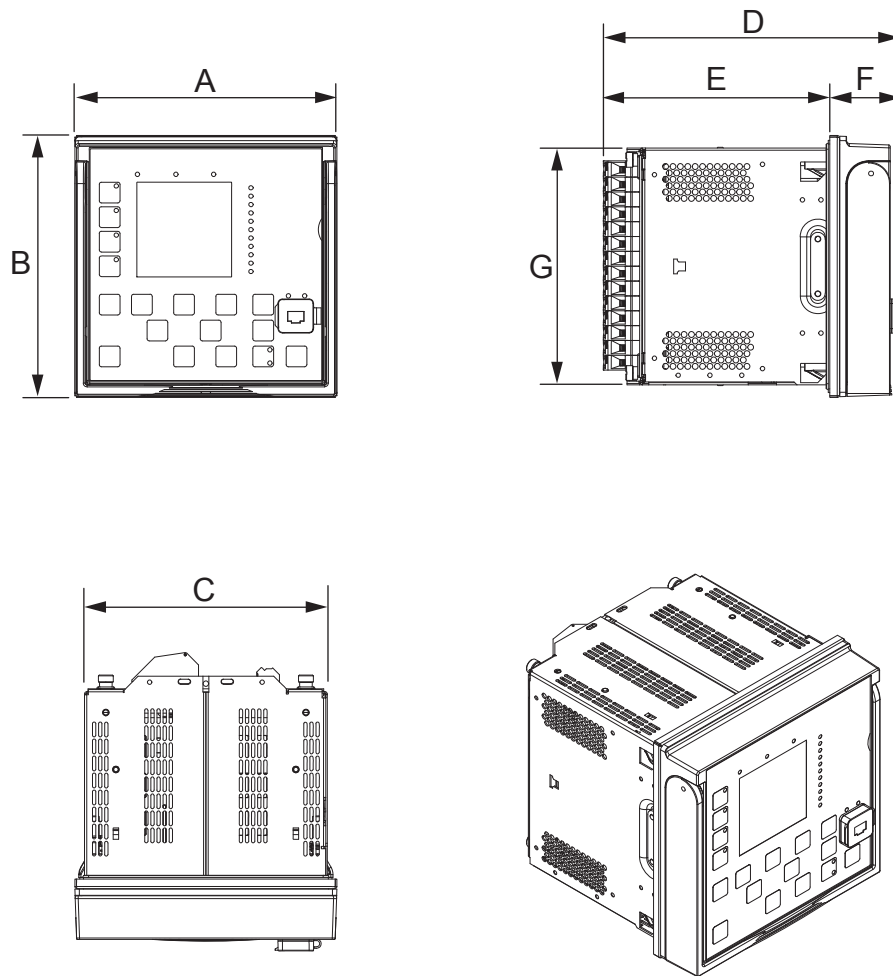


Figure 95: Rear view of REX615 wide with two wide analog input modules (X120 and X115)

7.2 Dimensions



A 177 mm (6.97 inches)
 B 177 mm (6.97 inches), 4U
 C 164 mm (6.46 inches)
 D 201 mm (7.91 inches)

E 153 mm (6.02 inches)
 F 48 mm (1.89 inches)
 G 160 mm (6.30 inches)

Figure 96: Standard size relay's main dimensions

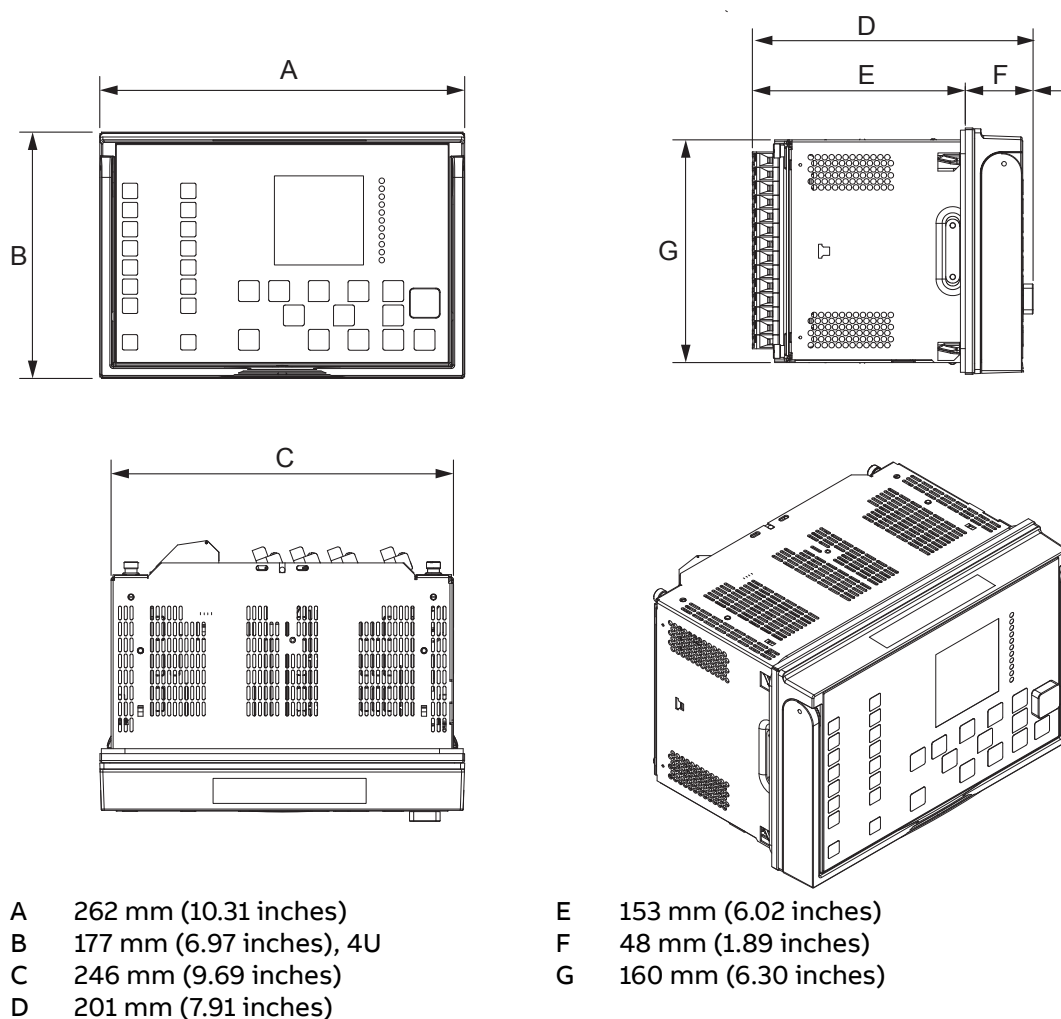


Figure 97: Wide size relay's main dimensions

Table 6: Dimensions

Description	Value	
Width (standard case)	Frame	177 mm (6.96 in)
	Case	164 mm (6.46 in)
Width (wide case)	Frame	262 mm (10.31 in)
	Case	246 mm (9.69 in)
Height	Frame	177 mm (6.97 in) (4U)
	Case	160 mm (6.30 in)
Depth	201 mm (153 mm + 48 mm) / 7.91 in (6.02 in + 1.89 in)	
Weight (standard case)	Complete protection relay	4.1 kg (9 lb)
	Plug-in unit only	2.1 kg (4.6 lb)
Weight (wide case)	Complete protection relay	max. 5.5 kg (12.1 lb)
	Plug-in unit only	max. 3.0 kg (86 lb)

7.3 Degree of protection of the protection relay

Table 7: Degree of protection of the protection relay

Description	Value
Front:	When flush mounted IP 54
Rear (Connector side):	IP 20 (with ring-lug signal connectors IP 00 or IP 10 depending on wiring)
Top and bottom:	IP 30
Sides:	IP 20

8 Accessories and ordering data

Table 8: Cables

Item	Order number
Optical sensor for arc protection, cable length 1.5 m (4.9 ft)	1MRS120534-1.5
Optical sensor for arc protection, cable length 3.0 m (9.8 ft)	1MRS120534-3
Optical sensor for arc protection, cable length 5.0 m (16.4 ft)	1MRS120534-5
Optical sensor for arc protection, cable length 7.0 m (23 ft)	1MRS120534-7
Optical sensor for arc protection, cable length 10.0 m (32.8 ft)	1MRS120534-10
Optical sensor for arc protection, cable length 15.0 m (49.2 ft)	1MRS120534-15
Optical sensor for arc protection, cable length 20.0 m (65.6 ft)	1MRS120534-20
Optical sensor for arc protection, cable length 25.0 m (82 ft)	1MRS120534-25
Optical sensor for arc protection, cable length 30.0 m (98.4 ft)	1MRS120534-30
Optical sensor for arc protection, cable length 40.0 m (131.2 ft)	1MRS120534-40
Optical sensor for arc protection, cable length 50.0 m (164 ft)	1MRS120534-50

Table 9: Mounting accessories for standard size relay

Item	Order number
Semi-flush mounting kit	1MRS050696
Inclined semi-flush mounting kit	1MRS050831
19" rack mounting kit with cut-out for one relay	1MRS050694
19" rack mounting kit with cut-out for two relays	1MRS050695
Wall mounting kit	1MRS050697
Mounting bracket for one relay with test switch RTXP in 4U Combiflex (RHGT 19" variant C)	2RCA022642P0001
Mounting bracket for one relay in 4U Combiflex (RHGT 19" variant C)	2RCA022643P0001
19" rack mounting kit for one relay and one RTXP18 test switch (the test switch is not included in the delivery)	2RCA021952A0003
19" rack mounting kit for one relay and one RTXP24 test switch (the test switch is not included in the delivery)	2RCA022561A0003
Functional earthing flange for RTD modules ³	2RCA036978A0001
Replacement kit for a Strömberg SP_J40 series relay (cut-out in the center of the installation plate)	2RCA027871A0001
Replacement kit for a Strömberg SP_J40 series relay (cut-out on the left or the right of the installation plate)	2RCA027874A0001
Replacement kit for two Strömberg SP_J3 series relays	2RCA027880A0001

Table continues on the next page

³ Cannot be used when the protection relay is mounted with the Combiflex 19" equipment frame (2RCA032826A0001)

Item	Order number
19" rack replacement kit for Strömberg SP_J3/J6 series relays (one cut-out)	2RCA027894A0001
19" rack replacement kit for Strömberg SP_J3/J6 series relays (two cut-outs)	2RCA027897A0001
Replacement kit for a Strömberg SP_J6 series relay	2RCA027881A0001
Replacement kit for three BBC S_ series relays	2RCA027882A0001
Replacement kit for a SPA 300 series relay	2RCA027885A0001
Protection cover kit	2RCA027998A0001

Table 10: Mounting accessories for wide size relay

Item	Order number
Semi-flush mounting kit	2RCA030573A0001
Inclined semi-flush mounting kit	2RCA054775A0001
19" rack mounting kit with cut-out for one relay	2RCA031135A0001
Wall mounting kit	2RCA030894A0001
19" rack mounting kit for one relay and one RTXP24 test switch (the test switch and wire harness are not included in the delivery)	2RCA032818A0001
Mounting bracket for one relay with test switch RTXP in 4U Combiflex (RHGT 19" variant C) (the test switch, wire harness and Combiflex RGHT 19" variant C are not included in the delivery)	2RCA032826A0001
Functional earthing flange for RTD modules ⁴	2RCA036978A0001
Protection cover kit	2RCA030963A0001

⁴ Cannot be used when the protection relay is mounted with the Combiflex 19" equipment frame (2RCA032826A0001)

9 Glossary

AIM	Analog input module
BI/O	Binary input/output
BIO	Binary input and output
CT	Current transformer
EMC	Electromagnetic compatibility
HW	Hardware
IEC	International Electrotechnical Commission
LCD	Liquid crystal display
LCP	Liquid crystal polymer
LED	Light-emitting diode
LHMI	Local human-machine interface
PA	Polyamide
PBT	Polybutylene terephthalate
PC	1. Personal computer 2. Polycarbonate
PSM	Power supply module
RoHS	Restriction of hazardous substances
RTD	Resistance temperature detector
SW	Software
VT	Voltage transformer



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